

MA1251- LINEAR ALGEBRA
I year ADS and AML - II Semester
Question Bank

UNIT I MATRICES AND SYSTEM OF LINEAR EQUATIONS

Matrices - Row echelon form - Rank - System of linear equations - Consistency - Gauss elimination method - Gauss Jordan method.

PART – A

1.	Determine the values of x, y and z for which $\begin{bmatrix} x & -2 \\ 1 & y \end{bmatrix} \begin{bmatrix} 4 & -1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -z \\ 13 & x+y \end{bmatrix}$
	Solution $\begin{bmatrix} x & -2 \\ 1 & y \end{bmatrix} \begin{bmatrix} 4 & -1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -z \\ 13 & x+y \end{bmatrix}$ $\begin{bmatrix} 4x-6 & -x-4 \\ 4+3y & -1+2y \end{bmatrix} = \begin{bmatrix} 2 & -z \\ 13 & x+y \end{bmatrix}$ $4x-6=2 \Rightarrow x=2$ $-x-4=-z \Rightarrow z=6$ $4+3y=13 \Rightarrow y=3$
2.	Define row echelon form of a matrix
	Solution A matrix is in row echelon form (ref) when it satisfies the following conditions. - The first non-zero element in each row, called the leading entry, is 1. - Each leading entry is in a column to the right of the leading entry in the previous row. - Rows with all zero elements, if any, are below rows having a non-zero element. Each of the matrices shown below are examples of matrices in row echelon form. $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 1 & 0 & 3 & 4 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$ are examples of echelon matrices while $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 0 \end{bmatrix}$ and $F = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ are examples of matrices that are not echelon matrices.
3.	State the Elementary row operations.
	Statement E_1: Interchange the i^{th} and the j^{th} rows. $R_i \leftrightarrow R_j$ E_2: Replace the i^{th} row by a non-zero scalar multiple of itself. $R_i \rightarrow k R_i$, where $k \neq 0$. E_3: Replace the i^{th} row by k times the j^{th} row plus the i^{th} row. $R_i \rightarrow k R_j + R_i$. E_1, E_2 and E_3 are called Elementary row operations.
4.	Reduce $A = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 7 & 8 \\ 5 & 2 & 3 \end{bmatrix}$ to an echelon form (Apr/May 2022, Nov/Dec 2022)

	Solution $A = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 7 & 8 \\ 5 & 2 & 3 \end{bmatrix}$ $\sim \begin{bmatrix} 1 & 3 & 4 \\ 0 & 1 & 0 \\ 0 & 13 & 17 \end{bmatrix} \begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow 5R_1 - R_3 \end{matrix}$ $\sim \begin{bmatrix} 1 & 3 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 17 \end{bmatrix} R_3 \rightarrow R_3 - 13R_2$ Therefore, $\begin{bmatrix} 1 & 3 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 17 \end{bmatrix}$ is an echelon form of matrix A.	
5.	Transform $A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 6 \\ 0 & -5 & 4 \end{pmatrix}$ to row-echelon form	(April/May 2023)
	Solution $A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 6 \\ 0 & -5 & 4 \end{pmatrix}$ $\sim \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 6 \\ 0 & 0 & 34 \end{pmatrix} R_3 \rightarrow R_3 + 5R_2$ $\sim \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 6 \\ 0 & 0 & 1 \end{pmatrix} R_3 \rightarrow \frac{R_3}{34}$	
6.	Find the rank of $\begin{pmatrix} 2 & 2 & 1 \\ 4 & 4 & 2 \\ 6 & 6 & 3 \end{pmatrix}$	
	Solution $A = \begin{pmatrix} 2 & 2 & 1 \\ 4 & 4 & 2 \\ 6 & 6 & 3 \end{pmatrix}$ $\sim \begin{pmatrix} 2 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{matrix} \sim \begin{pmatrix} 1 & 1 & 0.5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} R_1 \rightarrow \frac{R_1}{2}$ $R(A) = 1$	
7.	Do the equations $x - 3y - 8z = 0$, $3x + y = 0$ and $2x + 5y + 6z = 0$ have a non trivial solution? Why?	(Apr/May 2022)
	Solution	

	$ A = \begin{vmatrix} 1 & -3 & -8 \\ 3 & 1 & 0 \\ 2 & 5 & 6 \end{vmatrix} = 1(6) + 3(18) - 8(15 - 2) = -44 \neq 0$ A is non singular, therefore there is no nontrivial solution
8.	If the equations $x + 2y + z = 0$, $5x + y - z = 0$ and $x + 5y + \lambda z = 0$ have a non trivial solution, find the value of λ.
	Solution If A is singular, then the system has nontrivial solution $ A = 0 \Rightarrow \begin{vmatrix} 1 & 2 & 1 \\ 5 & 1 & -1 \\ 1 & 5 & \lambda \end{vmatrix} = 0$ $\Rightarrow (\lambda + 5) - 2(5\lambda + 1) + (25 - 1) = 0 \Rightarrow -9\lambda + 27 = 0 \Rightarrow \lambda = 3$
9.	State the necessary and sufficient condition for the consistency of a system of linear equations.
	Statement: The system of equations $AX = B$ is consistent, if and only if the coefficient matrix A and the augmented matrix $[A, B]$ are of the same rank.
10.	Test the consistency of $2x - 3y = 5$; $-4x + 6y = -10$ (April/May 2023)
	Solution $[A : B] = \left(\begin{array}{cc c} 2 & -3 & 5 \\ -4 & 6 & -10 \end{array} \right) R_2 \rightarrow R_2 + 2R_1$ $\sim \left(\begin{array}{cc c} 2 & -3 & 5 \\ 0 & 0 & 0 \end{array} \right)$ If $R[A : B] = R(A) = 1 < 2 =$ the number of variables, then system is consistent with infinite number solutions.
11.	If the augmented matrix of a system of equations is equivalent to $\begin{pmatrix} 1 & 1 & 2 & 3 \\ 0 & -3 & -1 & -2 \\ 0 & 0 & \lambda - 8 & \mu - 11 \end{pmatrix}$, find the values of λ and μ for which the system has no solution
	Solution If $R[A : B] \neq R(A)$, then system is inconsistent with no solution. $\Rightarrow \lambda - 8 = 0$ and $\mu - 11 \neq 0 \Rightarrow \lambda = 8$ and $\mu \neq 11$
12.	If the augmented matrix of a system of equations is equivalent to $\begin{pmatrix} 1 & -2 & 3 & 4 \\ 0 & 5 & 4 & 2 \\ 0 & 0 & \lambda - 2 & \mu - 3 \end{pmatrix}$, find the values of λ and μ for which the system has many solutions (Nov/Dec 2022)
	Solution If $R[A : B] = R(A) = 2 < 3 =$ the number of variables, then system is consistent with many solutions. $\Rightarrow \lambda - 2 = 0$ and $\mu - 3 = 0 \Rightarrow \lambda = 2$ and $\mu = 3$
13.	Given the linear system $2x_1 - 6ax_2 = 3$; $3ax_1 - x_2 = 3/2$, Find value(s) of a for which the system has no solutions.

	Solution $[A : B] = \begin{pmatrix} 2 & -6\alpha & 3 \\ 3\alpha & -1 & 3/2 \end{pmatrix}$ $\sim \begin{pmatrix} 1 & -3\alpha & 3/2 \\ 3\alpha & -1 & 3/2 \end{pmatrix} R_1 \rightarrow \frac{R_1}{2} \sim \begin{pmatrix} 1 & -3\alpha & 3/2 \\ 0 & 9\alpha^2 - 1 & 3(1-3\alpha)/2 \end{pmatrix} R_2 \rightarrow R_2 - 3\alpha R_1$ If $R[A : B] = 2$, $R(A) = 1$, $R[A : B] \neq R(A)$ then system is inconsistent with no solution. $\Rightarrow 9\alpha^2 - 1 = 0$ and $(3/2)(1-3\alpha) \neq 0 \Rightarrow \alpha = -(1/3)$
14.	Given the linear system $2x_1 - 6\alpha x_2 = 3$; $3\alpha x_1 - x_2 = 3/2$, Find value(s) of α for which the system has an infinite number of solutions.
	Solution $[A : B] = \begin{pmatrix} 2 & -6\alpha & 3 \\ 3\alpha & -1 & 3/2 \end{pmatrix}$ $\sim \begin{pmatrix} 1 & -3\alpha & 3/2 \\ 3\alpha & -1 & 3/2 \end{pmatrix} R_1 \rightarrow \frac{R_1}{2} \sim \begin{pmatrix} 1 & -3\alpha & 3/2 \\ 0 & 9\alpha^2 - 1 & 3(1-3\alpha)/2 \end{pmatrix} R_2 \rightarrow R_2 - 3\alpha R_1$ If $R[A : B] = R(A) = 1 < 2 =$ the number of variables, then system is consistent with infinite number solutions. $\Rightarrow 9\alpha^2 - 1 = 0$ and $(3/2)(1-3\alpha) = 0 \Rightarrow \alpha = 1/3$
15.	Given the linear system $2x_1 - 6\alpha x_2 = 3$; $3\alpha x_1 - x_2 = 3/2$, find the value of α, assuming a unique solution exists.
	Solution $[A : B] = \begin{pmatrix} 2 & -6\alpha & 3 \\ 3\alpha & -1 & 3/2 \end{pmatrix}$ $\sim \begin{pmatrix} 1 & -3\alpha & 3/2 \\ 3\alpha & -1 & 3/2 \end{pmatrix} R_1 \rightarrow \frac{R_1}{2} \sim \begin{pmatrix} 1 & -3\alpha & 3/2 \\ 0 & 9\alpha^2 - 1 & 3(1-3\alpha)/2 \end{pmatrix} R_2 \rightarrow R_2 - 3\alpha R_1$ If $R[A : B] = R(A) = 2 =$ the number of variables, then system is consistent with unique solution. $\Rightarrow 9\alpha^2 - 1 \neq 0 \Rightarrow \alpha \neq \pm(1/3)$
16.	Write the set of simultaneous equation that corresponds to the augmented matrix $[A : B] = \begin{pmatrix} 1 & 2/3 & 1/3 & -4/3 & : & 1/3 \\ 0 & 1 & -2/5 & 1 & : & -1 \\ 0 & 0 & 0 & 0 & : & 0 \end{pmatrix}$
	Solution $x_1 + \frac{2}{3}x_2 + \frac{1}{3}x_3 - \frac{4}{3}x_4 = \frac{1}{3}$ $x_2 - \frac{2}{5}x_3 + x_4 = -1$
17.	Solve the set of equations associated with the augmented matrix $\begin{pmatrix} 1 & -2 & 3 & 17 \\ 0 & 1 & 2 & -3 \\ 0 & 0 & 1 & -4 \end{pmatrix}$
	Solution The equations are $x_1 - 2x_2 + 3x_3 = 17$ $x_2 + 2x_3 = -3$ $x_3 = -4$

	By backward substitution, $x_3 = -4$, $x_2 + 2(-4) = -3 \Rightarrow x_2 = 5$ $x_1 - 2x_2 + 3x_3 = 17 \Rightarrow x_1 - 2(5) + 3(-4) = 17 \Rightarrow x_1 = 39$												
18.	For solving a linear system of equations, compare Gauss Elimination method and Gauss Jordan method.												
	<table><tr><th>S.No.</th><th>Gauss-Elimination method</th><th>Gauss – Jordan method</th></tr><tr><td>1.</td><td>Coefficient matrix is transformed into upper triangular matrix</td><td>Coefficient matrix is transformed into upper diagonal matrix</td></tr><tr><td>2.</td><td>Direct method</td><td>Direct method</td></tr><tr><td>3.</td><td>We obtain the solutions by back substitution method</td><td>No need of back substitution method</td></tr></table>	S.No.	Gauss-Elimination method	Gauss – Jordan method	1.	Coefficient matrix is transformed into upper triangular matrix	Coefficient matrix is transformed into upper diagonal matrix	2.	Direct method	Direct method	3.	We obtain the solutions by back substitution method	No need of back substitution method
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2.	Direct method	Direct method											
3.	We obtain the solutions by back substitution method	No need of back substitution method											
19.	By Gauss elimination method solve $x + y = 2$, $2x + 3y = 5$												
	Solution The given system equations can be written as $AX = B \Rightarrow \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$ $\Rightarrow [A, B] = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 3 & 5 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix} R_2 \rightarrow R_2 - 2R_1$ we have $x + y = 2$, $\Rightarrow y = 1$. Hence $x = y = 1$.												
20.	Determine the inverse of $A = \begin{pmatrix} 5 & 3 \\ 2 & 1 \end{pmatrix}$ by Jordan method												
	Solution $[A : I] = \begin{pmatrix} 5 & 3 & : & 1 & 0 \\ 2 & 1 & : & 0 & 1 \end{pmatrix}$ $\sim \begin{pmatrix} 1 & 0.6 & : & 0.2 & 0 \\ 2 & 1 & : & 0 & 1 \end{pmatrix} R_1 \rightarrow \frac{R_1}{5}$ $\sim \begin{pmatrix} 1 & 0.6 & : & 0.2 & 0 \\ 0 & 0.2 & : & -0.4 & 1 \end{pmatrix} R_2 \rightarrow R_2 - 2R_1$ $\sim \begin{pmatrix} 1 & 0.6 & : & 0.2 & 0 \\ 0 & 1 & : & 2 & -5 \end{pmatrix} R_2 \rightarrow R_2 / 0.2$ $\sim \begin{pmatrix} 1 & 0 & : & -1 & 3 \\ 0 & 1 & : & 2 & -5 \end{pmatrix} R_1 \rightarrow R_1 - 0.6R_2$ $A^{-1} = \begin{pmatrix} -1 & 3 \\ 2 & -5 \end{pmatrix}$												
PART-B													
1	Find the row reduced echelon form of matrix A and determine its rank, where $A = \begin{bmatrix} 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 5 & 6 & 7 \\ 4 & 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 & 13 \\ 14 & 15 & 16 & 17 & 18 \end{bmatrix}$ (Apr/May 2022, April/May 2023)												

Solution:

$$A = \begin{bmatrix} 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 5 & 6 & 7 \\ 4 & 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 & 13 \\ 14 & 15 & 16 & 17 & 18 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 4 & 5 & 6 \\ 0 & -1 & -2 & -3 & -4 \\ 0 & -1 & -2 & -3 & -4 \\ 0 & -7 & -14 & -21 & -28 \\ 0 & -6 & -12 & -18 & -24 \end{bmatrix} \begin{array}{l} R_2 \Rightarrow 2R_2 - 3R_1 \\ R_3 \Rightarrow R_3 - 2R_1 \\ R_4 \Rightarrow 2R_4 - 9R_1 \\ R_5 \Rightarrow R_5 - 7R_1 \end{array}$$

$$= \begin{bmatrix} 2 & 3 & 4 & 5 & 6 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 & 4 \end{bmatrix} \begin{array}{l} R_2 \Rightarrow R_2 / -1 \\ R_3 \Rightarrow R_3 / -1 \\ R_4 \Rightarrow R_4 / -7 \\ R_5 \Rightarrow R_5 / -6 \end{array}$$

$$= \begin{bmatrix} 1 & 1.5 & 2 & 2.5 & 3 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} R_1 \Rightarrow R_1 / 2 \\ R_3 \Rightarrow R_3 - R_2 \\ R_4 \Rightarrow R_4 - R_2 \\ R_5 \Rightarrow R_5 - R_2 \end{array}$$

Rank is 2

2**Find the row reduced echelon form of matrix A and determine its rank, where**

$$A = \begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{bmatrix}$$

(Nov/Dec 2022)**Solution** Now reduce matrix A to row reduced Echelon form.

$$A = \begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 0 & 1 & 3 & 2 & 0 \\ 0 & 5 & 15 & 10 & 0 \\ 0 & 10 & 18 & 8 & 0 \end{bmatrix} \begin{array}{l} R_2 \rightarrow 2R_1 - R_2 \\ R_4 \rightarrow -2R_1 + R_4 \end{array}$$

$$\sim \begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 0 & 1 & 3 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 12 & 12 & 0 \end{bmatrix} \begin{array}{l} R_3 \rightarrow R_3 - 5R_2 \\ R_4 \rightarrow 10R_2 - R_4 \end{array}$$

$$\sim \begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 0 & 1 & 3 & 2 & 0 \\ 0 & 0 & 12 & 12 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad R_3 \leftrightarrow R_4$$

	$\sim \begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 0 & 1 & 3 & 2 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad R_3 \rightarrow \frac{R_3}{12}$ Therefore, the row reduced echelon form of A is $\begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 0 & 1 & 3 & 2 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ and rank (A) = 3.
3	Find the rank of the matrix A= $\begin{pmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 3 & 1 & 1 & 3 \end{pmatrix}$
	Solution: $A = \begin{pmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 3 & 1 & 1 & 3 \end{pmatrix}$ $\sim \begin{pmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 3 & 1 & 1 & 3 \end{pmatrix} \quad R_1 \Leftrightarrow R_2$ $\sim \begin{pmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & -5 & -8 & -3 \end{pmatrix} \quad R_3 \Leftrightarrow R_3 - 3R_1$ $\sim \begin{pmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 2 & 2 \end{pmatrix} \quad R_3 \Leftrightarrow R_3 + 5R_2$ It is the row echelon form Therefore rank = 3
4	Test the consistency of the system $x_1 - 2x_2 - 3x_3 = 2; 3x_1 - 2x_2 = -1; -2x_2 - 3x_3 = 2; x_2 + 2x_3 = 1$
	Solution : In matrix notation, the system is $AX = B$, where $A = \begin{pmatrix} 1 & -2 & -3 \\ 3 & -2 & 0 \\ 0 & -2 & -3 \\ 0 & 1 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 2 \\ -1 \\ 2 \\ 1 \end{pmatrix}$. $[A : B] = \begin{pmatrix} 1 & -2 & -3 & : & 2 \\ 3 & -2 & 0 & : & -1 \\ 0 & -2 & -3 & : & 2 \\ 0 & 1 & 2 & : & 1 \end{pmatrix}$ $\sim \begin{pmatrix} 1 & -2 & -3 & : & 2 \\ 0 & 4 & 9 & : & -7 \\ 0 & -2 & -3 & : & 2 \\ 0 & 1 & 2 & : & 1 \end{pmatrix} \quad R_2 \rightarrow R_2 - 3R_1$

	$\sim \begin{pmatrix} 1 & -2 & -3 & : & 2 \\ 0 & 4 & 9 & : & -7 \\ 0 & 0 & 3 & : & -3 \\ 0 & 0 & -1 & : & 11 \end{pmatrix} \begin{matrix} R_3 \rightarrow 2R_3 + R_2 \\ R_4 \rightarrow 4R_4 - R_2 \end{matrix}$ $\sim \begin{pmatrix} 1 & -2 & -3 & : & 2 \\ 0 & 4 & 9 & : & -7 \\ 0 & 0 & 3 & : & -3 \\ 0 & 0 & 0 & : & 30 \end{pmatrix} R_4 \rightarrow 3R_4 + R_3$ $\therefore R(A : B) = 4; R(A) = 3 \Rightarrow R(A : B) \neq R(A)$. Therefore the system is inconsistent.
5	Investigate for what values of λ and μ the system of linear equation $x + 2y + z = 7, x + y + \lambda z = \mu, x + 3y - 5z = 5$ has (i) a unique solution (ii) infinite number of solutions (iii) no solution (Apr/May 2022)
	Solution: $[A, B] = \left[\begin{array}{ccc c} 1 & 2 & 1 & 7 \\ 1 & 1 & \lambda & \mu \\ 1 & 3 & -5 & 5 \end{array} \right] \sim \left[\begin{array}{ccc c} 1 & 2 & 1 & 7 \\ 0 & -1 & \lambda-1 & \mu-7 \\ 0 & 1 & -6 & -2 \end{array} \right] \begin{matrix} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{matrix}$ $\sim \left[\begin{array}{ccc c} 1 & 2 & 1 & 7 \\ 0 & -1 & \lambda-1 & \mu-7 \\ 0 & 0 & \lambda-7 & \mu-9 \end{array} \right] R_3 \rightarrow R_3 + R_2$ (i) a unique solution If $R[A : B] = R(A) = 3 =$ the number of variables, then system is consistent with unique solution. $\Rightarrow \lambda \neq 7$ and $\mu \neq 9$ (ii) infinite number of solutions If $R[A : B] = R(A) = 2 < 3 =$ the number of variables, then system is consistent with many solutions $\Rightarrow \lambda = 7$ and $\mu = 9$ (iii) no solution If $R[A : B] \neq R(A)$, then system is inconsistent with no solution. $\Rightarrow \lambda = 7$ and $\mu \neq 9$
6	Find the value of c for which the system $x_1 + x_2 + x_3 = 2; 2x_1 + x_2 + 2x_3 = 5; 4x_1 + 3x_2 + 4x_3 = c$ is (i) consistent (ii) inconsistent
	Solution : In matrix notation our system is $AX = B$, where $A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 4 & 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 2 \\ 5 \\ c \end{pmatrix}$. $(A : b) = \begin{pmatrix} 1 & 1 & 1 & 2 \\ 2 & 1 & 2 & 5 \\ 4 & 3 & 4 & c \end{pmatrix}$ Then $\sim \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & -1 & 0 & 1 \\ 0 & -1 & 0 & c-8 \end{pmatrix} \begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 4R_1 \end{matrix}$ $\sim \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & c-8 \end{pmatrix} R_2 \rightarrow -R_2$

	$\sim \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & c-9 \end{pmatrix} R_3 \rightarrow R_3 + R_2$ If $c \neq 9$ then $R(A) = 2$ but $R[A : B] = 3$ and the equations are inconsistent. If $c = 9$ then $R(A) = R[A : B] = 2 < 3$, the system of equations is consistent and the system has infinitely many solutions and it is given by $x_1 + x_2 + x_3 = 2$ $x_2 = -1.$ Let $x_3 = \lambda$, then $x_1 = 3 - \lambda$; $x_2 = -1$
7	Solve the following system of equations, if consistent $x_1 + 2x_2 - x_3 - 5x_4 = 4$; $x_1 + 3x_2 - 2x_3 - 7x_4 = 5$; $2x_1 - x_2 + 3x_3 = 3$
	Solution $[A : B] = \begin{pmatrix} 1 & 2 & -1 & -5 & : & 4 \\ 1 & 3 & -2 & -7 & : & 5 \\ 2 & -1 & 3 & 0 & : & 3 \end{pmatrix}$ $\sim \begin{pmatrix} 1 & 2 & -1 & -5 & : & 4 \\ 0 & 1 & -1 & -2 & : & 1 \\ 0 & -5 & 5 & 10 & : & -5 \end{pmatrix} R_2 \rightarrow R_2 - R_1$ $R_3 \rightarrow R_3 - 2R_1$ $\sim \begin{pmatrix} 1 & 2 & -1 & -5 & : & 4 \\ 0 & 1 & -1 & -2 & : & 1 \\ 0 & 0 & 0 & 0 & : & 0 \end{pmatrix} R_3 \rightarrow R_3 + 5R_2$ $\therefore R[A : B] = 2$; $R(A) = 2 < 4 =$ the number of variables. Therefore, the system is consistent with many solutions. From the first two rows of the equivalent matrix $x_1 + 2x_2 - x_3 - 5x_4 = 4$ $x_2 - x_3 - 2x_4 = 1$ As there are only two equations, we can solve for only two variables. Let $x_3 = k$ & $x_4 = k'$ $x_2 - x_3 - 2x_4 = 1 \Rightarrow x_2 - k - 2k' = 1 \Rightarrow x_2 = 1 + k + 2k'$ $x_1 + 2x_2 - x_3 - 5x_4 = 4 \Rightarrow x_1 + 2(1 + k + 2k') - k - 5k' = 4 \Rightarrow x_1 = 2 - k + k'$
8	Find the values of k for which the equations $x + y + z = 1$; $x + 2y + 3z = k$ & $x + 5y + 9z = k^2$ have a solution. For these values of k, find the solution also. (Nov/Dec 2022)
	Solution $[A : B] = \begin{pmatrix} 1 & 1 & 1 & : & 4 \\ 1 & 2 & 3 & : & k \\ 1 & 5 & 9 & : & k^2 \end{pmatrix}$ $\sim \begin{pmatrix} 1 & 1 & 1 & : & 4 \\ 0 & 1 & 2 & : & k-1 \\ 0 & 4 & 8 & : & k^2-1 \end{pmatrix} R_2 \rightarrow R_2 - R_1$ $R_3 \rightarrow R_3 - R_1$ $\sim \begin{pmatrix} 1 & 1 & 1 & : & 4 \\ 0 & 1 & 2 & : & k-1 \\ 0 & 0 & 0 & : & k^2-4k+3 \end{pmatrix} R_3 \rightarrow R_3 - 4R_2$ $R(A) = 2$. If the system possesses a solution. $R[A : B]$ must also be 2.

	Therefore, the last row of the last equivalent matrix must contain only zeros. (i.e) $k^2 - 4k + 3 = 0$ (i.e) $k = 1$ or 3 For these values of k, $R[A : B] = 2$; $R(A) = 2 < 3 =$ the number of variables. Therefore, the system is consistent with many solutions. <u>Case1:</u> $k = 1$ From the first two rows of the equivalent matrix $x + y + z = 1$; $y + 2z = 0$ Let $z = \lambda$ $y + 2z = 0 \Rightarrow y = -2\lambda$ $x + y + z = 1 \Rightarrow x - 2\lambda + \lambda = 1 \Rightarrow x = 1 + \lambda$ <u>Case2:</u> $k = 3$ From the first two rows of the equivalent matrix $x + y + z = 1$; $y + 2z = 2$ Let $z = \mu$ $y + 2z = 2 \Rightarrow y = 2 - 2\mu$ $x + y + z = 1 \Rightarrow x + 2 - 2\mu + \mu = 1 \Rightarrow x = -1 + \mu$
9	Find the value of k such that the following system of equations has (i) a unique solution (ii) many solutions and (iii) no solution $kx + y + z = 1$; $x + ky + z = 1$; $x + y + kz = 1$; Solution $[A : B] = \begin{pmatrix} k & 1 & 1 & : & 1 \\ 1 & k & 1 & : & 1 \\ 1 & 1 & k & : & 1 \end{pmatrix} \begin{matrix} R_2 \Rightarrow kR_2 - R_1, \quad k \neq 0 \\ R_3 \Rightarrow kR_3 - R_1 \end{matrix}$ $\sim \begin{pmatrix} k & 1 & 1 & : & 1 \\ 0 & k^2 - 1 & k - 1 & : & k - 1 \\ 0 & k - 1 & k^2 - 1 & : & k - 1 \end{pmatrix} \begin{matrix} R_2 \Rightarrow \frac{R_2}{k-1}, \quad k \neq 1 \\ R_3 \Rightarrow \frac{R_3}{k-1} \end{matrix}$ $\sim \begin{pmatrix} k & 1 & 1 & : & 1 \\ 0 & k+1 & 1 & : & 1 \\ 0 & 1 & k+1 & : & 1 \end{pmatrix} R_3 \Rightarrow (k+1)R_3 - R_2$ $\sim \begin{pmatrix} k & 1 & 1 & : & 1 \\ 0 & k+1 & 1 & : & 1 \\ 0 & 0 & (k+1)^2 - 1 & : & k \end{pmatrix} = \begin{pmatrix} k & 1 & 1 & : & 1 \\ 0 & k+1 & 1 & : & 1 \\ 0 & 0 & k^2 + 2k & : & k \end{pmatrix}$ (i) a unique solution If $R[A : B] = R(A) = 3 =$ the number of variables, then system is consistent with unique solution. $\Rightarrow k^2 + 2k \neq 0$ and $k \neq 0 \Rightarrow k \neq 0, 1, -2$ (ii) many solutions If $R[A : B] = R(A) = 2 < 3 =$ the number of variables, then system is consistent with many solutions. $\Rightarrow k^2 + 2k = 0$ and $k = 0 \Rightarrow k = 0$ But $k \neq 0$. Clearly if $k = 1$, then the system of equations reduces to single equation $x + y + z = 1$ Thus the system is consistent with many solutions. (iii) no solution If $R[A : B] \neq R(A)$, then system is inconsistent with no solution. $\Rightarrow k^2 + 2k = 0$ and $k \neq 0 \Rightarrow k = -2$
10	Solve by Gauss Jordan method, $2x + y + 2z = 10$, $x + 2y + z = 8$, $3x + y - z = 2$ (Apr/May 2022)

Solution:

$$\begin{aligned}
[A, B] &= \left[\begin{array}{ccc|c} 2 & 1 & 2 & 10 \\ 1 & 2 & 1 & 8 \\ 3 & 1 & -1 & 2 \end{array} \right] \sim \left[\begin{array}{ccc|c} 2 & 1 & 2 & 10 \\ 0 & 3 & 0 & 6 \\ 0 & -1 & -8 & -34 \end{array} \right] \begin{array}{l} R_2 \rightarrow 2R_2 - R_1 \\ R_3 \rightarrow 2R_3 - 3R_1 \end{array} \\
&\sim \left[\begin{array}{ccc|c} 2 & 1 & 2 & 10 \\ 0 & 3 & 0 & 6 \\ 0 & 0 & -24 & -96 \end{array} \right] \begin{array}{l} R_3 \rightarrow 3R_3 + R_2 \end{array} \sim \left[\begin{array}{ccc|c} 2 & 1 & 2 & 10 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 / 3 \\ R_3 \rightarrow R_3 / -24 \end{array} \\
&\sim \left[\begin{array}{ccc|c} 2 & 1 & 0 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{array} \right] \begin{array}{l} R_1 \rightarrow R_1 - 2R_3 \\ R_1 \rightarrow R_1 - 2R_2 \end{array} \sim \left[\begin{array}{ccc|c} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{array} \right] \\
&\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{array} \right] \begin{array}{l} R_1 \rightarrow R_1 / 2 \end{array}
\end{aligned}$$

$$x = 0; y = 2; z = 4$$

11 Solve the system of the following equations using Gauss Jordan method

$$2x_1 + 2x_2 - x_3 + x_4 = 4, 4x_1 + 3x_2 - x_3 + 2x_4 = 6, 8x_1 + 5x_2 - 3x_3 + 4x_4 = 12 \text{ and}$$

$$3x_1 + 3x_2 - 2x_3 + 2x_4 = 6$$

(Apr/May 2022, April/May 2023)**Solution:**

$$\begin{aligned}
(A, B) &= \left[\begin{array}{cccc|c} 2 & 2 & -1 & 1 & 4 \\ 4 & 3 & -1 & 2 & 6 \\ 8 & 5 & -3 & 4 & 12 \\ 3 & 3 & -2 & 2 & 6 \end{array} \right] \\
&= \left[\begin{array}{cccc|c} 1 & 1 & -0.5 & 0.5 & 2 \\ 4 & 3 & -1 & 2 & 6 \\ 8 & 5 & -3 & 4 & 12 \\ 3 & 3 & -2 & 2 & 6 \end{array} \right] R_1 \rightarrow R_1 / 2 \\
&= \left[\begin{array}{cccc|c} 1 & 1 & -0.5 & 0.5 & 2 \\ 0 & -1 & 1 & 0 & -2 \\ 0 & -3 & 1 & 0 & -4 \\ 0 & 0 & -0.5 & 0.5 & 0 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 4R_1 \\ R_3 \rightarrow R_3 - 8R_1 \\ R_4 \rightarrow R_4 - 3R_1 \end{array} \\
&= \left[\begin{array}{cccc|c} 1 & 1 & -0.5 & 0.5 & 2 \\ 0 & 1 & -1 & 0 & 2 \\ 0 & -3 & 1 & 0 & -4 \\ 0 & 0 & -0.5 & 0.5 & 0 \end{array} \right] R_2 \rightarrow R_2(-1) \\
&= \left[\begin{array}{cccc|c} 1 & 0 & -0.5 & 0.5 & 0 \\ 0 & 1 & -1 & 0 & 2 \\ 0 & 0 & -2 & 0 & 2 \\ 0 & 0 & -0.5 & 0.5 & 0 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_1 - R_2 \\ R_3 \rightarrow R_3 + 3R_2 \end{array}
\end{aligned}$$

	$= \begin{bmatrix} 1 & 0 & 0 & 0.5 & 0.5 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0.5 & -0.5 \end{bmatrix} \begin{array}{l} R_1 \rightarrow R_1 - 0.5R_3 \\ R_2 \rightarrow R_2 + R_3 \\ R_4 \rightarrow R_4 + 0.5R_3 \end{array}$ $= \begin{bmatrix} 1 & 0 & 0 & 0.5 & 0.5 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \begin{array}{l} R_1 \rightarrow R_1 - 0.5R_4 \\ R_4 \rightarrow R_4 / 0.5 \end{array}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$ $\therefore x_1 = 1, x_2 = 1, x_3 = -1, x_4 = -1$
12	Solve the following system of equations by Gauss Jordan method $27x + 16y - z = 85,$ $x + y + 54z = 110, 6x + 15y + 2z = 72$
	$[A, B] = \left[\begin{array}{ccc c} 27 & 16 & -1 & 85 \\ 1 & 1 & 54 & 110 \\ 6 & 15 & 2 & 72 \end{array} \right]$ $\sim \left[\begin{array}{ccc c} 1 & 1 & 54 & 110 \\ 27 & 16 & -1 & 85 \\ 6 & 15 & 2 & 72 \end{array} \right] \quad R_1 \leftrightarrow R_2$ $\sim \left[\begin{array}{ccc c} 1 & 1 & 54 & 110 \\ 0 & -11 & -1459 & -2885 \\ 0 & 9 & -322 & -588 \end{array} \right] \quad R_2 \rightarrow R_2 - 27R_1; \quad R_3 \rightarrow R_3 - 6R_1$ $\sim \left[\begin{array}{ccc c} 1 & 1 & 54 & 110 \\ 0 & -11 & -1459 & -2885 \\ 0 & 0 & -16673 & -32433 \end{array} \right] \quad R_3 \rightarrow 11R_3 + 9R_2$ $\sim \left[\begin{array}{ccc c} 1 & 16679 & 0 & 1678240 \\ 0 & -183403 & 0 & -23775698 \\ 0 & 0 & -16673 & -32433 \end{array} \right] \quad R_1 \rightarrow 16673R_1 + 54R_3; \quad R_2 \rightarrow 16673R_2 - 1459R_3$ $\sim \left[\begin{array}{ccc c} 183403 & 0 & 0 & -887606162 \\ 0 & -183403 & 0 & -23775698 \\ 0 & 0 & -16673 & -32433 \end{array} \right] \quad R_1 \rightarrow 183403R_1 + 16679R_2$ By backward subs. $x = 0.6939$ $y = 4.26306$ $z = 1.9452$
13	Apply Gauss Jordan method to find the solution of the following system: $x + 3y + 3z = 16, x + 4y + 3z = 18, x + 3y + 4z = 19.$

Solution:

$$\text{Given } A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 16 \\ 18 \\ 19 \end{bmatrix}$$

$$(A, B) = \left[\begin{array}{ccc|c} 1 & 3 & 3 & 16 \\ 1 & 4 & 3 & 18 \\ 1 & 3 & 4 & 19 \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 1 & 3 & 3 & 16 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right] \begin{array}{l} R_2 - R_1 \\ R_3 - R_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 3 & 10 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right] R_1 - 3R_2$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right] R_1 - 3R_3$$

Therefore x= 1,y= 2,z=3.

14 Solve Gauss Jordan method, the equations $2x + y + 4z = 12$, $8x - 3y + 2z = 20$, $4x + 11y - z = 33$

Solution:

$$(A, B) = \left[\begin{array}{ccc|c} 2 & 1 & 4 & 12 \\ 8 & -3 & 2 & 20 \\ 4 & 11 & -1 & 33 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 2 & 1 & 4 & 12 \\ 0 & -7 & -14 & -28 \\ 0 & 9 & -9 & 9 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 4R_1, R_3 \rightarrow R_3 - 2R_1 \end{array}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 2 & 1 & 4 & 12 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & -1 & 1 \end{array} \right] \begin{array}{l} R_2 \rightarrow \frac{R_2}{-7}, R_3 \rightarrow \frac{R_3}{9} \end{array}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 2 & 1 & 4 & 12 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 3 & 3 \end{array} \right] R_3 \rightarrow R_2 - R_3$$

$$\Rightarrow \left[\begin{array}{ccc|c} 6 & 3 & 0 & 24 \\ 0 & 3 & 0 & 6 \\ 0 & 0 & 3 & 3 \end{array} \right] \begin{array}{l} R_1 \rightarrow 3R_1 - 4R_3, R_2 \rightarrow 3R_2 - 2R_3 \end{array}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 6 & 0 & 0 & 18 \\ 0 & 3 & 0 & 6 \\ 0 & 0 & 3 & 3 \end{array} \right] R_1 \rightarrow R_1 - R_2$$

$$3z = 3; 2y = 6; 6x = 18$$

Hence the solution is, $x = 1, y = 2, z = 3$

15 Solve the following linear system of equations by Gauss elimination method $2x + 3y + z = -1$, $5x + y + z = 9$, $3x + 2y + 4z = 11$ (April/May 2023)

Solution:

$$[A, B] = \left[\begin{array}{ccc|c} 2 & 3 & 1 & -1 \\ 5 & 1 & 1 & 9 \\ 3 & 2 & 4 & 11 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 2 & 3 & 1 & -1 \\ 0 & -13 & -3 & 23 \\ 0 & -5 & 5 & 25 \end{array} \right] \quad R_2 \rightarrow 2R_2 - 5R_1; R_3 \rightarrow 2R_3 - 3R_1$$

$$\sim \left[\begin{array}{ccc|c} 2 & 3 & 1 & -1 \\ 0 & -13 & -3 & 23 \\ 0 & 0 & 80 & 210 \end{array} \right] \quad R_3 \rightarrow 13R_3 - 5R_2$$

By Backward Subs.

$$80z = 210 \Rightarrow z = \frac{210}{80} \Rightarrow z = 2.625$$

$$-13y - 3z = 23 \Rightarrow y = 2.375$$

$$2x + 3y + z = -1 \Rightarrow x = -5.375$$

16 Solve the system of equations by Gauss Elimination method
 $3x+y-z=3, 2x-8y+z=-5, x-2y+9z=8.$

Solution:The given system can be written as matrix form $AX=B$,

$$\text{where the coefficient matrix } A = \begin{pmatrix} 1 & -2 & 9 \\ 3 & 1 & -1 \\ 2 & -8 & 1 \end{pmatrix}, X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, B = \begin{pmatrix} 8 \\ 3 \\ -5 \end{pmatrix}$$

$$\text{The augmented matrix is } [A | I] = \left(\begin{array}{ccc|c} 1 & -2 & 9 & 8 \\ 3 & 1 & -1 & 3 \\ 2 & -8 & 1 & -5 \end{array} \right)$$

by Gauss elimination method,

$$[A | I] = \left(\begin{array}{ccc|c} 1 & -2 & 9 & 8 \\ 0 & 7 & -28 & -21 \\ 0 & -4 & -17 & -21 \end{array} \right) \quad \begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array}$$

$$[A | I] = \left(\begin{array}{ccc|c} 1 & -2 & 9 & 8 \\ 0 & 7 & -28 & -21 \\ 0 & 0 & -231 & -231 \end{array} \right) \quad R_3 \rightarrow 4R_2 + 7R_3$$

$$-231z = -231 \Rightarrow z = 1$$

$$7y - 28z = -21 \Rightarrow 7y = 7 \Rightarrow y = 1$$

$$x - 2y + 9z = 8 \Rightarrow x - 2 + 9 = 8 \Rightarrow x = 1$$

Hence the solution is $x=1, y=1, z=1$

17	Solve the system of equations by Gauss Elimination method $5x_1 + x_2 + x_3 + x_4 = 4; x_1 + 7x_2 + x_3 + x_4 = 12; x_1 + x_2 + 6x_3 + x_4 = -5; x_1 + x_2 + x_3 + 4x_4 = -6$
	Solution: $(A, B) = \left[\begin{array}{cccc c} 5 & 1 & 1 & 1 & 4 \\ 1 & 7 & 1 & 1 & 12 \\ 1 & 1 & 6 & 1 & -5 \\ 1 & 1 & 1 & 4 & -6 \end{array} \right] \sim \left[\begin{array}{cccc c} 5 & 1 & 1 & 1 & 4 \\ 0 & 34 & 4 & 4 & 56 \\ 0 & 4 & 29 & 4 & -29 \\ 0 & 4 & 4 & 19 & -34 \end{array} \right] \begin{array}{l} R_2 \rightarrow 5R_2 - R_1 \\ R_3 \rightarrow 5R_3 - R_1 \\ R_4 \rightarrow 5R_4 - R_1 \end{array}$ $\sim \left[\begin{array}{cccc c} 5 & 1 & 1 & 1 & 4 \\ 0 & 34 & 4 & 4 & 56 \\ 0 & 0 & 970 & 120 & -1210 \\ 0 & 0 & 120 & 630 & -1380 \end{array} \right] \begin{array}{l} R_3 \rightarrow 34R_3 - 4R_2 \\ R_4 \rightarrow 34R_4 - 4R_2 \end{array}$ $\sim \left[\begin{array}{cccc c} 5 & 1 & 1 & 1 & 4 \\ 0 & 17 & 2 & 2 & 28 \\ 0 & 0 & 97 & 12 & -121 \\ 0 & 0 & 0 & 5967 & -11934 \end{array} \right] \begin{array}{l} R_4 \rightarrow 970R_4 - 120R_3 \end{array}$ Hence $5967x_4 = -11934$ $x_4 = -2$ $97x_3 + 12x_4 = -121$ $x_3 = -1$ $17x_2 + 2x_3 + 2x_4 = 28$ $x_2 = 2$ $5x_1 + x_2 + x_3 + x_4 = 4$ $x_1 = 1$ Hence the solution is $x_1 = 1; x_2 = 2; x_3 = -1; x_4 = -2$
18	Solve the following linear system of equations by Gauss elimination method $x + 2y - 5z = -9,$ $3x - y + 2z = 5, 2x + 3y - z = 3$
	Solution: $[A, B] = \left[\begin{array}{ccc c} 1 & 2 & -5 & -9 \\ 3 & -1 & 2 & 5 \\ 2 & 3 & -1 & 3 \end{array} \right]$ $\sim \left[\begin{array}{ccc c} 1 & 2 & -5 & -9 \\ 0 & -7 & 17 & 32 \\ 0 & -1 & 9 & 21 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 3R_1; R_3 \rightarrow R_3 - 2R_1 \end{array}$ $\sim \left[\begin{array}{ccc c} 1 & 2 & -5 & -9 \\ 0 & -7 & 17 & 32 \\ 0 & 0 & 46 & 115 \end{array} \right] \begin{array}{l} R_3 \rightarrow 7R_3 - R_2 \end{array}$ By Backward Subs.

	$46z = 115 \Rightarrow z = \frac{115}{46} \Rightarrow z = 2.5$ $-7y + 17z = 32 \Rightarrow y = 1.5$ $x + 2y - 5z = -9 \Rightarrow x = 0.5$	
19	Solve the system of equations by Gauss Elimination method $x_1 - 2x_2 - 3x_3 = 2, \quad 3x_1 - 2x_2 = -1, \quad -2x_2 - 3x_3 = 2$	(Nov/Dec 2022)
	Solution: $[A, B] = \left[\begin{array}{ccc c} 1 & -2 & -3 & 2 \\ 3 & -2 & 0 & -1 \\ 0 & -2 & -3 & 2 \end{array} \right]$ $\sim \left[\begin{array}{ccc c} 1 & -2 & -3 & 2 \\ 0 & 4 & 9 & -7 \\ 0 & -2 & -3 & 2 \end{array} \right] \quad R_2 \rightarrow R_2 - 3R_1$ $\sim \left[\begin{array}{ccc c} 1 & -2 & -3 & 2 \\ 0 & 4 & 9 & -7 \\ 0 & 0 & 3 & -3 \end{array} \right] \quad R_3 \rightarrow 2R_3 + R_2$ By Backward Subs. $x_1 - 2x_2 - 3x_3 = 2$ $4x_2 + 9x_3 = -7$ $3x_3 = -3$ $\therefore x_1 = 0 ; x_2 = 0.5 ; x_3 = -1$	
UNIT II VECTOR SPACES Vector spaces, Subspaces, Linear combinations, Linear independence and linear dependence, Bases and dimensions.		
PART – A		
1.	Define Vector space.	(Nov/Dec 2022)
	Solution: A vector space (or linear space) V over a field F consists of a set on which two operations (called addition and scalar multiplication, respectively) are defined so that for each pair of elements x, y in V there is a unique element x + y in V, and for each element a in F and each element x in V there is a unique element ax in V, such that the following conditions hold. 1. $x + y = y + x, \forall x, y \in V$ (commutativity of addition). 2. $(x + y) + z = x + (y + z), \forall x, y, z \in V$ (associativity of addition). 3. There exists an element in V denoted by 0 such that $x + 0 = x, \forall x \in V$. 4. There exists an element y in V such that $x + y = 0, \forall x \in V$. 5. $1x = x, \forall x \in V$ 6. $(ab)x = a(bx), \forall x \in V \text{ and } a, b \in F$ 7. $a(x + y) = ax + ay, \forall x, y \in V \text{ and } a \in F$ 8. $(a + b)x = ax + bx, \forall x \in V \text{ and } a, b \in F$	
2.	Let W be the union of the first and third quadrant in the xy-plane i.e. $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix}, xy \geq 0 \right\}$ (a) If u is in W and c is any scalar, then is cu in W? why? (b) Find the specific vectors u and v in W such that u + v is not in W.	

	Solution: (a) Let $u = \begin{bmatrix} x \\ y \end{bmatrix} \in W$, $xy \geq 0$ Let c be any scalar (+ve or -ve) then $cu = c \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} cx \\ cy \end{bmatrix} = c^2 xy \geq 0 \therefore cu \in W$ (b) Let $u = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, v = \begin{bmatrix} -1 \\ -5 \end{bmatrix}, u, v \in W$ but $u + v = \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} -1 \\ -5 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \notin W$
3.	Define Subspace. Give some examples. (Apr/May 2022) Solution: A subspace of a vector space V is a subset H of V that has three properties: 1. The zero vector of V is in H. 2. H is closed under vector addition. i.e., $u + v \in H, \forall u, v \in H$ 3. H is closed under multiplication by scalars. i.e., $cu \in H, \forall c \in F, u \in H$ Examples: 1. The set of all square matrices $M_{n \times n}(R)$ is a subspace of set of all matrices $M_{m \times n}(R)$. 2. The set of all polynomials of degree at most n with real coefficient $P_n(R)$ is a subspace of $P(R)$.
4.	If $V = R^3$, then verify whether $W = \{(a_1, a_2, a_3) / 2a_1 - 7a_2 + a_3 = 0\}$ is a subspace or not. Solution: Let $w_1 = (a_1, a_2, a_3), w_2 = (b_1, b_2, b_3); w_1, w_2 \in W$ $2a_1 - 7a_2 + a_3 = 0$ & $2b_1 - 7b_2 + b_3 = 0$ $\alpha w_1 + \beta w_2 = \alpha(a_1, a_2, a_3) + \beta(b_1, b_2, b_3)$ $= (\alpha a_1, \alpha a_2, \alpha a_3) + (\beta b_1, \beta b_2, \beta b_3) = (\alpha a_1 + \beta b_1, \alpha a_2 + \beta b_2, \alpha a_3 + \beta b_3)$ Consider $2(\alpha a_1 + \beta b_1) - 7(\alpha a_2 + \beta b_2) + (\alpha a_3 + \beta b_3) = \alpha(2a_1 - 7a_2 + a_3) + \beta(2b_1 - 7b_2 + b_3)$ $= \alpha \cdot 0 + \beta \cdot 0 = 0$ $\Rightarrow \alpha w_1 + \beta w_2 \in W \therefore W \text{ is a subspace of } V.$
5.	What are the possible subspaces of R^2? Solution: Subspaces of R^2 are (i) $\{(0, 0)\}$ is the subspace of R^2 (ii) R^2 is the subspace of R^2 (iii) set of lines through the origin.
6.	Determine whether the following sets are subspaces of R^3 (a) R^2 (b) $W = \{(a_1, a_2, a_3) \in R^3 : a_1 = a_3 + 2\}$. Solution: (a) Since the vectors in R^3 have three entries, whereas the vectors in R^2 have only two, The vector space R^2 is not a subspace of R^3, because R^2 is not even a subset of R^3. (b) Let $x = (a_1, a_2, a_3) \in W$, where $a_1 = a_3 + 2$ $y = (b_1, b_2, b_3) \in W, \text{ where } b_1 = b_3 + 2.$ $x + y = (a_1, a_2, a_3) + (b_1, b_2, b_3)$ $= (a_1 + b_1, a_2 + b_2, a_3 + b_3)$ Now, $a_1 + b_1 = (a_3 + 2) + (b_3 + 2) = a_3 + b_3 + 4 \neq a_3 + b_3 + 2$ $\Rightarrow x + y \notin W. \therefore [I \text{ coordinate of } x + y \neq (III \text{ coordinate of } x + y) + 2]$ $\therefore W$ is not a subspace of V.

7.	Define linear combination of vectors.
	Solution: Let V be a vector space over a field F and S is the subset of V. Linear combination of vectors: A vector $v \in V$ is a linear combination of vectors of S if and only if there exists a finite number of vectors $u_1, \dots, u_n \in S$ and $a_1, \dots, a_n \in F$ such that $v = a_1u_1 + \dots + a_nu_n$.
8.	Define span of subset and generator set in vector space.
	Span of subset S of V: The set consisting of all linear combinations of vectors in S. It is denoted by $\text{span}(S)$. Generator set: A subset S of a vector space V is said to be generator set of V if $\text{span}(S) = V$.
9.	Show that the matrices $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ generate $M_{2 \times 2}(F)$. (April/May 2023)
	Solution: Let $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ be any arbitrary matrix in $M_{2 \times 2}(F)$. Let a, b, c and d be any scalars so that $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ $= \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ c & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & d \end{pmatrix}$ $= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow a = a_{11}, b = a_{12}, c = a_{21}, \text{ and } d = a_{22}.$ $\therefore$ The given matrices generate $M_{2 \times 2}(F)$.
10.	For what values of h will y be in the subspace of $\mathbb{R}^3$ spanned by v_1, v_2, v_3 if $v_1 = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}, v_2 = \begin{bmatrix} 5 \\ -4 \\ -7 \end{bmatrix}, v_3 = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$ and $y = \begin{bmatrix} -4 \\ 3 \\ h \end{bmatrix}$.
	Solution: Let Y is the subspace spanned by v_1, v_2, v_3 If $y \in Y$, then $y = av_1 + bv_2 + cv_3$ $\begin{bmatrix} -4 \\ 3 \\ h \end{bmatrix} = a \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix} + b \begin{bmatrix} 5 \\ -4 \\ -7 \end{bmatrix} + c \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$ $-4 = a + 5b - 3c \rightarrow (1)$ $3 = -a - 4b + c \rightarrow (2)$ $h = -2a - 7b \rightarrow (3)$ $(1) + (2) \times 3 \Rightarrow 5 = -2a - 7b \rightarrow (4)$ From (3) and (4), we have $h = 5$.
11.	Define linear dependent set and Independent set of a vector space.
	Solution: Linear dependent set: A subset $\{u_1, u_2, \dots, u_n\}$ of a vector space V over F is called linearly dependent set if there exist a finite number of scalars $a_1, a_2, \dots, a_n$ in F <u>not all zero</u>, such that $a_1u_1 + a_2u_2 + \dots + a_nu_n = 0$. Linear independent set: A subset $\{u_1, u_2, \dots, u_n\}$ of a vector space V over F is called linearly independent set if there exist a finite number of scalars $a_1, a_2, \dots, a_n$ in F <u>all are zero</u>, such that $a_1u_1 + a_2u_2 + \dots + a_nu_n = 0$.

12.	Determine whether the vectors $v_1 = (1, -2, 3)$, $v_2 = (5, 6, -1)$, $v_3 = (3, 2, 1)$ form a linearly dependent or linearly independent set in $\mathbb{R}^3$.
	Solution: $\begin{vmatrix} 1 & 5 & 3 \\ -2 & 6 & 2 \\ 3 & -1 & 1 \end{vmatrix} = 1[6+2] + 5[-2-6] + 3[2-18] = 8 - 40 - 48 = 0$ Since the determinant value is zero, the vectors form linearly dependent set in $\mathbb{R}^3$.
13.	Show that the vectors $u = (1, 2, 3)$, $v = (2, 5, 7)$, $w = (1, 3, 5)$ are linearly independent. (Nov/Dec 2022)
	Solution: $\begin{vmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 3 & 7 & 5 \end{vmatrix} = 1[25-21] - 2[10-9] + 1[14-15] = 4 - 2 - 1 = 1 \neq 0$ Since the determinant value is non zero, the vectors are linearly independent.
14.	Given v_1 & v_2 in a vector space V and let $H = \text{span}(\{v_1, v_2\})$. Show that H is a subspace of V.
	Solution: The zero vector is in H, since $0 = 0v_1 + 0v_2$. To show that H is closed under vector addition, take two arbitrary vectors in H, say, $u = s_1v_1 + s_2v_2$ and $w = t_1v_1 + t_2v_2$ For the vector space V, $\begin{aligned} u + w &= (s_1v_1 + s_2v_2) + (t_1v_1 + t_2v_2) \\ &= (s_1 + t_1)v_1 + (s_2 + t_2)v_2 \end{aligned}$ So $u + w$ is in H. Furthermore, if c is any scalar, $cu = c(s_1v_1 + s_2v_2) = (cs_1)v_1 + (cs_2)v_2$ which shows that cu is in H and H is closed under scalar multiplication. Thus H is a subspace of V.
15.	Define Basis of a vector space (April/May 2023)
	Solution: A subset S of a vector space is said to be a basis of V if (i) S is a linearly independent set and (ii) S generates V i.e. $\text{span}(S) = V$ (i.e) a basis S for a vector space V is a linearly independent subset of V that generates V
16.	Write down the standard basis for the vector space $\mathbb{R}^n$, $P_n(\mathbb{R})$ and $M_{2 \times 2}(\mathbb{R})$.
	(1) For the vector space $\mathbb{R}^n$, $\{e_1, e_2, \dots, e_n\}$ is a standard basis where $e_1 = (1, 0, 0, \dots, 0)$, $e_2 = (0, 1, 0, \dots, 0)$, ..., $e_n = (0, 0, \dots, 0, 1)$; (2) For the vector space of all polynomials of degree $\leq n$, $P_n(\mathbb{R})$, $\{1, x, x^2, \dots, x^n\}$ is a standard basis. (3) For the vector space $M_{2 \times 2}(\mathbb{R})$, $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$ is a standard basis.
17.	Define finite dimensional vector space and dimension of vector space with examples.
	Solution: If the basis of a vector space V consists of finite number of vectors, then the vector space V is called finite dimensional. The unique number of vectors in each basis for V is called the dimension of V and is denoted by $\dim(V)$. A vector space that is not finite dimensional is called infinite dimensional. Example: 1) Vector space F^n has dimension n. 2) Vector space $P_n(F)$ has dimension $n+1$. 3) Vector space $M_{m \times n}(F)$ has dimension mn.
18.	Show that $\dim(\mathbb{C}) = 2$
	$C = \{a + ib : a, b \in \mathbb{R}\}$ So, any element in C is a linear combination of $\{1, i\}$. So $C = L(\{1, i\})$

	Further $a + ib = 0 \Rightarrow a = 0, b = 0$. $\therefore \{1, i\}$ is L.I. Hence $\{1, i\}$ is a basis of C over R $\therefore \dim C(R) = 2$.
19.	If $V = A \oplus B$, then show that $\dim V = \dim A + \dim B$. Proof : We know that, $\dim(A+B) = \dim A + \dim B - \dim(A \cap B)$ Since $V = A \oplus B$ then $V = A + B$ and $A \cap B = \{0\} \Rightarrow \dim(A \cap B) = 0$. $\therefore \dim(A+B) = \dim A + \dim B$. Hence, $\dim V = \dim A + \dim B$.
20.	Find the dimension of W, where $\{(x_1, x_2, x_3) / x_1 + x_2 + x_3 = 0\}$. (Apr/May 2022) Solution: $x_1 + x_2 + x_3 = 0$ $x_3 = -x_1 - x_2$ $(x_1, x_2, -x_1 - x_2) = (x_1, 0, -x_1) + (0, x_2, -x_2) = x_1(1, 0, -1) + x_2(0, 1, -1)$ $\therefore B = \{(1, 0, -1), (0, 1, -1)\}$ is a basis of W. $\therefore \dim(W) = 2$.
PART B	
1	Prove that F^n, the set of all ordered n-tuples with entries from F is a vector space over F under coordinate wise addition and scalar multiplication. (April/May 2023) Solution: Given $F^n = \{(a_1, a_2, \dots, a_n) : \text{each } a_i \in F\}$. Vector addition defined as coordinate wise addition: $(a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n) = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$. Scalar multiplication defined as coordinate wise scalar multiplication: $a(a_1, a_2, \dots, a_n) = (aa_1, aa_2, \dots, aa_n)$. <u>To check vector addition of two vectors is in F^n :</u> Let $x = (a_1, a_2, \dots, a_n) \in F^n$, where each $a_i \in F$ and $y = (b_1, b_2, \dots, b_n) \in F^n$, where each $b_i \in F$. $\therefore x + y = (a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n) = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n) \in F^n$ ($\because$ each $a_i + b_i \in F$) <u>To check scalar multiplication of scalar and vector is in F^n :</u> Let $a \in F$ and $x = (a_1, a_2, \dots, a_n) \in F^n$, where each $a_i \in F$ $\therefore a x = (aa_1, aa_2, \dots, aa_n) \in F^n$ ($\because$ each $aa_i \in F$) (i) $x + y = (a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n) = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$ $y + x = (b_1, b_2, \dots, b_n) + (a_1, a_2, \dots, a_n) = (b_1 + a_1, b_2 + a_2, \dots, b_n + a_n)$ $= (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$ $\therefore x + y = y + x$. (ii) Take $z = (c_1, c_2, \dots, c_n) \in F^n$, where each $c_i \in F$. $(x + y) + z = ((a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n)) + (c_1, c_2, \dots, c_n) = (a_1 + b_1 + c_1, a_2 + b_2 + c_2, \dots, a_n + b_n + c_n)$ $x + (y + z) = (a_1, a_2, \dots, a_n) + ((b_1, b_2, \dots, b_n) + (c_1, c_2, \dots, c_n)) = (a_1 + b_1 + c_1, a_2 + b_2 + c_2, \dots, a_n + b_n + c_n)$ $\therefore (x + y) + z = x + (y + z)$. (iii) $\forall x = (a_1, a_2, \dots, a_n) \in F^n$ there exist a vector $O = (0, 0, \dots, 0) \in F^n$ such that $x + O = (a_1, a_2, \dots, a_n) + (0, 0, \dots, 0) = (a_1 + 0, a_2 + 0, \dots, a_n + 0) = (a_1, a_2, \dots, a_n) = x$ Similarly, $O + x = O$. $\therefore x + O = x = O + x$.

Here, $O=(0,0,...,0)$ is a zero vector of F^n .

(iv) For every $x=(a_1, a_2, ..., a_n) \in F^n$ there exist a vector $-x=(-a_1, -a_2, ..., -a_n) \in F^n$ such that

$$x + (-x) = (a_1, a_2, ..., a_n) + (-a_1, -a_2, ..., -a_n) = (a_1 - a_1, a_2 - a_2, ..., a_n - a_n) = (0, 0, ..., 0) = O$$

Similarly, $(-x) + x = O$.

$$\therefore x + (-x) = O = (-x) + x.$$

Here, $-x=(-a_1, -a_2, ..., -a_n)$ is a additive inverse vector of F^n .

(v) $1 \in F$ and $x=(a_1, a_2, ..., a_n) \in F^n$, where each $a_i \in F$.

$$1x = (1a_1, 1a_2, ..., 1a_n) = (a_1, a_2, ..., a_n) = x$$

(vi) Let $a, b \in F$ and $x=(a_1, a_2, ..., a_n) \in F^n$, where each $a_i \in F$.

$$(ab)x = (ab)(a_1, a_2, ..., a_n) = (aba_1, aba_2, ..., aba_n). a(bx) = a(ba_1, ba_2, ..., ba_n) = (aba_1, aba_2, ..., aba_n).$$

$$\therefore (ab)x = a(bx).$$

(vii) Let $a \in F$ and $x=(a_1, a_2, ..., a_n) \in F^n$, where each $a_i \in F$

$$a(x+y) = a((a_1, a_2, ..., a_n) + (b_1, b_2, ..., b_n)) = a(a_1 + b_1, a_2 + b_2, ..., a_n + b_n)$$

$$= (aa_1 + ab_1, aa_2 + ab_2, ..., aa_n + ab_n)$$

$$ax + ay = a(a_1, a_2, ..., a_n) + a(b_1, b_2, ..., b_n) = (aa_1, aa_2, ..., aa_n) + (ab_1, ab_2, ..., ab_n)$$

$$= (aa_1 + ab_1, aa_2 + ab_2, ..., aa_n + ab_n)$$

$$\therefore a(x+y) = ax + ay.$$

(viii) Let $a, b \in F$ and $x=(a_1, a_2, ..., a_n) \in F^n$, where each $a_i \in F$.

$$(a+b)x = (a+b)(a_1, a_2, ..., a_n) = ((a+b)a_1, (a+b)a_2, ..., (a+b)a_n)$$

$$= (aa_1 + ba_1, aa_2 + ba_2, ..., aa_n + ba_n).$$

$$ax + bx = a(a_1, a_2, ..., a_n) + b(a_1, a_2, ..., a_n) = (aa_1, aa_2, ..., aa_n) + (ba_1, ba_2, ..., ba_n)$$

$$= (aa_1 + ba_1, aa_2 + ba_2, ..., aa_n + ba_n).$$

$$\therefore (a+b)x = ax + bx.$$

Hence F^n is a vector space.

2 Prove that $P_n(R)$, the set of all polynomials of degree at most n with real coefficient is a vector space under usual addition and constant multiplication of polynomial.

Solution: Given $P_n(R) = \{a_0 + a_1t + a_2t^2 + \dots + a_nt^n : \text{polynomial of degree } \leq n \text{ and each } a_i \in R\}$.

Vector addition defined as addition of polynomial

$$\begin{aligned} p(t) + q(t) &= a_0 + a_1t + a_2t^2 + \dots + a_nt^n + b_0 + b_1t + b_2t^2 + \dots + b_nt^n \\ &= (a_0 + b_0) + (a_1 + b_1)t + (a_2 + b_2)t^2 + \dots + (a_n + b_n)t^n \end{aligned}$$

Scalar multiplication defined as constant multiplication of polynomial:

$$ap(t) = a(a_0 + a_1t + a_2t^2 + \dots + a_nt^n) = aa_0 + aa_1t + aa_2t^2 + \dots + aa_nt^n$$

To check vector addition of two vectors is in $P_n(R)$

Let $p(t) = a_0 + a_1t + a_2t^2 + \dots + a_nt^n$, where each $a_i \in R$ and

$q(t) = b_0 + b_1t + b_2t^2 + \dots + b_nt^n$, where each $b_i \in R$ then

$$(p+q)(t) = p(t) + q(t) = (a_0 + b_0) + (a_1 + b_1)t + (a_2 + b_2)t^2 + \dots + (a_n + b_n)t^n \in P_n(R)$$

$\therefore (p+q)(t)$ is a polynomial of degree $\leq n$ and each $(a_i + b_i) \in R$ is a polynomial of degree $\leq n$ and

To check scalar multiplication of scalar and vector is in

$$P_n(R): ap(t) = a(a_0 + a_1t + a_2t^2 + \dots + a_nt^n) = aa_0 + aa_1t + aa_2t^2 + \dots + aa_nt^n \in P_n(R)$$

$\therefore (ap)(t)$ is a polynomial of degree $\leq n$ and each $aa_i \in R$

$$\begin{aligned} \text{(i)} \quad (p+q)(t) &= [(a_0 + b_0) + (a_1 + b_1)t + (a_2 + b_2)t^2 + \dots + (a_n + b_n)t^n] \\ &= [(b_0 + a_0) + (b_1 + a_1)t + (b_2 + a_2)t^2 + \dots + (b_n + a_n)t^n] \\ &= (q+p)(t) \end{aligned}$$

$$\text{(ii)} \quad \text{Let } r(t) = c_0 + c_1t + c_2t^2 + \dots + c_nt^n.$$

$$\begin{aligned} ((p+q)+r)(t) &= (p+q)(t) + r(t) \\ &= [(a_0 + b_0) + (a_1 + b_1)t + (a_2 + b_2)t^2 + \dots + (a_n + b_n)t^n] + (c_0 + c_1t + c_2t^2 + \dots + c_nt^n) \\ &= (a_0 + b_0 + c_0) + (a_1 + b_1 + c_1)t + (a_2 + b_2 + c_2)t^2 + \dots + (a_n + b_n + c_n)t^n \\ &= (a_0 + (b_0 + c_0)) + (a_1 + (b_1 + c_1))t + (a_2 + (b_2 + c_2))t^2 + \dots + (a_n + (b_n + c_n))t^n \\ &= (a_0 + a_1t + a_2t^2 + \dots + a_nt^n) + [(b_0 + c_0) + (b_1 + c_1)t + (b_2 + c_2)t^2 + \dots + (b_n + c_n)t^n] \\ &= p(t) + (q+r)(t) \\ &= (p+(q+r))(t) \quad \forall p(t), q(t), r(t) \in P_n(R) \end{aligned}$$

(iii) If all the coefficients are zero, $O(t) = 0 + 0t + 0t^2 + \dots + 0t^n$ is a zero polynomial.

The zero polynomial is included in P_n even though its degree is not defined.

$$\text{(iv)} \quad -p(t) = (-1)p(t) = (-1)(a_0 + a_1t + a_2t^2 + \dots + a_nt^n) = -a_0 - a_1t - a_2t^2 - \dots - a_nt^n$$

$$(p+(-p))(t) = (a_0 - a_0) + (a_1 - a_1)t + (a_2 - a_2)t^2 + \dots + (a_n - a_n)t^n = 0 + 0t + 0t^2 + \dots + 0t^n = O(t)$$

$-p(t)$ acts as a additive inverse.

$$\text{(v)} \quad 1 p(t) = p(t) \quad \forall p(t) \in P_n(R)$$

$$\begin{aligned} \text{(vi)} \quad (ab)p(t) &= (ab)(a_0 + a_1t + a_2t^2 + \dots + a_nt^n) \\ &= (aba_0) + (aba_1)t + (aba_2)t^2 + \dots + (aba_n)t^n \\ &= (a)(ba_0 + ba_1t + ba_2t^2 + \dots + ba_nt^n) \\ &= (a)(bp(t)) \quad \forall p(t) \in P_n(R), a, b \in R \end{aligned}$$

$$\begin{aligned} \text{(viii)} \quad (a+b)(p(t)) &= (a+b)(a_0 + a_1t + a_2t^2 + \dots + a_nt^n) \\ &= (aa_0 + aa_1t + aa_2t^2 + \dots + aa_nt^n) + (ba_0 + ba_1t + ba_2t^2 + \dots + ba_nt^n) \\ &= ap(t) + bp(t) \quad \forall p(t) \in P_n(R), a, b \in R \end{aligned}$$

Hence, $P_n(R)$ is the vector space.

3 Prove that the set of all $m \times n$ matrices over F denoted by $M_{m \times n}(F)$ is a vector space over F with respect to the operation matrix addition and scalar multiplication of matrix.

Solution:

Let $A, B, C \in M(F)$ and $\alpha, \beta \in F$.

Here $A = (a_{ij})_{m \times n}$

$$B = (b_{ij})_{m \times n}$$

$$C = (c_{ij})_{m \times n}$$

(i) Closure:

Let $A, B \in M(F)$

Then $A + B = (a_{ij}) + (b_{ij}) = (d_{ij}) \in M(F)$

(ii) Associative:

Let $A, B, C \in M(F)$

Then $(A + B) + C = A + (B + C)$

LHS :

$$\begin{aligned}(A + B) + C &= \{(a_{ij}) + (b_{ij})\} + (c_{ij}) \\ &= (d_{ij}) + (c_{ij}) \quad \because (d_{ij}) = (a_{ij}) + (b_{ij}) \\ &= (e_{ij}) \in M(F)\end{aligned}$$

RHS :

$$\begin{aligned}A + (B + C) &= (a_{ij}) + \{(b_{ij}) + (c_{ij})\} \\ &= (a_{ij}) + (d_{ij}) \quad \because (d_{ij}) = (b_{ij}) + (c_{ij}) \\ &= (e_{ij}) \in M(F)\end{aligned}$$

(iii) Identity:

Let $A \in M(F)$

Then $A + 0 = A$

$$\text{LHS: } A + 0 = (a_{ij}) + (0_{ij}) = a_{ij} \in M(F)$$

(iv) Inverse :

Let $A \in M(F)$

Then $A - A = 0$

$$A + (-A) = 0$$

$$(a_{ij}) - (a_{ij}) = 0_{ij} \Rightarrow (-a_{ij}) \in M(F)$$

(v) Commutative :

Let $A, B \in M(F)$

Then $A + B = B + A$

$$(a_{ij}) + (b_{ij}) = (b_{ij}) + (a_{ij})$$

(vi) $\alpha A = A, \alpha \in F, A \in M(F)$

$$\alpha A = (\alpha(a_{ij})) = (\alpha a_{ij}) = (d_{ij}) \in M(F)$$

(vii) $\alpha(A + B) = \alpha A + \alpha B$

$$\text{LHS : } \alpha(A + B) = \alpha((a_{ij}) + (b_{ij})) = \alpha(d_{ij}) = (e_{ij}) \in M(F)$$

$$\text{RHS : } \alpha A + \alpha B = (\alpha(a_{ij})) + (\alpha(b_{ij})) = (\alpha a_{ij}) + (\alpha b_{ij}) = \alpha(d_{ij}) = (e_{ij}) \in M(F)$$

(viii) $(\alpha + \beta)A = \alpha A + \beta A$

$$\text{LHS : } (\alpha + \beta)A = (\alpha + \beta)(a_{ij})$$

$$= \alpha(a_{ij}) + \beta(a_{ij}) = (\alpha a_{ij}) + (\beta a_{ij}) = (e_{ij}) \in M(F)$$

$$\text{RHS : } \alpha A + \beta A = \alpha(a_{ij}) + \beta(a_{ij})$$

$$= \alpha(a_{ij}) + \beta(a_{ij}) = (\alpha a_{ij}) + (\beta a_{ij}) = (e_{ij}) \in M(F)$$

(ix) $\alpha\beta(A) = \alpha(\beta A)$

$$(\alpha\beta)(A) = \alpha\beta(a_{ij})$$

$$= (\alpha\beta a_{ij}) = (e_{ij}) \in M(F)$$

(x) $IA = A$

$$(I_{ij})(a_{ij}) = (a_{ij})$$

$\therefore M_{m \times n}(F)$ is a vector space under F .

4

Determine whether the set of all 2×2 matrices of the form $\begin{pmatrix} a & a+b \\ a+b & b \end{pmatrix}, a, b \in \mathbb{R}$ with respect to standard matrix addition and scalar multiplication is a vector space or not? If not list all the axioms that fail to hold.

Solution:

Let $V =$ Set of all 2×2 matrices of the form $\begin{pmatrix} a & a+b \\ a+b & b \end{pmatrix}, a, b \in \mathbb{R}$

(To check vector addition of two vectors is in V)

$$\text{Let } x = \begin{pmatrix} a_1 & a_1+b_1 \\ a_1+b_1 & b_1 \end{pmatrix} \in V, y = \begin{pmatrix} a_2 & a_2+b_2 \\ a_2+b_2 & b_2 \end{pmatrix} \in V$$

$$x + y = \begin{pmatrix} a_1 & a_1+b_1 \\ a_1+b_1 & b_1 \end{pmatrix} + \begin{pmatrix} a_2 & a_2+b_2 \\ a_2+b_2 & b_2 \end{pmatrix} = \begin{pmatrix} a_1+a_2 & a_1+b_1+a_2+b_2 \\ a_1+b_1+a_2+b_2 & b_1+b_2 \end{pmatrix} \in V$$

To check scalar multiplication of scalar and vector is in V :

$$\text{Let } c \in F \text{ and } x = \begin{pmatrix} a_1 & a_1+b_1 \\ a_1+b_1 & b_1 \end{pmatrix} \in V$$

$$\therefore cx = c \begin{pmatrix} a_1 & a_1+b_1 \\ a_1+b_1 & b_1 \end{pmatrix} = \begin{pmatrix} ca_1 & ca_1+cb_1 \\ ca_1+cb_1 & cb_1 \end{pmatrix} \in V$$

$$(i) \text{ Let } x = \begin{pmatrix} a_1 & a_1+b_1 \\ a_1+b_1 & b_1 \end{pmatrix} \in V, y = \begin{pmatrix} a_2 & a_2+b_2 \\ a_2+b_2 & b_2 \end{pmatrix} \in V$$

$$x + y = \begin{pmatrix} a_1 & a_1+b_1 \\ a_1+b_1 & b_1 \end{pmatrix} + \begin{pmatrix} a_2 & a_2+b_2 \\ a_2+b_2 & b_2 \end{pmatrix} \\ = \begin{pmatrix} a_1+a_2 & a_1+b_1+a_2+b_2 \\ a_1+b_1+a_2+b_2 & b_1+b_2 \end{pmatrix}$$

$$y + x = \begin{pmatrix} a_2 & a_2+b_2 \\ a_2+b_2 & b_2 \end{pmatrix} + \begin{pmatrix} a_1 & a_1+b_1 \\ a_1+b_1 & b_1 \end{pmatrix} \\ = \begin{pmatrix} a_2+a_1 & a_2+b_2+a_1+b_1 \\ a_2+b_2+a_1+b_1 & b_2+b_1 \end{pmatrix} = \begin{pmatrix} a_1+a_2 & a_1+b_1+a_2+b_2 \\ a_1+b_1+a_2+b_2 & b_1+b_2 \end{pmatrix}$$

$$\therefore x + y = y + x.$$

$$(ii) \text{ Take } z = \begin{pmatrix} a_3 & a_3+b_3 \\ a_3+b_3 & b_3 \end{pmatrix} \in V$$

$$(x + y) + z = \left[\begin{pmatrix} a_1 & a_1+b_1 \\ a_1+b_1 & b_1 \end{pmatrix} + \begin{pmatrix} a_2 & a_2+b_2 \\ a_2+b_2 & b_2 \end{pmatrix} \right] + \begin{pmatrix} a_3 & a_3+b_3 \\ a_3+b_3 & b_3 \end{pmatrix} \\ = \begin{pmatrix} a_1+a_2+a_3 & a_1+b_1+a_2+b_2+a_3+b_3 \\ a_1+b_1+a_2+b_2+a_3+b_3 & b_1+b_2+b_3 \end{pmatrix}$$

$$x + (y + z) = \begin{pmatrix} a_1 & a_1+b_1 \\ a_1+b_1 & b_1 \end{pmatrix} + \left[\begin{pmatrix} a_2 & a_2+b_2 \\ a_2+b_2 & b_2 \end{pmatrix} + \begin{pmatrix} a_3 & a_3+b_3 \\ a_3+b_3 & b_3 \end{pmatrix} \right] \\ = \begin{pmatrix} a_1+a_2+a_3 & a_1+b_1+a_2+b_2+a_3+b_3 \\ a_1+b_1+a_2+b_2+a_3+b_3 & b_1+b_2+b_3 \end{pmatrix}$$

$$\therefore (x + y) + z = x + (y + z).$$

(iii) For all $x = \begin{pmatrix} a_1 & a_1 + b_1 \\ a_1 + b_1 & b_1 \end{pmatrix} \in V$ there exist a matrix $O = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \in V$ such that

$$x + O = \begin{pmatrix} a_1 & a_1 + b_1 \\ a_1 + b_1 & b_1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} a_1 & a_1 + b_1 \\ a_1 + b_1 & b_1 \end{pmatrix} = x$$

$$\text{Similarly, } O + x = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} a_1 & a_1 + b_1 \\ a_1 + b_1 & b_1 \end{pmatrix} = \begin{pmatrix} a_1 & a_1 + b_1 \\ a_1 + b_1 & b_1 \end{pmatrix} = x$$

Here, $O = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ is a zero vector (identity vector) of V .

(iv) For every $x = \begin{pmatrix} a_1 & a_1 + b_1 \\ a_1 + b_1 & b_1 \end{pmatrix} \in V$ there exist a vector $-x = \begin{pmatrix} -a_1 & -(a_1 + b_1) \\ -(a_1 + b_1) & -b_1 \end{pmatrix} \in V$ such

$$\text{that } x + (-x) = \begin{pmatrix} a_1 & a_1 + b_1 \\ a_1 + b_1 & b_1 \end{pmatrix} + \begin{pmatrix} -a_1 & -(a_1 + b_1) \\ -(a_1 + b_1) & -b_1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = O$$

Similarly, $(-x) + x = O$.

$$\therefore x + (-x) = O = (-x) + x.$$

Here, $-x = \begin{pmatrix} -a_1 & -(a_1 + b_1) \\ -(a_1 + b_1) & -b_1 \end{pmatrix}$ is an additive inverse vector of V .

(v) $1 \in F$ and $x = \begin{pmatrix} a_1 & a_1 + b_1 \\ a_1 + b_1 & b_1 \end{pmatrix} \in V$

$$1x = 1 \begin{pmatrix} a_1 & a_1 + b_1 \\ a_1 + b_1 & b_1 \end{pmatrix} = \begin{pmatrix} a_1 & a_1 + b_1 \\ a_1 + b_1 & b_1 \end{pmatrix} = x$$

(vi) Let $a, b \in F$ and $x = \begin{pmatrix} a_1 & a_1 + b_1 \\ a_1 + b_1 & b_1 \end{pmatrix} \in V$

$$(ab)x = (ab) \begin{pmatrix} a_1 & a_1 + b_1 \\ a_1 + b_1 & b_1 \end{pmatrix} = \begin{pmatrix} ab a_1 & ab(a_1 + b_1) \\ ab(a_1 + b_1) & abb_1 \end{pmatrix} = a \begin{pmatrix} b a_1 & b(a_1 + b_1) \\ b(a_1 + b_1) & b b_1 \end{pmatrix} = a(bx)$$

$$\therefore (ab)x = a(bx).$$

(vii) Let $a \in F$, $x = \begin{pmatrix} a_1 & a_1 + b_1 \\ a_1 + b_1 & b_1 \end{pmatrix} \in V$ and $y = \begin{pmatrix} a_2 & a_2 + b_2 \\ a_2 + b_2 & b_2 \end{pmatrix} \in V$

$$\begin{aligned} a(x+y) &= a \left[\begin{pmatrix} a_1 & a_1 + b_1 \\ a_1 + b_1 & b_1 \end{pmatrix} + \begin{pmatrix} a_2 & a_2 + b_2 \\ a_2 + b_2 & b_2 \end{pmatrix} \right] \\ &= \begin{pmatrix} a(a_1 + a_2) & a(a_1 + b_1 + a_2 + b_2) \\ a(a_1 + b_1 + a_2 + b_2) & a(b_1 + b_2) \end{pmatrix} \\ &= a \begin{pmatrix} a_1 & a_1 + b_1 \\ a_1 + b_1 & b_1 \end{pmatrix} + a \begin{pmatrix} a_2 & a_2 + b_2 \\ a_2 + b_2 & b_2 \end{pmatrix} \\ &= ax + ay \end{aligned}$$

$$\therefore a(x+y) = ax + ay.$$

(viii) Let $a, b \in F$ and $x = \begin{pmatrix} a_1 & a_1 + b_1 \\ a_1 + b_1 & b_1 \end{pmatrix}.$

	$(a+b)x = (a+b) \begin{pmatrix} a_1 & a_1+b_1 \\ a_1+b_1 & b_1 \end{pmatrix} = \begin{pmatrix} (a+b)a_1 & (a+b)(a_1+b_1) \\ (a+b)(a_1+b_1) & (a+b)b_1 \end{pmatrix}$ $= \begin{pmatrix} aa_1+ba_1 & a(a_1+b_1)+b(a_1+b_1) \\ a(a_1+b_1)+b(a_1+b_1) & ab_1+bb_1 \end{pmatrix}$ $= a \begin{pmatrix} a_1 & a_1+b_1 \\ a_1+b_1 & b_1 \end{pmatrix} + b \begin{pmatrix} a_1 & a_1+b_1 \\ a_1+b_1 & b_1 \end{pmatrix}$ $\therefore (a+b)x = ax + bx.$ Hence V is a vector space.
5	Let V be the set of all positive real numbers. Define the vector addition and scalar multiplication as follows: $x + y = xy$ and $kx = x^k$. Determine whether V is a vector space over $\mathbb{R}$ with respect to above operations. Solution: Let $V = \{x \in \mathbb{R}^+ / x + y = xy \text{ and } kx = x^k, \text{ where } x, y \in \mathbb{R}^+ = (0, \infty)\}$ <u>To check vector addition of two vectors is in V</u> Take $x_1, y_1 \in V \Rightarrow x_1 + y_1 = x_1 y_1 \in V$ ($\because$ product of two positive real numbers is positive). <u>To check scalar multiplication of scalar and vector is in V</u> Take $x_1 \in V$ and k is any scalar. $k x_1 = x_1^k \in V$. (i) $x_1 + y_1 = x_1 y_1 = y_1 x_1 = y_1 + x_1$ (ii) $(x_1 + y_1) + z_1 = (x_1 y_1) + z_1$ $= (x_1 y_1) z_1$ $= x_1 (y_1 z_1)$ $= x_1 + (y_1 z_1)$ $= x_1 + (y_1 + z_1)$ (iii) Let $e \in V, x_1 \in V$ $x_1 + e = x_1 e = x_1 \Rightarrow e = 1 \in V$ (iv) $(-x_1) + x_1 = (-x_1) x_1$ $= (-x_1)(x_1 e)$ $= (-x_1 x_1) e$ $= e \Rightarrow -x_1 = \frac{1}{x_1} \in V$ (v) Let $1 \in F$ and $x \in V$. $1x = x^1 = x$ (vi) Let $a, b \in F$ and $x \in V$. $(ab)x = x^{ab} = (x^b)^a = a(x^b) = a(bx).$ $\therefore (ab)x = a(bx).$ (vii) Let $a \in F$ and $x, y \in V$. $a(x + y) = a(xy) = (xy)^a = (x^a)(y^a) = ax + ay$ $\therefore a(x + y) = ax + ay.$

	(viii) Let $a, b \in F$ and $x \in V$. $(a+b)x = x^{a+b} = x^a x^b = x^a + x^b = ax + bx.$ $\therefore (a+b)x = ax + bx.$ Hence V is a vector space.
6	Let W_1 and W_2 be subspaces of a vector space V. Prove that $W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$. (Apr/May 2022, Nov/Dec 2022) Proof: Assume that $W_1 \cup W_2$ is a subspace of V. To prove that $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$. Assume the contrary that $W_1 \not\subseteq W_2$ and $W_2 \not\subseteq W_1$. Then there exist elements $x \in W_1$ but $x \notin W_2$ and $y \in W_2$ but $y \notin W_1$. Therefore, x and $y \in W_1 \cup W_2$. Since $W_1 \cup W_2$ is a subspace of V then $x + y \in W_1 \cup W_2$. Case 1: Take $x + y \in W_1$. Now $x + y \in W_1$ and $-x \in W_1$ then $-x + (x + y) \in W_1 \Rightarrow (-x + x) + y \in W_1 \Rightarrow y \in W_1$. This is a contradiction. Case 2: Take $x + y \in W_2$. Now $x + y \in W_2$ and $-y \in W_2$ then $(x + y) + (-y) \in W_2 \Rightarrow x \in W_2$. This is a contradiction. Therefore, $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$. Conversely, assume that $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$. To prove that $W_1 \cup W_2$ is a subspace of V. Let $W_1 \subseteq W_2$. Then $W_1 \cup W_2 = W_2$, a subspace of V. Let $W_2 \subseteq W_1$. Then $W_1 \cup W_2 = W_1$, a subspace of V.
7	Prove that intersection of two subspace is again a subspace over a vector space. Proof: Let V be a vector space over a field F. Let W_1, W_2 be two subspace of a vector space $V(F)$. Let $0 \in W_1$ & $0 \in W_2$ Then $0 \in W_1 \cap W_2$ $W_1 \cap W_2$ is not empty. Let W_1 be a subspace Let $\alpha, \beta \in F, u, v \in W_1$ Then $\alpha u + \beta v \in W_1$ -----(1) Let W_2 be a subspace Let $\alpha, \beta \in F, u, v \in W_2$ Then $\alpha u + \beta v \in W_2$ -----(2) From (1) and (2) $\alpha u + \beta v \in W_1 \cap W_2$ $W_1 \cap W_2$ is a subspace of $V(F)$.

8	Prove that the vector $(2,6,8)$ can be expressed as a linear combinations of $(1,2,1), (-2,-4,-2), (0,2,3), (2,0,-3), (-3,8,16)$. (April/May 2022, April/May 2023)
	Solution: Let $u_1 = (1,2,1); u_2 = (-2,-4,-2); u_3 = (0,2,3); u_4 = (2,0,-3); u_5 = (-3,8,16)$. Determine the scalars a_1, a_2, a_3, a_4, a_5 such that $(2,6,8) = a_1u_1 + a_2u_2 + a_3u_3 + a_4u_4 + a_5u_5$ $= a_1(1,2,1) + a_2(-2,-4,-2) + a_3(0,2,3) + a_4(2,0,-3) + a_5(-3,8,16)$ $= (a_1 - 2a_2 + 2a_4 - 3a_5, 2a_1 - 4a_2 + 2a_3 + 8a_5, a_1 - 2a_2 + 3a_3 - 3a_4 + 16a_5)$ Therefore, $a_1 - 2a_2 + 0a_3 + 2a_4 - 3a_5 = 2$ $2a_1 - 4a_2 + 2a_3 + 0a_4 + 8a_5 = 6$ $a_1 - 2a_2 + 3a_3 - 3a_4 + 16a_5 = 8$ Solving the above system by elimination method $\left(\begin{array}{ccccc c} 1 & -2 & 0 & 2 & -3 & 2 \\ 2 & -4 & 2 & 0 & 8 & 6 \\ 1 & -2 & 3 & -3 & 16 & 8 \end{array} \right)$ $\sim \left(\begin{array}{ccccc c} 1 & -2 & 0 & 2 & -3 & 2 \\ 0 & 0 & 2 & -4 & 14 & 2 \\ 0 & 0 & 3 & -5 & 19 & 6 \end{array} \right) \begin{array}{l} R_2 \leftrightarrow R_2 - 2R_1 \\ R_3 \leftrightarrow R_3 - R_1 \end{array}$ $\sim \left(\begin{array}{ccccc c} 1 & -2 & 0 & 2 & -3 & 2 \\ 0 & 0 & 1 & -2 & 7 & 1 \\ 0 & 0 & 0 & 2 & -4 & 6 \end{array} \right) \begin{array}{l} R_3 \leftrightarrow 2R_3 - 3R_2 \end{array}$ The reduced equations are $a_1 - 2a_2 + 2a_4 - 3a_5 = 2$ $a_3 - 2a_4 + 7a_5 = 1$ $2a_4 - 4a_5 = 6$ Take $a_2 = 0$ and $a_5 = 0$ we get, $a_1 = -4, a_3 = 7, a_4 = 3$. $\therefore (2,6,8) = -4(1,2,1) + 7(0,2,3) + 3(2,0,-3)$ Thus, $(2,6,8)$ can be expressed as a linear combinations of $(1,2,1), (-2,-4,-2), (0,2,3), (2,0,-3), (-3,8,16)$ Take $a_2 = 0$ and $a_5 = 0$ we get $a_1 + 2a_4 = 2$ ---- (1) $2a_1 + 2a_3 = 6$ ---- (2) $a_1 + 3a_3 - 3a_4 = 8$ ---- (3) (3) - (1) $\Rightarrow 3a_3 - 5a_4 = 6$ ---- (4) (2) - 2 $\times$ (1) $\Rightarrow 2a_3 - 4a_4 = 2$ ---- (5) Solving (4) and (5) $2 \times (4) - 3 \times (5) \Rightarrow 2a_4 = 6 \Rightarrow a_4 = 3$ $\therefore a_3 = 7, a_1 = -4$.
9	Prove that the vector $(2,-5,4)$ can be expressed as a linear combinations of $(1,-3,2), (2,-1,1)$
	Solution: Let $\alpha_1, \alpha_2 \in F$ Let $(2,-5,4) = \alpha_1(1,-3,2) + \alpha_2(2,-1,1)$ $= (\alpha_1, -3\alpha_1, 2\alpha_1) + (2\alpha_2, -\alpha_2, \alpha_2)$ $= (\alpha_1 + \alpha_2, -3\alpha_1 - \alpha_2, 2\alpha_1 + \alpha_2)$

	$\alpha_1 + 2\alpha_2 = 2$ ----- (1) $-3\alpha_1 - \alpha_2 = -5$ ----- (2) $2\alpha_1 + \alpha_2 = 4$ ----- (3) $\alpha_1 = 1$ $(1) \Rightarrow 1 + 2\alpha_2 = 2 \Rightarrow \alpha_2 = \frac{1}{2}$
10	Determine the given set in $P_4(R)$ is linearly dependent or linearly independent for $x^4 - x^3 + 5x^2 - 8x + 6, -x^4 + x^3 - 5x^2 + 5x - 3, x^4 + 3x^2 - 3x + 5$ and $2x^4 + x^3 + 4x^2 + 8x$. (Apr/May 2022, Nov/Dec 2022)
	Solution: Let $v_1 = x^4 - x^3 + 5x^2 - 8x + 6, v_2 = -x^4 + x^3 - 5x^2 + 5x - 3, v_3 = x^4 + 3x^2 - 3x + 5$ $v_4 = 2x^4 + x^3 + 4x^2 + 8x$ Let $a_1v_1 + a_2v_2 + a_3v_3 + a_4v_4 = 0$ where a_1, a_2, a_3, a_4 are scalars $\Rightarrow a_1(x^4 - x^3 + 5x^2 - 8x + 6) + a_2(-x^4 + x^3 - 5x^2 + 5x - 3)$ $+ a_3(x^4 + 3x^2 - 3x + 5) + a_4(2x^4 + x^3 + 4x^2 + 8x) = 0$ $x^4(a_1 - a_2 + a_3 + 2a_4) + x^3(-a_1 + a_2 + a_4) + x^2(5a_1 - 5a_2 + 3a_3 + 4a_4)$ $+ x(-8a_1 + 5a_2 - 3a_3 + 8a_4) + (6a_1 - 3a_2 + 5a_3) = 0$ Equating the coefficients $\Rightarrow a_1 - a_2 + a_3 + 2a_4 = 0,$ $-a_1 + a_2 + a_4 = 0,$ $5a_1 - 5a_2 + 3a_3 + 4a_4 = 0,$ $-8a_1 + 5a_2 - 3a_3 + 8a_4 = 0,$ $6a_1 - 3a_2 + 5a_3 = 0$ By solving the above equations $a_1 = 4, a_2 = 3, a_3 = -3, a_4 = 1.$ Hence the given set in $P_4(R)$ is linearly dependent.
11	Verify whether the first polynomial can be expressed as a linear combination of the other two in $P_3(R)$ for the given $x^3 - 8x^2 + 4x, x^3 - 2x^2 + 3x - 1$ and $x^3 - 2x + 3$.
	Solution: Let $x^3 - 8x^2 + 4x = a(x^3 - 2x^2 + 3x - 1) + b(x^3 - 2x + 3)$ $x^3 - 8x^2 + 4x = x^3(a + b) + x^2(-2a) + x(3a - 2b) + (-a + 3b)$ $\Rightarrow 1 = a + b, -8 = -2a, 4 = 3a - 2b, 0 = -a + 3b$ $\Rightarrow a = 4, b = -3$ from equations $a + b = 1$ and $2a = 8$. But $3a - 2b = 4$ & $-a + 3b = 0$ are not satisfied for $a=4$ and $b=-3$. Therefore $x^3 - 8x^2 + 4x$ is not a linear combination $x^3 - 2x^2 + 3x - 1$ and $x^3 - 2x + 3$.
12	In $M_{2 \times 3}(R)$, show that the set $\left\{ \begin{pmatrix} 1 & -3 & 2 \\ -4 & 0 & 5 \end{pmatrix}, \begin{pmatrix} -3 & 7 & 4 \\ 6 & -2 & -7 \end{pmatrix}, \begin{pmatrix} -2 & 3 & 11 \\ -1 & -3 & 2 \end{pmatrix} \right\}$ is linearly dependent
	Solution: Consider, $a \begin{bmatrix} 1 & -3 & 2 \\ -4 & 0 & 5 \end{bmatrix} + b \begin{bmatrix} -3 & 7 & 4 \\ 6 & -2 & -7 \end{bmatrix} + c \begin{bmatrix} -2 & 3 & 11 \\ -1 & -3 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\begin{bmatrix} a & -3a & 2a \\ -4a & 0 & 5a \end{bmatrix} + \begin{bmatrix} -3b & 7b & 4b \\ 6b & -2b & -7b \end{bmatrix} + \begin{bmatrix} -2c & 3c & 11c \\ -c & -3c & 2c \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

	Equating $a - 3b - 2c = 0 \text{ --- (1)}$ $-3a + 7b + 3c = 0 \text{ --- (2)}$ $2a + 4b + 11c = 0 \text{ --- (3)}$ $-4a + 6b - c = 0 \text{ --- (4)}$ $-2b - 3c = 0 \text{ --- (5)}$ $5a - 7b + 2c = 0 \text{ --- (6)}$ From (5) $c = \frac{-2b}{3}$ Substitute in (1) $a - 3b - 2\left(\frac{-2b}{3}\right) = 0 \Rightarrow a - 3b + \left(\frac{4b}{3}\right) = 0 \Rightarrow a - \left(\frac{5b}{3}\right) = 0 \Rightarrow a = \left(\frac{5b}{3}\right)$ Take $b=3$ we get, $a=5$, $c=-2$. (they satisfies all the above equations) Hence the given set is linearly dependent.
13	Show that the polynomials $x^2 + 3x - 2$, $2x^2 + 5x - 3$ and $-x^2 - 4x + 4$ generate $P_2(R)$. Solution: Let $a_2x^2 + a_1x + a_0$ be any arbitrary 2nd degree polynomial in $P_2(R)$. Determine the scalars a, b, c such that $\begin{aligned} a_2x^2 + a_1x + a_0 &= a(x^2 + 3x - 2) + b(2x^2 + 5x - 3) + c(-x^2 - 4x + 4) \\ &= (ax^2 + 3ax - 2a) + (2bx^2 + 5bx - 3b) + (-cx^2 - 4cx + 4c) \\ &= (a + 2b - c)x^2 + (3a + 5b - 4c)x + (-2a - 3b + 4c) \end{aligned}$ On equating we get, $a + 2b - c = a_2$ ---(1) $3a + 5b - 4c = a_1 \text{ ---(2)}$ $-2a - 3b + 4c = a_0 \text{ ---(3)}$ $2 \times (1) + (3)$ gives $b + 2c = 2a_2 + a_0$ ---(4) $3 \times (1) - (2)$ gives $b + c = 3a_2 - a_1$ ---(5) $(4) - (5)$ gives $c = -a_2 + a_1 + a_0$ Substitute c in (5) we get, $b + (-a_2 + a_1 + a_0) = 3a_2 - a_1$ i.e. $b = 4a_2 - 2a_1 - a_0$ Substitute b and c in (1) we get, $a + 2(4a_2 - 2a_1 - a_0) - (-a_2 + a_1 + a_0) = a_2$ i.e. $a = -8a_2 + 5a_1 + 3a_0$ $\begin{aligned} \therefore a_2x^2 + a_1x + a_0 &= (-8a_2 + 5a_1 + 3a_0)(x^2 + 3x - 2) + (4a_2 - 2a_1 - a_0)(2x^2 + 5x - 3) \\ &\quad + (-a_2 + 2a_1 + a_0)(-x^2 - 4x + 4) \end{aligned}$ Hence, the given polynomials generate $P_2(R)$.
14	Prove that the vectors $u_1=(2,-3,1)$, $u_2=(1,4,-2)$, $u_3=(-8,12,-4)$, $u_4=(1,37,-17)$ and $u_5=(-3,-5,8)$ generate R^3. Find a subset of the set $\{u_1, u_2, u_3, u_4, u_5\}$ that is a basis for R^3. (Nov/Dec 2022) Solution: Let $u_1 = (2, -3, 1)$, $u_2 = (1, 4, -2)$, $u_3 = (-8, 12, -4)$, $u_4 = (1, 37, -17)$ and $u_5 = (-3, -5, 8)$ Here $\text{Dim } R^3 = 3$.

We know that

If $\dim V = n$ then any set of more than n is L.D

$$\therefore S = \{u_1, u_2, \dots, u_5\}$$

Here we have to find subset of three L.I vectors.

$$\text{Choose } S_1 = \{u_1, u_2\}$$

$$\text{Then } \alpha_1 u_1 + \alpha_2 u_2 = 0$$

$$\alpha_1(2, -3, 1) + \alpha_2(1, 4, -2) = 0$$

$$(2\alpha_1 + \alpha_2, -3\alpha_1 + 4\alpha_2, \alpha_1 - 2\alpha_2) = (0, 0, 0)$$

Solving we get $\alpha_1 = 0, \alpha_2 = 0$.

$\therefore S_1$ is L.I

Choose u_3 which is not in S

$$\text{Choose } u_3 \in L(S_1)$$

$$(-8, 12, -4) = \alpha_1 u_1 + \alpha_2 u_2$$

$$(-8, 12, -4) = \alpha_1(2, -3, 1) + \alpha_2(1, 4, -2)$$

$$(-8, 12, -4) = (2\alpha_1 + \alpha_2, -3\alpha_1 + 4\alpha_2, \alpha_1 - 2\alpha_2)$$

$$2\alpha_1 + \alpha_2 = -8 \text{ --- (1)}$$

$$-3\alpha_1 + 4\alpha_2 = 12 \text{ --- (2)}$$

$$\alpha_1 - 2\alpha_2 = -4 \text{ --- (3)}$$

Solving (1), (2) & (3) we get

$$\alpha_1 = -4, \alpha_2 = 0$$

We know that If S is L.I set then $S \cup \{v\}$ is L.D iff $V \notin L(S)$

$\therefore S_1 \cup \{u_3\}$ is L.D iff $u_3 \in L(S)$

u_3 not belongs to the basis

$$\text{Choose } u_4 \in L(S_1)$$

$$(1, 37, -17) = \alpha_1 u_1 + \alpha_2 u_2$$

$$(1, 37, -17) = \alpha_1(2, -3, 1) + \alpha_2(1, 4, -2)$$

$$(1, 37, -17) = (2\alpha_1 + \alpha_2, -3\alpha_1 + 4\alpha_2, \alpha_1 - 2\alpha_2)$$

$$2\alpha_1 + \alpha_2 = 1 \text{ --- (4)}$$

$$-3\alpha_1 + 4\alpha_2 = 37 \text{ --- (5)}$$

$$\alpha_1 - 2\alpha_2 = -17 \text{ --- (6)}$$

Solving (4), (5) & (6) we get

$$\alpha_1 = -3, \alpha_2 = 7$$

$\therefore S_1 \cup \{u_4\}$ is L.D iff $u_4 \in L(S)$

u_4 not belongs to the basis.

$$\text{Choose } u_5 \in L(S_1)$$

$$(-3, -3, 8) = \alpha_1 u_1 + \alpha_2 u_2$$

$$(-3, -3, 8) = \alpha_1(2, -3, 1) + \alpha_2(1, 4, -2)$$

$$(-3, -3, 8) = (2\alpha_1 + \alpha_2, -3\alpha_1 + 4\alpha_2, \alpha_1 - 2\alpha_2)$$

$$2\alpha_1 + \alpha_2 = -3 \text{ --- (7)}$$

$$-3\alpha_1 + 4\alpha_2 = -3 \text{ --- (8)}$$

$$\alpha_1 - 2\alpha_2 = 8 \text{ --- (9)}$$

	Solving (4),(5)& (6) we get $\alpha_1 = -3, \alpha_2 = 7$ $\therefore S_1 \cup \{u_5\}$ is L.D iff $u_5 \notin L(S)$ u_5 belongs to the basis. $\therefore S = \{u_1, u_2, \dots, u_5\}$ form a basis.
15	Let $S = \{v_1, v_2, v_3\}$ where $v_1 = (1, -3, -2), v_2 = (-3, 1, 3), v_3 = (-2, -10, -2)$. Verify whether S form a basis or not. Solution: Let $a_1 v_1 + a_2 v_2 + a_3 v_3 = 0$ where a_1, a_2, a_3, a_4 are scalars $a_1(1, -3, -2) + a_2(-3, 1, 3) + a_3(-2, -10, -2) = 0$ $(a_1 - 3a_2 - 2a_3, -3a_1 + a_2 - 10a_3, -2a_1 + 3a_2 - 2a_3) = 0$ $\Rightarrow a_1 - 3a_2 - 2a_3 = 0, -3a_1 + a_2 - 10a_3 = 0, -2a_1 + 3a_2 - 2a_3 = 0$ $\begin{vmatrix} 1 & -3 & -2 \\ -3 & 1 & -10 \\ -2 & 3 & -2 \end{vmatrix} = 0$ Hence the given vectors are linearly dependent Therefore, the given set of vectors does not form a basis The set of vectors is not a basis.
16	Check whether the sets are basis of R^3 or not $\{(1, 0, -1), (2, 5, 1), (0, -4, 3)\}$ Solution: $v_1 = (1, 0, -1), v_2 = (2, 5, 1), v_3 = (0, -4, 3)$ Let $S = \{(1, 0, -1), (2, 5, 1), (0, -4, 3)\}$ To prove A is basis or not: (i.e) (i) S is linearly independent (ii) $V=L(S)$ $(i) A = \begin{vmatrix} 1 & 0 & -1 \\ 2 & 5 & 1 \\ 0 & -4 & 3 \end{vmatrix}$ $= 1(15 + 4) - 0 - 1(-8 - 0) = 27 \neq 0$ It is linearly independent. Let $(x, y, z) \in R^3$ Consider $a(1, 0, -1) + b(2, 5, 1) + c(0, -4, 3) = (x, y, z)$ $a + 2b = x, 5b - 4c = y, -a + b + 3c = z$ $[A/B] \sim \begin{pmatrix} 1 & 2 & 0 & x \\ 0 & 5 & -4 & y \\ -1 & 1 & 3 & z \end{pmatrix}$ $\sim \begin{pmatrix} 1 & 2 & 0 & x \\ 0 & 5 & -4 & y \\ 0 & 3 & 3 & z+x \end{pmatrix} R_3 \rightarrow R_3 + R_1$ $\sim \begin{pmatrix} 1 & 2 & 0 & x \\ 0 & 5 & -4 & y \\ 0 & 0 & 27 & 5z+5x-3y \end{pmatrix} R_3 \rightarrow 5R_3 - 3R_2$

	Using back substitution method, $27c = 5z + 5x - 3y$ $c = \frac{5z + 5x - 3y}{27}$ $5b - 4c = y$ $5b - 4\left(\frac{5z + 5x - 3y}{27}\right) = y$ $5b = y + \frac{20z + 20x - 12y}{27}$ $5b = \frac{27y + 20z + 20x - 12y}{27} = \frac{15y + 20z + 20x}{27}$ $b = \frac{3y + 4z + 4x}{27}$ $a + 2b = x$ $a = x - 2\left(\frac{3y + 4z + 4x}{27}\right) = \frac{19x - 6y - 8z}{27}$ $\therefore V = L(S)$
17	State and prove Completion of basis theorem Statement: Let $\dim V = n$ then any linearly independent subset of V can be extended to form a basis of V. Proof: Let $S = \{v_1, v_2, \dots, v_n\}$ be a linearly independent set of vectors in V. Given: $\dim V = n$ If $L(S) = V$, then S is a basis of V otherwise $L(S) \neq V$, then there is an element $V_{r+1} \in V$ such that $V_{r+1} \notin L(S)$ We know that $S \cup \{v\}$ is linearly dependent iff $V \in L(S)$ where S be an independent set in $V(F)$ and v be a vector in $V(F)$ that is not in S. $\therefore S \cup \{v_{r+1}\}$ is linearly independent if $V_{r+1} \notin L(S)$ $\therefore S_1 = \{v_1, v_2, \dots, v_r, v_{r+1}\}$ is linearly independent If $L(S_1) = V$, then S_1 is basis of V otherwise $L(S_1) \neq V$, then there is element $V_{r+2} \in V$ such that $V_{r+2} \notin L(S_1)$ $\Rightarrow S_2 = \{S_1 \cup V_{r+2}\}$ is linearly independent if $V_{r+2} \notin L(S_1)$. $\therefore S_2$ is linearly independent Proceed until the dimension value. Since $\dim V = n$ $S = (v_1, v_2, \dots, v_n)$ spans V It is a basis of V.
18	Let S be a linearly independent set in $V(F)$ and let v be a vector in $V(F)$ that is not in S. Then $S \cup \{v\}$ is linearly dependent iff $v \in L(S)$.
	Proof: Let S be a linearly independent set in $V(F)$. Let V be a vector in $V(F)$ that is not in S. Also given $S \cup (V)$ is linearly dependent To prove $V \in L(S)$ If $S \cup (V)$ is linearly dependent then there exists a vector $v_1, v_2, v_3, \dots, v_n \in S$ and $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n \in F$

	Then $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n = 0$ not all α_i are zero. Since S is linearly independent one of v_i's is v. (i.e) $v_i = v$ $\alpha_1 v_1 = \alpha_2 v_2 - \alpha_3 v_3 - \dots - \alpha_n v_n$ $v = \alpha_2 \alpha_1^{-1} v_2 - \alpha_3 \alpha_1^{-1} v_3 - \dots - \alpha_n \alpha_1^{-1} v_n$ v is a linear combination of element in S. $\therefore v \in L(S)$ Conversely, Let $v \in L(S)$ To prove: $S \cup \{v\}$ is linearly dependent then there exists a finite vectors $v_1, v_2, v_3, \dots, v_n \in S \text{ and } \alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n \in F$ such that $v = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n$ v can be expressed as a linear combination of other vectors (i.e) $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n - v = 0$ This is the linear combination with not all coefficient as zero. $\therefore v_1, v_2, v_3, \dots, v_n$ are linearly dependent over R $\therefore S \cup \{v\}$ is linearly dependent.
UNIT III LINEAR TRANSFORMATION Linear transformation - Rank space and null space - Rank and nullity - Dimension theorem - Matrix representation of linear transformation - Eigenvalues and eigenvectors of linear transformation.	
PART A	
1.	Define Linear transformation on a vector space.
	Solution: Let V and W be vector spaces over F. We call a function $T : V \rightarrow W$ a linear transformation from V to W if for all $x, y \in V$ and $c \in F$, we have $T(x + y) = T(x) + T(y)$ and $T(cx) = cT(x)$.
2.	Show that T: $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (2a_1 + a_2, a_1)$ is linear.
	Solution: Let $X, Y \in \mathbb{R}^2$ and $c \in \mathbb{R}$, where $X = (b_1, b_2)$, $Y = (d_1, d_2)$ Since we know that T is linear if and only if $T(cX + Y) = cT(X) + T(Y)$ Now $(cX + Y) = (cb_1 + d_1, cb_2 + d_2)$ $T(cX + Y) = (2(cb_1 + d_1) + cb_2 + d_2, cb_1 + d_1)$ Also $cT(X) + T(Y) = c(2b_1 + b_2, b_1) + (2d_1 + d_2, d_1)$ $= (2cb_1 + cb_2 + 2d_1 + d_2, cb_1 + d_1)$ $= (2(cb_1 + d_1) + cb_2 + d_2, cb_1 + d_1)$ $T(cX + Y) = cT(X) + T(Y)$ Therefore T is linear.
3.	Verify that $T : \mathbb{R}^3 \rightarrow \mathbb{R}$ and $T(u) = \ u\$ is a linear transformation or not. (Apr/May 2022)
	Solution: $T(u + v) = \ u + v\ \leq \ u\ + \ v\ , \text{ by Triangle inequality}$ $\Rightarrow T(u + v) \leq T(u) + T(v) \Rightarrow T(u + v) \neq T(u) + T(v) \text{ So it is not a linear transformation.}$
4.	Determine whether the function T: $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (x^2, y)$ is linear? (Nov/Dec 2022)
	Solution: $T((x, y) + (z, w)) = T(x + z, y + w) = ((x + z)^2, y + w)$ $\neq (x^2, y) + (z^2, w) = T(x, y) + T(z, w)$ Therefore T is not linear.
5.	Let T: $\mathbb{R} \rightarrow \mathbb{R}$ be defined by $T(x) = 2^x \forall x \in \mathbb{R}$. Is T linear? (April/May 2023)
	Solution: No. For example, if $x = 1, y = 2$ then $T(x) + T(y) = T(1) + T(2) = 2 + 2^2 = 6$ But $T(x + y) = T(1 + 2) = T(3) = 2^3 = 8$

	$\therefore T(x+y) \neq T(x) + T(y) \quad \forall x,y \in \mathbb{R} \Rightarrow T$ is not linear.
6.	Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation such that $T(1, 0, 0) = (2, 4, -1)$, $T(0, 1, 0) = (1, 3, -2)$, $T(0, 0, 1) = (0, -2, 2)$. Compute $T(-2, 4, -1)$.
	Solution: $(-2, 4, -1) = -2(1, 0, 0) + 4(0, 1, 0) - (0, 0, 1)$ $T(-2, 4, -1) = -2T(1, 0, 0) + 4T(0, 1, 0) - T(0, 0, 1)$ since T is linear $= -2(2, 4, -1) + 4(1, 3, -2) - (0, -2, 2) = (3, 5, -1)$
7.	Define null space and range of T. (Nov/Dec 2022)
	Solution: Let V and W be vector spaces and let $T: V \rightarrow W$ be linear. The null space or kernel, $N(T)$ is the set of all vectors x in V such that $T(x) = 0$ i.e, $N(T) = \{x \in V : T(x) = 0\}$. The Range or image $R(T)$ is the subset of W consisting of all images under T of vectors in V i.e, $R(T) = \{T(x) : x \in V\}$.
8.	Define Rank and Nullity of T.
	Solution: Let V, W are vector spaces over F and $T: V \rightarrow W$ be a linear transformation. If $R(T)$ and $N(T)$ are finite-dimensional, then $\text{rank}(T) = \dim[R(T)]$ and $\text{nullity}(T) = \dim[N(T)]$
9.	State Dimension theorem. (Apr/May 2022)
	Statement: Let V and W be vector spaces and let $T: V \rightarrow W$ be linear. If V is finite dimensional, then $\text{Nullity}(T) + \text{rank}(T) = \dim(V)$.
10.	Determine whether $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$ is one to one or onto.
	Solution: T is one to one or onto only if $N(T) = \{0\}$ $N(T) = \{x : T(x) = 0\}$ $T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3) = (0, 0)$ $\Rightarrow a_1 - a_2 = 0$ and $2a_3 = 0$ $\Rightarrow a_1 = a_2$ and $a_3 = 0$ $N(T) = x = (a_1, a_1, 0) \neq \{0\}$ Therefore, T is not one to one. By the Dimension Theorem, we obtained that $\dim(R(T)) = 3 - \dim(N(T)) = 2$. This implies that $\dim$. of range of T coincides with $\dim$. of $\mathbb{R}^2$, i.e. $R(T) = \mathbb{R}^2$, i.e. T is onto.
11.	Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x,y,z) = (x - y, 2z)$ is a linear transformation. Find the nullity of T. Is T is onto?
	Solution: $N(T) = \{(x,y,z): T(x,y,z) = 0\}$ $T(x,y,z) = 0 \Rightarrow x - y = 0, 2z = 0 \Rightarrow z = 0; x = y$ $\therefore N(T) = \{(x,x,0): x \in \mathbb{R}\} = \{x(1,1,0): x \in \mathbb{R}\}$ $\therefore N(T)$ is spanned by single nonzero element $(1,1,0)$ and $\{(1,1,0)\}$ is linearly independent and hence a basis of $N(T)$. $\therefore \text{nullity} = \dim N(T) = 1 \Rightarrow \dim \mathbb{R}^3 = 3$ $\text{Rank}(T) = 3 - 1 = 2 = \dim \mathbb{R}^2 \Rightarrow \therefore T$ is onto.
12.	For the following linear operator T on a vector space V and ordered basis β compute $[T]_\beta$ $V = \mathbb{R}^2, T \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 10a - 6b \\ 17a - 10b \end{pmatrix}$ and $\beta = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right\}$. (April/May 2023)
	Solution: $T \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 10a - 6b \\ 17a - 10b \end{pmatrix}$

	$T \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 10-12 \\ 17-20 \end{pmatrix} = \begin{pmatrix} -2 \\ -3 \end{pmatrix}$ $T \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 20-18 \\ 34-30 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$ $[T]_{\beta} = \begin{bmatrix} -2 & 2 \\ -3 & 4 \end{bmatrix}$
13.	Let $T : P_3(R) \rightarrow P_2(R)$ be a linear transformation defined by $T(f(x)) = f'(x)$. Let B_1 and B_2 be the standard bases for $P_3(R)$ and $P_2(R)$ respectively. Then find $[T]$
	Solution: $T(f(x)) = f'(x)$ $T(1) = 0 = 0.1 + 0.x + 0.x^2$ $T(x) = 1 = 1.1 + 0.x + 0.x^2$ $T(x^2) = 2x = 0.1 + 2.x + 0.x^2$ $T(x^3) = 3x^2 = 0.1 + 0.x + 3.x^2$ $[T]_{B_2}^{B_1} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$
14.	Obtain the matrix representing the linear transformation $T : V_3(R) \rightarrow V_3(R)$ given by $T(a, b, c) = (3a, a-b, 2a+b+c)$ with respect to the standard basis $\{e_1, e_2, e_3\}$.
	Solution: $T(e_1) = T(1, 0, 0) = (3, 1, 2) = 3e_1 + e_2 + 2e_3$ $T(e_2) = T(0, 1, 0) = (0, -1, 1) = -e_2 + e_3$ $T(e_3) = T(0, 0, 1) = (0, 0, 1) = e_3$ The matrix representing T is $\begin{pmatrix} 3 & 1 & 2 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$
15.	Define eigen value and eigen vector of a linear transformation T.
	Solution: Let T be a linear operator in a vector space V. A non zero vector $v \in V$ is called an eigen vector of T if there exists a scalar λ such that $T(v) = \lambda v$. The scalar λ is called the eigen value corresponding to the eigen vector v.
16.	Find the eigen values of $A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \in M_{2 \times 2}(R)$.
	Solution: $A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$ $\det(A - tI_2) = \begin{vmatrix} 1-t & 1 \\ 4 & 1-t \end{vmatrix}$ $= (1-t)^2 - 4 = t^2 - 2t + 1 - 4$ $= t^2 - 2t - 3 = (t-3)(t+1)$ Eigen values of A are 3 and -1.

17.	Find the characteristic equation and the eigen values of $\begin{bmatrix} 1 & -4 \\ 4 & 2 \end{bmatrix}$.
	Solution: The characteristic polynomial is $\det(A - tI_2) = 0$ $\det(A - tI_2) = \begin{vmatrix} 1-t & -4 \\ 4 & 2-t \end{vmatrix} = 0$ $\Rightarrow (1-t)(2-t) + 16 = 0$ $t^2 - 3t + 18 = 0 \quad \Rightarrow \quad t = \frac{3 \pm \sqrt{9-72}}{2} = \frac{3 \pm \sqrt{-63}}{2}$ Since t is complex, A has no real eigen values. The matrix A is acting on a real vector space R^2 and there is no non-zero vector X in R^2 such that $AX = \lambda X$.
18.	Find the characteristic equation of $\begin{bmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & -8 & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.
	Solution: The characteristic equation is $\det(A - tI_4) = 0$ $\det(A - tI_4) = \begin{vmatrix} 5-t & -2 & 6 & -1 \\ 0 & 3-t & -8 & 0 \\ 0 & 0 & 5-t & 4 \\ 0 & 0 & 0 & 1-t \end{vmatrix} = 0$ $(5-t)(3-t)(5-t)(1-t) = 0$ $t^4 - 14t^3 + 68t^2 - 130t + 75 = 0$
19.	Let $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$, $u = \begin{bmatrix} 6 \\ -5 \end{bmatrix}$, $v = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$. Are u and v eigen vectors of A ?
	Solution: $Au = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ -5 \end{bmatrix} = \begin{bmatrix} -24 \\ 20 \end{bmatrix} = -4 \begin{bmatrix} 6 \\ -5 \end{bmatrix} = -4u$ $Av = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} -9 \\ 11 \end{bmatrix} \neq \lambda v$ u is the eigen vector of A corresponding to an eigen value -4 but v is not an eigen vector of A because Av is not a multiple of v.
20.	Let $A = \begin{bmatrix} -3 & 12 \\ -2 & 7 \end{bmatrix}$, $v_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Suppose v_1 and v_2 are eigen vectors of A , find the eigen values of A .
	Solution: $Av_1 = \begin{bmatrix} -3 & 12 \\ -2 & 7 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 1 \cdot \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ $Av_2 = \begin{bmatrix} -3 & 12 \\ -2 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix} = 3 \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ Eigen values are 1 and 3.
PART B	
1.	State and prove Dimension theorem. (Nov/Dec 2022) (April/May 2023)

Statement:

Let V and W be vector spaces, and let $T : V \rightarrow W$ be linear. If V is finite-dimensional, then $\text{nullity}(T) + \text{rank}(T) = \dim(V)$.

Proof:

Suppose that $\dim(V) = n$, $\dim(N(T)) = k$, and $\{v_1, v_2, \dots, v_k\}$ is a basis for $N(T)$.

We know that if W is a subspace of a finite-dimensional vector space V , then any basis for W can be extended to a basis for V . So we may extend $\{v_1, v_2, \dots, v_k\}$ to a basis $\beta = \{v_1, v_2, \dots, v_n\}$ for V .

We claim that $S = \{T(v_{k+1}), T(v_{k+2}), \dots, T(v_n)\}$ is a basis for $R(T)$.

First we prove that S generates $R(T)$. Since $T(v_i) = 0$ for $1 \leq i \leq k$, we have

$$\begin{aligned} R(T) &= \text{span}(\{T(v_1), T(v_2), \dots, T(v_n)\}) \\ &= \text{span}(\{T(v_{k+1}), T(v_{k+2}), \dots, T(v_n)\}) = \text{span}(S). \end{aligned}$$

Now we prove that S is linearly independent.

Suppose that $\sum_{i=k+1}^n b_i T(v_i) = 0$ for $b_{k+1}, b_{k+2}, \dots, b_n \in F$.

Since T is linear, we have

$$T\left(\sum_{i=k+1}^n b_i v_i\right) = 0.$$

So $\sum_{i=k+1}^n b_i v_i \in N(T)$.

Hence there exist $c_1, c_2, \dots, c_k \in F$ such that

$$\sum_{i=k+1}^n b_i v_i = \sum_{i=1}^k c_i v_i \text{ or } \sum_{i=1}^k (-c_i) v_i + \sum_{i=k+1}^n b_i v_i = 0.$$

Since β is a basis for V , we have $b_i = 0$ for all i . Hence S is linearly independent. This shows that $T(v_{k+1}), T(v_{k+2}), \dots, T(v_n)$ are distinct; $\therefore \text{rank}(T) = n - k$. Hence the proof.

2 Let V and W be vector spaces and let $T : V \rightarrow W$ be linear. Then prove that T is one to one if and only if $N(T) = \{0\}$. (Nov/Dec 2022)

Solution:

Suppose that T is one-to-one and $x \in N(T)$.

Then $T(x) = 0 = T(0)$. Since T is one-to-one, we have $x = 0$. Hence $N(T) = \{0\}$.

Now assume that $N(T) = \{0\}$, and suppose that $T(x) = T(y)$.

$$\text{Then } 0 = T(x) - T(y) = T(x - y)$$

$$\therefore x - y \in N(T) = \{0\}.$$

So $x - y = 0$, or $x = y$. This means that T is one-to-one.

**3 Let V and W be vector spaces of equal (finite) dimension and let $T : V \rightarrow W$ be linear. Then prove that the following are equivalent
(i) T is one to one (ii) T is onto (iii) $\text{Rank}(T) = \dim(V)$ (Apr/May 2022)**

Solution:

From the dimension theorem, we have

$$\text{nullity}(T) + \text{rank}(T) = \dim(V) \text{ -----(1)}$$

Now by Q2(i), we have that T is one-to-one $\Leftrightarrow N(T) = \{0\}$,

$$\Leftrightarrow \text{nullity}(T) = 0,$$

$$\Leftrightarrow \text{rank}(T) = \dim(V) \text{ using (1)}$$

$$\Leftrightarrow \text{rank}(T) = \dim(W), \text{ by hypothesis}$$

$$\Leftrightarrow \dim(R(T)) = \dim(W), \text{ by the definition of rank}$$

$$\Leftrightarrow R(T) = W,$$

$$\Leftrightarrow T \text{ is onto}$$

	Thus T is one to one $\Leftrightarrow \text{Rank}(T) = \dim(V) \Leftrightarrow T$ is onto
4	For the following transformation $T : R^3 \rightarrow R^2$ defined by $T(x, y, z) = (2x - y, 3z)$. Verify whether T is linear or not. Find $N(T)$, $R(T)$ and hence verify the dimension theorem. Solution: $T(x_1 + x_2, y_1 + y_2, z_1 + z_2) = (2x_1 + 2x_2 - y_1 - y_2, 3z_1 + 3z_2)$ $= (2x_1 - y_1, 3z_1) + (2x_2 - y_2, 3z_2)$ $= T(x_1, y_1, z_1) + T(x_2, y_2, z_2)$ $T(\alpha(x_1, y_1, z_1)) = T(\alpha x_1, \alpha y_1, \alpha z_1)$ $= (2\alpha x_1 - \alpha y_1, 3\alpha z_1)$ $= \alpha(2x_1 - y_1, 3z_1)$ $\Rightarrow T \text{ is linear}$ $N(T) = \{(x, y, z) / T(x, y, z) = (0, 0)\}$ $= \{(x, y, z) / 2x - y = 0, 3z = 0\}$ $= \{(x, y, 0) / 2x = y\} \text{ i.e., The line } y = 2x \text{ on } xy\text{-plane.}$ $\dim(N(T)) = 1$ $R(T) = \text{span}\{T(1, 0, 0), T(0, 1, 0), T(0, 0, 1)\}$ $= \text{span}\{(2, 0), (-1, 0), (0, 3)\}$ $= \text{span}\{(2, 0), (0, 3)\}$ $\dim(R(T)) = 2$ By Dimension Theorem, $\dim(R^3) = \dim N(T) + \dim R(T)$ LHS=3; RHS=1+2=3 Hence Verified
5	For the following transformation $T : R^2 \rightarrow R^3$ defined by $T(x, y) = (x - y, x, 2x + y)$. Let $B = \{e_1, e_2\}$; $e_1 = (1, 2)$; $e_2 = (2, 3)$ be the basis for R^2 and $B' = \{(1, 1, 0), (0, 1, 1), (2, 2, 3)\}$ be the basis for R^3. Find the matrix of the transformation with respect to these bases. (Apr/May 2022, Nov/Dec 2022) Solution: $T(e_1) = T(1, 2) = (-1, 1, 4) = -\frac{7}{3}(1, 1, 0) + 2(0, 1, 1) + \frac{2}{3}(1, 1, 0)$ $T(e_2) = T(2, 3) = (-1, 2, 7) = -\frac{11}{3}(1, 1, 0) + 3(0, 1, 1) + \frac{4}{3}(1, 1, 0)$ $T = \begin{bmatrix} -\frac{7}{3} & 2 & \frac{2}{3} \\ -\frac{11}{3} & 3 & \frac{4}{3} \end{bmatrix}^T = \begin{bmatrix} -\frac{7}{3} & -\frac{11}{3} \\ 2 & 3 \\ \frac{2}{3} & \frac{4}{3} \end{bmatrix}$
6	Let $T : P_2(R) \rightarrow P_2(R)$ be defined by $T(f(x)) = f(x) + (x+1)f'(x)$. Find the eigen values and corresponding eigen vectors of T with respect to standard basis of $P_2(R)$. (Apr/May 2022) Solution: Let T be the linear operator on $P_2(R)$ defined by $T(f(x)) = f(x) + (x+1)f'(x)$. Let $\beta = \{1, x, x^2\}$ be the standard ordered basis for $P_2(R)$ $T(1) = 1.1 + 0.x + 0.x^2$

$$T(x) = x + (x + 1) = 1 + 2x = 1.1 + 2.x + 0.x^2$$

$$T(x^2) = x^2 + (x + 1)2x = 2x + 3x^2 = 0.1 + 2.x + 3.x^2$$

Let $A = [T]_{\beta}$. Then $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{pmatrix}$

The characteristic polynomial of T is

$$\det(A - tI_3) = \det \begin{pmatrix} 1-t & 1 & 0 \\ 0 & 2-t & 2 \\ 0 & 0 & 3-t \end{pmatrix}$$

$$= (1-t)(2-t)(3-t)$$

$$= -(t-1)(t-2)(t-3)$$

Hence λ is an eigenvalue of T (or A) if and only if $\lambda = 1, 2$, or 3 .

Eigenvector for $\lambda_1 = 1$:

$$(A - \lambda_1 I)X = 0$$

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$x_2 = 0$$

$$x_2 + 2x_3 = 0$$

$$2x_3 = 0$$

$$x_3 = 0, x_2 = 0 \Rightarrow X_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Eigenvector for $\lambda_2 = 2$:

$$(A - \lambda_2 I)X = 0$$

$$\begin{pmatrix} -1 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$-x_1 + x_2 = 0 \Rightarrow x_2 = x_1$$

$$2x_3 = 0 \Rightarrow x_3 = 0$$

$$x_3 = 0, \text{ take } x_1 = x_2 = 1 \Rightarrow X_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

Eigenvector for $\lambda_3 = 3$:

$$(A - \lambda_3 I)X = 0$$

$$\begin{pmatrix} -2 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

	$-2x_1 + x_2 = 0 \Rightarrow x_2 = 2x_1$ $-x_2 + 2x_3 = 0 \Rightarrow x_2 = 2x_3$ $x_1 = 1, x_2 = 2, x_3 = 1 \quad \therefore X_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$
7	Find the linear transformation $T: V_3(R) \rightarrow V_3(R)$ determined by the matrix $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$ with respect to the standard basis $\{e_1, e_2, e_3\}$.
	Solution: $T(e_1) = e_1 + 2e_2 + e_3 = (1, 2, 1)$ $T(e_2) = 0e_1 + e_2 + e_3 = (0, 1, 1)$ $T(e_3) = -e_1 + 3e_2 + 4e_3 = (-1, 3, 4)$ Now $(a, b, c) = a(1, 0, 0) + b(0, 1, 0) + c(0, 0, 1)$ $(a, b, c) = ae_1 + be_2 + ce_3$ $T(a, b, c) = T(ae_1 + be_2 + ce_3)$ $= aT(e_1) + bT(e_2) + cT(e_3)$ $= a(1, 2, 1) + b(0, 1, 1) + c(-1, 3, 4)$ $T(a, b, c) = (a - c, 2a + b + 3c, a + b + 4c) \quad \text{This is the required linear transformation.}$
8	For the following linear operator T on a vector space V and ordered basis β, compute $[T]_\beta$ and determine whether β is a basis consisting of eigen vectors of T. $V = R^3, T \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 3a + 2b - 2c \\ -4a - 3b + 2c \\ -c \end{pmatrix} \text{ and } \beta = \left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \right\}.$
	Solution: $T \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 3a + 2b - 2c \\ -4a - 3b + 2c \\ -c \end{pmatrix} \quad \& \quad \beta = \left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \right\}$

$$T \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3(0) + 2(1) - 2(1) \\ -4(0) - 3(1) + 2(1) \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ -1 \end{pmatrix} = -1 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + 0 \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

Similarly,

$$T \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = 0 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

$$T \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ -2 \end{pmatrix} = 0 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + 0 \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} - 1 \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

$$\therefore [T]_{\beta} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\text{Since } T \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ -1 \end{pmatrix} = -1 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad T \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \text{ and } T \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ -2 \end{pmatrix} = -1 \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

The eigen values are $-1, 1, -1$ and eigen vectors are $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$.

$\therefore \beta$ is a basis consisting of eigenvectors of T .

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Consider the basis $S = \{v_1, v_2, v_3\}$ for R^3 , where $v_1 = (1, 1, 1), v_2 = (1, 1, 0), v_3 = (1, 0, 0)$.

Let $T : R^3 \rightarrow R^2$ be the linear transformation such that $T(v_1) = (1, 0), T(v_2) = (2, -1)$ and $T(v_3) = (4, 3)$. Find the formulae for $T(x_1, x_2, x_3)$, then use this formulae to compute $T(2, -3, 5)$.

Solution:

We first express $x = (x_1, x_2, x_3)$ as a linear combination of v_1, v_2 and v_3 .

If we write $(x_1, x_2, x_3) = C_1(1, 1, 1) + C_2(1, 1, 0) + C_3(1, 0, 0)$

Then on equating corresponding components

$$C_1 + C_2 + C_3 = x_1$$

$$C_1 + C_2 = x_2$$

$$C_1 = x_3$$

By solving, $C_1 = x_3, C_2 = x_2 - x_3, C_3 = x_1 - x_2$.

$$(x_1, x_2, x_3) = x_3(1, 1, 1) + (x_2 - x_3)(1, 1, 0) + (x_1 - x_2)(1, 0, 0)$$

$$= x_3 v_1 + (x_2 - x_3) v_2 + (x_1 - x_2) v_3$$

$$\text{Thus } T(x_1, x_2, x_3) = x_3 T(v_1) + (x_2 - x_3) T(v_2) + (x_1 - x_2) T(v_3)$$

$$= x_3(1, 0) + (x_2 - x_3)(2, -1) + (x_1 - x_2)(4, 3)$$

$$= 4x_1 - 2x_2 - x_3, 3x_1 - 4x_2 + x_3$$

From this we obtain, $T(2, -3, 5) = (9, 2, 3)$.

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$T : R^2 \rightarrow R^3$ is a linear transformation such that $T(1, 1) = (1, 0, 2); T(2, 3) = (1, -1, 4)$.

(i) Determine T

(ii) Find $T(2, 5), T(8, 11)$

(iii) Find nullity of T

(iv) Find rank of T

(Apr/May 2022, Nov/Dec 2022)

	Solution: (i) $T(1,1) = (1,0,2) \Rightarrow T(1,0) + T(0,1) = (1,0,2)$ $T(2,3) = (1,-1,4) \Rightarrow 2T(1,0) + 3T(0,1) = (1,-1,4)$ Solving, $T(1,0) = (2,1,2)$ and $T(0,1) = (-1,-1,0)$ $T(x,y) = xT(1,0) + yT(0,1)$ $= x(2,1,2) + y(-1,-1,0)$ $T(x,y) = (2x-y, x-y, 2x)$ (ii) $T(2,5) = (-1,-3,4)$ and $T(8,11) = (5,-3,16)$ (iii) $N(T) = \{(x,y) / T(x,y) = (0,0,0)\}$ $(2x-y, x-y, 2x) = (0,0,0) \Rightarrow x = 0; y = 0$ $N(T) = \{(0,0)\} \Rightarrow \text{nullity}(T) = 0$ (iv) By Dimension theorem, $\text{nullity}(T) + \text{rank}(T) = \dim(R^2)$ $0 + \text{rank}(T) = 2 \Rightarrow \text{rank}(T) = 2$
11	Let $T : R^2 \rightarrow P_2(R)$ is a linear transformation such that $T(1,1) = 2 - 3x + x^2, T(2,3) = 1 - x^2$ (i) Determine T (ii) Find $T(-1,2), T(-5,2)$ (iii) Find rank T (iii) Find nullity of T (April/May 2023)
	Solution: (i) $T(1,1) = 2 - 3x + x^2 \Rightarrow T(1,0) + T(0,1) = 2 - 3x + x^2$ $T(2,3) = 1 - x^2 \Rightarrow 2T(1,0) + 3T(0,1) = 1 - x^2$ Solving, $T(1,0) = 5 - 9x + 4x^2$ and $T(0,1) = -3 + 6x - 3x^2$ $T(a,b) = aT(1,0) + bT(0,1)$ $= a(5 - 9x + 4x^2) + b(-3 + 6x - 3x^2)$ $T(a,b) = (5a - 3b) + (-9a + 6b)x + (4a - 3b)x^2$ (ii) $T(-1,2) = -11 + 21x - 10x^2$ and $T(-5,2) = -31 + 57x - 26x^2$ (iv) $N(T) = \{(a,b) \in R^2 / T(a,b) = 0\}$ $(5a - 3b) + (-9a + 6b)x + (4a - 3b)x^2 = 0 + 0x + 0x^2 \Rightarrow a = 0; b = 0$ $N(T) = \{(0,0)\} \Rightarrow \text{nullity}(T) = 0$ (iii) By Dimension theorem, $\text{nullity}(T) + \text{rank}(T) = \dim(R^2)$ $0 + \text{rank}(T) = 2 \Rightarrow \text{rank}(T) = 2$
12	Find the eigen values and eigen vectors of $A = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix}$.
	Solution: <u>To find the eigen values:</u> $A = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix}$ We compute the characteristic polynomial: $\det(A - tI_2) = \det \begin{bmatrix} 1-t & 1 \\ 4 & 1-t \end{bmatrix}$ $= t^2 - 2t - 3 = (t-3)(t+1).$ $\therefore t$ is an eigen value of A if and only if $\det(A - tI_n) = 0$, $\det(A - tI_2) = 0 \Rightarrow t = 3, -1$ Hence the only eigen values of A are 3 and -1. <u>To find the eigen vectors:</u> Let $\lambda_1 = 3$ and $\lambda_2 = -1$.

	Eigenvector corresponding to $\lambda_1 = 3$. $(A - \lambda_1 I)X = 0$ $\left(\begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} - \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \right) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $\begin{aligned} -2x_1 + x_2 &= 0 \\ 4x_1 - 2x_2 &= 0 \end{aligned}$ Clearly the set of all solutions to this equation is $\left\{ t \begin{pmatrix} 1 \\ 2 \end{pmatrix} : t \neq 0 \in R \right\}$ Eigenvector corresponding to $\lambda_2 = -1$. $(A - \lambda_1 I)X = 0$ $\left(\begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} - \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \right) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $\begin{aligned} 2x_1 + x_2 &= 0 \\ 4x_1 + 2x_2 &= 0. \end{aligned}$ Clearly the set of all solutions to this equation is $\left\{ t \begin{pmatrix} 1 \\ -2 \end{pmatrix} : t \neq 0 \in R \right\}$
13	Find the eigen values and eigen vectors of $A = \begin{pmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{pmatrix}$
	Solution: The characteristic equation is $\lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0$. $S_1 =$ Sum of the main diagonal elements $= -1$; $S_2 =$ Sum of the minors of the main diagonal elements $= -21$; $S_3 = A = 45$. The characteristic equation is $\lambda^3 + \lambda^2 - 21\lambda - 45 = 0$. $(\lambda + 3)(\lambda + 3)(\lambda - 5) = 0$ $\lambda = -3, -3, 5$ Consider $\begin{pmatrix} -2 - \lambda & 2 & -3 \\ 2 & 1 - \lambda & -6 \\ -1 & -2 & 0 - \lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \quad \text{-----(1)}$ Case (1) : $\lambda = 5$ Substituting $\lambda = 5$ in the (1) and solving we get $-7x_1 + 2x_2 - 3x_3 = 0, 2x_1 - 4x_2 - 6x_3 = 0,$ $x_1 - 2x_2 - 3x_3 = 0$ Solving the above three equations we get $x_1 = -1, x_2 = -2, x_3 = 1$ Eigen vector is $X_1 = \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}$ Case (2) : $\lambda = -3$ Substituting $\lambda = -3$ in the (1) and solving we get $x_1 + 2x_2 - 3x_3 = 0, 2x_1 + 4x_2 - 6x_3 = 0,$ $x_1 + 2x_2 - 3x_3 = 0,$ The above three equations are same. $\Rightarrow x_1 + 2x_2 - 3x_3 = 0,$

Put $x_2 = 0$ we get the value, $x_1 = 3$ $x_3 = 1$ $\therefore X_2 = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$

Put $x_3 = 0$ we get the value, $x_1 = -2$ $x_2 = 1$ $\therefore X_3 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$

UNIT IV INNER PRODUCT SPACES

Inner product and norms - Properties - Orthogonal, Orthonormal vectors - Gram Schmidt orthonormalization process - Least square approximation

PART A

1.	Define Inner Product space. (Nov/Dec 2022)
	Solution: Let V be a vector space over F. An inner product on V is a function that assigns, to every ordered pair of vectors x and y in V then i). $\langle x+z, y \rangle = \langle x, y \rangle + \langle z, y \rangle$ ii). $\langle cx, y \rangle = c\langle x, y \rangle$ iii). $\langle \overline{x}, y \rangle = \langle y, x \rangle$ iv). $\langle x, x \rangle > 0$ if $x \neq 0$
2.	In an Euclidean inner product find cosine of the angle between the vectors $u = (2, 3, 5)$ and $v = (1, -4, 3)$.
	Solution: $\cos \theta = \frac{\langle u, v \rangle}{\ u\ \ v\ }$ $\langle u, v \rangle = 2 - 12 + 15 = 5, \ u\ = \sqrt{4 + 9 + 25} = \sqrt{38}, \ v\ = \sqrt{1 + 16 + 9} = \sqrt{26}$ $\cos \theta = \frac{\langle u, v \rangle}{\ u\ \ v\ } = \frac{5}{\sqrt{38} \sqrt{26}}.$
3.	Compute the angle between two vectors (x, y) and $(-y, x)$ in an Euclidean inner product space R^2. (April/May 2023)
	Solution: $\cos \theta = \frac{\langle u, v \rangle}{\ u\ \ v\ }$ $\langle u, v \rangle = 0, \ u\ = \sqrt{x^2 + y^2}, \ v\ = \sqrt{x^2 + y^2}$ $\cos \theta = \frac{\langle u, v \rangle}{\ u\ \ v\ } = \frac{0}{x^2 + y^2} = 0 \Rightarrow \theta = \frac{\pi}{2}.$
4.	Find norm and distance between the vectors $u = (1, 0, 1)$ and $v = (-1, 1, 0)$. (Nov/Dec 2022)
	Solution: $\ u\ ^2 = 1^2 + 0^2 + 1^2 = 2, \ v\ ^2 = (-1)^2 + 1^2 + 0^2 = 2$ $\langle u, v \rangle = \langle v, u \rangle = -1 + 0 + 0 = -1$ The distance between two vectors u and v is defined by $d(u, v) = \ u - v\$ $u - v = (2, -1, 1)$ $\ u - v\ ^2 = 4 + 1 + 1 = 6 \Rightarrow \ u - v\ = \sqrt{6}$

5.	Consider $f(t) = 3t - 5$ and $g(t) = t^2$ in the polynomial space $P(t)$ with inner product $\langle f, g \rangle = \int_0^1 f(t)g(t)dt$ hence find $\ f\$ and $\ g\$.
	Solution: $f(t)f(t) = 9t^2 - 30t + 25$, $g(t)g(t) = t^4$ $\ f\ ^2 = \langle f, f \rangle = \int_0^1 (9t^2 - 30t + 25)dt = \left[3t^3 - 15t^2 + 25t \right]_0^1 = 13 \Rightarrow \ f\ = \sqrt{13}$ $\ g\ ^2 = \langle g, g \rangle = \int_0^1 t^4 dt = \left[\frac{t^5}{5} \right]_0^1 = \frac{1}{5} \Rightarrow \ g\ = \frac{1}{\sqrt{5}}$
6.	Find the value of a if the vectors $(2, a)$ and $(6, 4)$ are orthogonal vectors in R^2.
	Solution: Let $u = (2, a)$ and $v = (6, 4)$ Since the vectors are orthogonal $\langle u, v \rangle = 0$ $12 + 4a = 0$ $4a = -12 \Rightarrow a = -3$
7.	Find k so that $u = (1, 2, k, 3)$ and $v = (3, k, 7, -5)$ in R^4 are orthogonal. (Apr/May 2022)
	Solution: $\langle u, v \rangle = (1, 2, k, 3) \cdot (3, k, 7, -5) = 3 + 2k + 7k - 15 = 9k - 12$ Then set $\langle u, v \rangle = 9k - 12 = 0 \Rightarrow k = \frac{4}{3}$.
8.	If $v = (1, 2, 1)$ and $u = (2, 1, 2)$ find $\text{proj}(v, u)$.
	Solution: $\text{Proj}(v, u) = \frac{\langle v, u \rangle}{\langle u, u \rangle} u \Rightarrow \text{Proj}(v, u) = \frac{6}{9} (2, 1, 2) = \left(\frac{4}{3}, \frac{2}{3}, \frac{4}{3} \right)$
9.	Prove that in an inner product space V, for any $u, v \in V$. $\ u + v\ ^2 + \ u - v\ ^2 = 2\ u\ ^2 + 2\ v\ ^2$.
	Solution: $\ u + v\ ^2 + \ u - v\ ^2 = \langle u + v, u + v \rangle + \langle u - v, u - v \rangle$ $= \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle + \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle$ $= \ u\ ^2 + \ v\ ^2 + \ u\ ^2 + \ v\ ^2 = 2\ u\ ^2 + 2\ v\ ^2$
10.	Let $V = R^2$ and $S = \{(1, 0), (0, 1)\}$. Check whether S is orthonormal basis or not.
	Solution: Let $x = (1, 0)$ and $y = (0, 1)$ $\langle x, y \rangle = \langle (1, 0), (0, 1) \rangle = 0$ $\ x\ = \sqrt{\langle x, x \rangle} = \sqrt{1} = 1$ $\ y\ = \sqrt{\langle y, y \rangle} = \sqrt{1} = 1$ $\ x\ = \ y\ = 1 \quad \therefore S \text{ is an orthonormal basis.}$
11.	If u and v are orthonormal vectors in an inner product space V then find $\ u + v\$.

	Solution: $\ u+v\ ^2 = \langle u+v, u+v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \quad \because \langle u, v \rangle = \langle v, u \rangle = 0$ $= \ u\ ^2 + \ v\ ^2 = 1+1=2$
12.	Let R^2 have the weighted Euclidean inner product defined as $\langle u, v \rangle = 2u_1v_1 + 3u_2v_2$ and let $u = (1,1), v = (3,2), w = (0,-1)$. compute the value of $\langle u+v, 3w \rangle$
	Solution: $u+v = (4,3), 3w = (0,-3)$ $\langle u+v, 3w \rangle = 2(4)(0) + 3(3)(-3) = -27$
13.	Find the norm of the vector $u = (1,1,-1)$ and $v = (-1,1,0)$ in R^3 with respect to the inner product defined by $\langle u, v \rangle = u_1v_1 + 2u_2v_2 + 3u_3v_3$ where $u = (u_1, u_2, u_3)$ and $v = (v_1, v_2, v_3)$. (Apr/May 2022)
	Solution: $\ u\ ^2 = 1+2+3=6, \ v\ ^2 = 1+2+0=3$ $\Rightarrow \ u\ = \sqrt{6}, \ v\ = \sqrt{3}$
14.	Write down the definition of Orthogonal & Orthonormal basis for inner product space V.
	Solution: Orthogonal vector: A vector $u \in V$ is said to be orthogonal to $v \in V$ if $\langle u, v \rangle = 0$. Orthonormal Basis: A basis of an inner product space that consists of mutually orthogonal unit vectors is called an Orthonormal basis.
15.	Define Orthogonal Complement. (April/May 2023)
	Solution: Let S be a subspace of V (an inner product space). Then the set $S^\perp = \{x \in V : \langle x, y \rangle = 0 \ \forall y \in S\}$ is called orthogonal Complement of S .
16.	Let $w = (1,2,3,1)$ be a vector in R^4. Find an orthogonal basis for $W^\perp$.
	Solution: Find a nonzero solution of $x+2y+3z+t=0$, say $v_1 = (0,0,1,-3)$ Now find a nonzero solution of the system $x+2y+3z+t=0, z-3t=0$ say $v_2 = (0,-5,3,1)$ Lastly find a nonzero solution of the system $x+2y+3z+t=0, -5y+3z+t=0, z-3t=0$ say $v_3 = (-14,2,3,1)$. Thus v_1, v_2, v_3 form an orthogonal basis of $W^\perp$.
17.	State Cauchy Schwarz inequality.
	For $x, y \in V$, $ \langle x, y \rangle \leq \ x\ \ y\ $
18.	Define Hermitian conjugate matrix or conjugate transpose of a matrix.
	Solution: Let $A = (a_{ij})$ be an $m \times n$ matrix with complex entries. Hermitian conjugate matrix of A or conjugate transpose of A is the $n \times m$ matrix $A^* = (b_{ij})$ such that $b_{ij} = \overline{a_{ji}}$. i.e. $A^H = A^* = \overline{A^T}$.
19.	Find the conjugate transpose of $A = \begin{pmatrix} i & 1+2i \\ 2 & 3+4i \end{pmatrix}$.

	Solution: $\bar{A} = \begin{pmatrix} -i & 1-2i \\ 2 & 3-4i \end{pmatrix};$ Conjugate transpose of $A = A^H = \begin{pmatrix} -i & 2 \\ 1-2i & 3-4i \end{pmatrix}.$
20.	Let P_2 have the inner product $\langle p, q \rangle = \int_{-1}^1 p(x)q(x)dx$. Find the angle between p and q where $p = x$ and $q = x^2$ with respect to the inner product on P_2.
	Solution: $\ p\ = [\langle p, p \rangle]^{\frac{1}{2}} = \left[\int_{-1}^1 x^2 dx \right]^{\frac{1}{2}} = \left[\left(\frac{x^3}{3} \right)_{-1}^1 \right]^{\frac{1}{2}} = \sqrt{\frac{2}{3}}$ $\ q\ = [\langle q, q \rangle]^{\frac{1}{2}} = \left[\int_{-1}^1 x^4 dx \right]^{\frac{1}{2}} = \left[\left(\frac{x^5}{5} \right)_{-1}^1 \right]^{\frac{1}{2}} = \sqrt{\frac{2}{5}}$ $\langle p, q \rangle = \int_{-1}^1 x \cdot x^2 dx = \int_{-1}^1 x^3 dx = 0.$ $\cos \theta = \frac{\langle p, q \rangle}{\ p\ \ q\ } = \frac{0}{\ p\ \ q\ } = 0 \Rightarrow \theta = \frac{\pi}{2}$
PART-B	
1	Let V be the set of all continuous real valued functions defined on the closed interval $[0,1]$, then prove that V is a real inner product space with $\langle f, g \rangle = \int_0^1 f(t)g(t)dt$. (Nov/Dec 2022)
	Proof: Let $f, g, h \in V$ and $\alpha \in R$ (i) $\langle f + g, h \rangle = \int_0^1 [f(t) + g(t)]h(t)dt$ $= \int_0^1 f(t)h(t)dt + \int_0^1 g(t)h(t)dt = \langle f, h \rangle + \langle g, h \rangle$ (ii) $\langle \alpha f, g \rangle = \int_0^1 \alpha f(t)g(t)dt = \alpha \int_0^1 f(t)g(t)dt = \alpha \langle f, g \rangle$ (iii) $\langle f, g \rangle = \int_0^1 f(t)g(t)dt = \int_0^1 g(t)f(t)dt = \langle g, f \rangle.$ (iv) $\langle f, f \rangle = \int_0^1 [f(t)]^2 dt \geq 0$ and $\langle f, f \rangle = 0$ iff $f = 0$.
2	State and prove Schwarz inequality. (Nov/Dec 2022)
	Statement: If V be an inner product space over F, For $x, y \in V$, $\langle x, y \rangle \leq \ x\ \ y\$ Proof: Case(i). Let $x = 0$ or $y = 0$, then $\langle x, y \rangle = 0$ and $\ x\ \ y\ = 0$. Therefore $\langle x, y \rangle = \ x\ \ y\$ Case(ii). Let $x \neq 0$ & $y \neq 0$.

	Consider the vector $w = x - \frac{\langle x, y \rangle}{\ x\ ^2} y$ Let $x, y \in V$, $\frac{\langle x, y \rangle}{\ y\ ^2} \in F$ and $c = \frac{\langle x, y \rangle}{\ y\ ^2}$, then we have $w = x - \frac{\langle x, y \rangle}{\ y\ ^2} y = x - cy$ $0 \leq \ w\ ^2 = \ x - cy\ ^2$ $= \langle x - cy, x - cy \rangle$ $= \langle x, x \rangle - \langle x, cy \rangle - \langle cy, x \rangle + \langle cy, cy \rangle$ $= \ x\ ^2 - \bar{c} \langle x, y \rangle - c \langle x, y \rangle + c\bar{c} \langle y, y \rangle$ Then $0 \leq \ x\ ^2 - \frac{\langle x, y \rangle}{\ y\ ^2} \langle x, y \rangle - \frac{\langle x, y \rangle}{\ y\ ^2} \overline{\langle x, y \rangle} + \frac{ \langle x, y \rangle ^2}{\ y\ ^4} \ y\ ^2$ $0 \leq \ x\ ^2 \ y\ ^2 - \langle x, y \rangle ^2 - \langle x, y \rangle ^2 + \langle x, y \rangle ^2$ $\Rightarrow \langle x, y \rangle ^2 \leq \ x\ ^2 \ y\ ^2$ $\Rightarrow \langle x, y \rangle \leq \ x\ \ y\ $
3	Using Euclidean inner product on R^3 show that $u = (-3, 1, 0)$ and $v = (2, -1, 3)$ satisfy (i) Cauchy Schwartz inequality (ii) Triangle inequality. (April/May 2023)
	Solution: $\ x\ ^2 = \langle x, x \rangle = 9 + 1 + 0 = 10; \quad \ x\ = \sqrt{10}$ $\ y\ ^2 = \langle y, y \rangle = 4 + 1 + 9 = 14; \quad \ y\ = \sqrt{14}$ $\langle x, y \rangle = (-6) + (-1) + 0 = -7$ $ \langle x, y \rangle = 7; \quad \ x\ \ y\ = \sqrt{10} \sqrt{14} = 11.8322$ $ \langle x, y \rangle < \ x\ \ y\ $ Cauchy Schwartz inequality is satisfied $x + y = (-1, 0, 3)$ $\ x + y\ ^2 = \langle x + y, x + y \rangle = 1 + 0 + 9 = 10$ $\ x + y\ = \sqrt{10} = 3.1623; \quad \ x\ + \ y\ = \sqrt{10} + \sqrt{14} = 6.9039$ $\Rightarrow \ x + y\ \leq \ x\ + \ y\ $ Triangle inequality is satisfied
4	If V be an inner product space over F. Then for all $x, y \in V$ and $c \in F$, prove the following $\ cx\ = c \ x\$ $\ x\ = 0$ if and only if $x = 0$. $\ x + y\ \leq \ x\ + \ y\$ (April/May 2022, April/May 2023)
	Solution: $\ cx\ ^2 = \langle cx, cx \rangle = c\bar{c} \langle x, x \rangle = c ^2 \ x\ ^2 \Rightarrow \ cx\ = c \ x\$ $\ x\ = 0 \Leftrightarrow \ x\ ^2 = 0$ $\Leftrightarrow \langle x, x \rangle = 0$ $\Leftrightarrow x = 0$ $\ x + y\ ^2 = \langle x + y, x + y \rangle$

$$\begin{aligned}
&= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\
&= \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle + \|y\|^2 \\
&\leq \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2 \\
&\leq \|x\|^2 + 2|\langle x, y \rangle| + \|y\|^2 \\
&\|x + y\|^2 \leq (\|x\| + \|y\|)^2 \\
&\Rightarrow \|x + y\| \leq \|x\| + \|y\|
\end{aligned}$$

5 State and prove Gram-Schmidt Orthogonalisation Process.

Statement: Every finite dimensional inner product space has an orthonormal set as a basis.

Proof:

Let V be a finite dimensional inner product space with $\dim V = n$.

Let $B = \{u_1, u_2, \dots, u_n\}$ be a basis of V .

The proof is by the principle of mathematical induction on n .

Step 1: $v_1 = u_1 \neq 0$

Step 2: $v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1$ then $v_2 \neq 0$

Suppose $v_2 = 0 \Rightarrow u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1 = 0$

$$u_2 = \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1$$

u_2 depends on v_1 (i.e) u_1

(i.e) u_1, u_2 are linearly dependent, which is a contradiction.

Consider $\langle v_2, v_1 \rangle = \left\langle u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1, v_1 \right\rangle$

$$= \langle u_2, v_1 \rangle - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} \langle v_1, v_1 \rangle = 0$$

$\Rightarrow v_1$ and v_2 are orthogonal

$\Rightarrow \{v_1, v_2\}$ is an orthogonal set.

Step 3: Assume that the statement is true for $n = k$

Consider $\{v_1, v_2, \dots, v_{k-1}, v_k, v_{k+1}\}$ and $\{v_1, v_2, \dots, v_{k-1}, v_k\}$ is an orthogonal set where

$v_{k+1} = u_{k+1} - \frac{\langle u_{k+1}, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_{k+1}, v_2 \rangle}{\|v_2\|^2} v_2 - \dots - \frac{\langle u_{k+1}, v_k \rangle}{\|v_k\|^2} v_k$, then $v_{k+1} \neq 0$

If $v_{k+1} = 0$, then $v_{k+1} = u_{k+1} - \frac{\langle u_{k+1}, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_{k+1}, v_2 \rangle}{\|v_2\|^2} v_2 - \dots - \frac{\langle u_{k+1}, v_k \rangle}{\|v_k\|^2} v_k = 0$

$$u_{k+1} = \frac{\langle u_{k+1}, v_1 \rangle}{\|v_1\|^2} v_1 + \frac{\langle u_{k+1}, v_2 \rangle}{\|v_2\|^2} v_2 + \dots + \frac{\langle u_{k+1}, v_k \rangle}{\|v_k\|^2} v_k$$

u_{k+1} depends on $v_1, v_2, v_3, \dots, v_k$

(i.e) u_{k+1} depends on $u_1, u_2, u_3, \dots, u_k$

(i.e) $\{u_1, u_2, u_3, \dots, u_k, u_{k+1}\}$ is linearly dependent, which is a contradiction.

Therefore $v_{k+1} \neq 0$

Consider $\langle v_{k+1}, v_i \rangle = \left\langle u_{k+1} - \frac{\langle u_{k+1}, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_{k+1}, v_2 \rangle}{\|v_2\|^2} v_2 - \dots - \frac{\langle u_{k+1}, v_k \rangle}{\|v_k\|^2} v_k, v_i \right\rangle$ for $i = 1, 2, 3, \dots, k$

$$= \langle u_{k+1}, v_i \rangle - \frac{\langle u_{k+1}, v_1 \rangle}{\|v_1\|^2} \langle v_1, v_i \rangle - \frac{\langle u_{k+1}, v_2 \rangle}{\|v_2\|^2} \langle v_2, v_i \rangle - \dots - \frac{\langle u_{k+1}, v_k \rangle}{\|v_k\|^2} \langle v_k, v_i \rangle$$

	$= \langle u_{k+1}, v_i \rangle - 0 - 0 - \dots - \frac{\langle u_{k+1}, v_i \rangle}{\ v_i\ ^2} \langle v_i, v_i \rangle - \dots - 0 = 0$ $\langle u_{k+1}, v_i \rangle = 0$ $\Rightarrow v_{k+1}$ and v_i are orthogonal $\Rightarrow \{v_1, v_2, \dots, v_{k-1}, v_k, v_{k+1}\}$ is an orthogonal set Therefore, by the principle of mathematical induction $\{v_1, v_2, \dots, v_{n-1}, v_n\}$ is an orthogonal basis. $w_1 = \frac{v_1}{\ v_1\ }; w_2 = \frac{v_2}{\ v_2\ }; w_3 = \frac{v_3}{\ v_3\ }; \dots w_n = \frac{v_n}{\ v_n\ }$ $\{w_1, w_2, \dots, w_n\}$ is an orthonormal basis.
6	Let R^3 have the Euclidean inner product. Use the Gram-Schmidt process to transform the basis $\{u_1, u_2, u_3\}$ into an orthonormal basis, where $u_1 = (1, 1, 1), u_2 = (0, 1, 1), u_3 = (0, 0, 1)$. (Apr/May 2022)
	To find Orthogonal Basis (v_1, v_2, v_3) $v_1 = u_1 = (1, 1, 1)$ $v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\ v_1\ ^2} v_1 = (0, 1, 1) - \frac{2}{3}(1, 1, 1) = \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$ $v_3 = u_3 - \frac{\langle u_3, v_1 \rangle}{\ v_1\ ^2} v_1 - \frac{\langle u_3, v_2 \rangle}{\ v_2\ ^2} v_2 = (0, 0, 1) - \left\{ \frac{1}{3}(1, 1, 1) + \frac{1}{2} \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right) \right\} = \left(0, -\frac{1}{2}, \frac{1}{2}\right)$ To find Orthonormal Basis (w_1, w_2, w_3) $w_1 = \frac{v_1}{\ v_1\ } = \frac{(1, 1, 1)}{\ (1, 1, 1)\ } = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ $w_2 = \frac{v_2}{\ v_2\ } = \frac{\left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)}{\left\ \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)\right\ } = \left(-\sqrt{\frac{2}{3}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$ $w_3 = \frac{v_3}{\ v_3\ } = \frac{\left(0, -\frac{1}{2}, \frac{1}{2}\right)}{\left\ \left(0, -\frac{1}{2}, \frac{1}{2}\right)\right\ } = \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
7	Let R^3 have the Euclidean inner product. Use the Gram-Schmidt process to convert basis $B = \{u_1, u_2, u_3\}$ where $u_1 = (1, 0, 1), u_2 = (-1, 1, 0), u_3 = (-3, 2, 0)$ into an orthonormal basis. (Nov/Dec 2022) (April/May 2023)
	Solution: To find Orthogonal Basis (v_1, v_2, v_3) $v_1 = u_1 = (1, 0, 1)$ $v_2 = u_2 - \sum_{j=1}^1 \frac{\langle u_2, v_j \rangle}{\ v_j\ ^2} v_j = u_2 - \frac{\langle u_2, v_1 \rangle}{\ v_1\ ^2} v_1$ $= (-1, 1, 0) + \frac{1}{2}(1, 0, 1) = \left(-\frac{1}{2}, 1, \frac{1}{2}\right)$ $v_3 = u_3 - \sum_{j=1}^2 \frac{\langle u_3, v_j \rangle}{\ v_j\ ^2} v_j = (-3, 2, 0) - \left\{ \frac{\langle u_3, v_1 \rangle}{\ v_1\ ^2} v_1 + \frac{\langle u_3, v_2 \rangle}{\ v_2\ ^2} v_2 \right\}$

$$\begin{aligned}
 &= (-3, 2, 0) - \left\{ -\frac{3}{2}(1, 0, 1) + \frac{7/2}{3/2} \left(-\frac{1}{2}, 1, \frac{1}{2} \right) \right\} \\
 &= (-3, 2, 0) + \left(\frac{3}{2}, 0, \frac{3}{2} \right) - \frac{7}{3} \left(-\frac{1}{2}, 1, \frac{1}{2} \right) \\
 &= (-3, 2, 0) + \left(\frac{3}{2}, 0, \frac{3}{2} \right) + \left(\frac{7}{6}, \frac{-7}{3}, \frac{-7}{6} \right) = \left(\frac{-1}{3}, \frac{-1}{3}, \frac{1}{3} \right)
 \end{aligned}$$

To find Orthonormal Basis (w_1, w_2, w_3)

$$w_1 = \frac{v_1}{\|v_1\|} = \frac{(1, 0, 1)}{\|(1, 0, 1)\|} = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right)$$

$$w_2 = \frac{v_2}{\|v_2\|} = \frac{\left(-\frac{1}{2}, 1, \frac{1}{2} \right)}{\left\| \left(-\frac{1}{2}, 1, \frac{1}{2} \right) \right\|} = \frac{\left(-\frac{1}{2}, 1, \frac{1}{2} \right)}{\sqrt{\frac{1}{4} + 1 + \frac{1}{4}}} = \frac{\left(-\frac{1}{2}, 1, \frac{1}{2} \right)}{\sqrt{\frac{3}{2}}} = \left(\frac{-\sqrt{2}}{2\sqrt{3}}, \frac{\sqrt{3}}{\sqrt{2}}, \frac{\sqrt{2}}{2\sqrt{3}} \right)$$

$$w_3 = \frac{v_3}{\|v_3\|} = \frac{\left(\frac{-1}{3}, \frac{-1}{3}, \frac{1}{3} \right)}{\left\| \left(\frac{-1}{3}, \frac{-1}{3}, \frac{1}{3} \right) \right\|} = \frac{\left(\frac{-1}{3}, \frac{-1}{3}, \frac{1}{3} \right)}{\sqrt{\frac{1}{9} + \frac{1}{9} + \frac{1}{9}}} = \frac{\left(\frac{-1}{3}, \frac{-1}{3}, \frac{1}{3} \right)}{\sqrt{\frac{1}{3}}} = \left(\frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$$

8

Let the vector space P_2 have the inner product $\langle p, q \rangle = \int_{-1}^1 p(x)q(x)dx$. Apply the Gram-schmidt process to transform the basis $S = \{u_1, u_2, u_3\} = \{1, x, x^2\}$ into an orthonormal basis.

Solution:

To find Orthogonal Basis (v_1, v_2, v_3)

Let $u_1 = 1, u_2 = x, u_3 = x^2$;

$$v_1 = u_1 = 1$$

$$\langle u_2, v_1 \rangle = \int_{-1}^1 x dx = \left(\frac{x^2}{2} \right)_{-1}^1 = 0; \quad \|v_1\|^2 = \langle v_1, v_1 \rangle = \int_{-1}^1 dx = 2$$

$$v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1 = x - 0 = x$$

$$\langle u_3, v_1 \rangle = \int_{-1}^1 x^2 dx = \left(\frac{x^3}{3} \right)_{-1}^1 = \frac{2}{3}; \quad \langle u_3, v_2 \rangle = \int_{-1}^1 x^3 dx = 0; \quad \|v_2\|^2 = \langle v_2, v_2 \rangle = \int_{-1}^1 t^2 dt = \left[\frac{t^3}{3} \right]_{-1}^1 = \frac{2}{3}$$

$$v_3 = u_3 - \frac{\langle u_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_3, v_2 \rangle}{\|v_2\|^2} v_2 = x^2 - \frac{2/3}{2} - 0 = x^2 - \frac{1}{3}$$

Therefore $\left\{ 1, x, x^2 - \frac{1}{3} \right\}$ is an orthonormal basis for $P_2(R)$

To find Orthonormal Basis (w_1, w_2, w_3)

$$w_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\|1\|} = \frac{1}{\sqrt{2}}$$

$$\|v_2\|^2 = \langle v_2, v_2 \rangle = \int_{-1}^1 t^2 dt = \left[\frac{t^3}{3} \right]_{-1}^1 = \frac{2}{3}$$

$$w_2 = \frac{v_2}{\|v_2\|} = \frac{x}{\|x\|} = \frac{x}{\sqrt{\frac{3}{2}}} = \sqrt{\frac{2}{3}}x$$

$$\|v_3\|^2 = \langle v_3, v_3 \rangle = \int_{-1}^1 (t^2 - \frac{1}{3})(t^2 - \frac{1}{3})dt$$

$$= \int_{-1}^1 (t^4 - \frac{2}{3}t^2 + \frac{1}{9})dt = \left[\frac{t^5}{5} - \frac{2}{3} \frac{t^3}{3} + \frac{1}{9}t \right]_{-1}^1 = \frac{2}{5} - \frac{2}{3} \left(\frac{2}{3} \right) + \frac{2}{9} = \frac{8}{45}$$

$$w_3 = \frac{v_3}{\|v_3\|} = \frac{x^2 - \frac{1}{3}}{\left\| x^2 - \frac{1}{3} \right\|} = \frac{x^2 - \frac{1}{3}}{\sqrt{\frac{8}{45}}} = \frac{\sqrt{45}}{\sqrt{8}} x^2 - \frac{\sqrt{5}}{\sqrt{8}}$$

9 Let C^3 have the Euclidean inner product. Use the Gram-Schmidt process to convert basis $B = \{u_1, u_2, u_3\}$ where $u_1 = (0, 1, -1), u_2 = (1+i, 1, 1), u_3 = (1-i, 1, 1)$ into an orthonormal basis.

Solution:

To find Orthogonal Basis (v_1, v_2, v_3)

$$v_1 = u_1 = (0, 1, -1)$$

$$\langle u_2, v_1 \rangle = (1+i)(0) + (1)(1) + (1)(-1) = 0; \quad \|v_1\|^2 = 0 + 1 + 1 = 2$$

$$v_2 = u_2 - \sum_{j=1}^1 \frac{\langle u_2, v_j \rangle}{\|v_j\|^2} v_j = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1$$

$$= (1+i, 1, 1) + \frac{0}{2}(0, 1, -1) = (1+i, 1, 1)$$

$$\langle u_3, v_1 \rangle = (1-i)(0) + (1)(1) + (1)(-1) = 0; \quad \|v_1\|^2 = 0 + 1 + 1 = 2$$

$$\langle u_3, v_2 \rangle = (1-i)(1+i) + (1)(1) + (1)(1) = 2 - 2i; \quad \|v_2\|^2 = (1+i)(1+i) + (1)(1) + (1)(1) = 4$$

$$v_3 = u_3 - \sum_{j=1}^2 \frac{\langle u_3, v_j \rangle}{\|v_j\|^2} v_j = (1-i, 1, 1) - \left\{ \frac{\langle u_3, v_1 \rangle}{\|v_1\|^2} v_1 + \frac{\langle u_3, v_2 \rangle}{\|v_2\|^2} v_2 \right\}$$

$$= (1-i, 1, 1) - \left\{ 0 + \frac{2(1-i)}{4}(1+i, 1, 1) \right\}$$

$$= (1-i, 1, 1) - \frac{(1-i)}{2}(1+i, 1, 1)$$

$$= (1-i, 1, 1) - \left(\frac{1+i}{2}, \frac{(1-i)}{2}, \frac{(1-i)}{2} \right)$$

$$= \left(-i, \frac{1+i}{2}, \frac{1+i}{2} \right)$$

To find Orthonormal Basis (w_1, w_2, w_3)

$$\|v_1\|^2 = 0 + 1 + 1 = 2$$

$$w_1 = \frac{v_1}{\|v_1\|} = \frac{(0, 1, -1)}{\|(0, 1, -1)\|} = \left(0, \frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}} \right)$$

$$\|v_2\|^2 = (1+i)(1+i) + (1)(1) + (1)(1) = 4$$

$$w_2 = \frac{v_2}{\|v_2\|} = \frac{(1+i, 1, 1)}{\|(1+i, 1, 1)\|} = \frac{(1+i, 1, 1)}{\sqrt{4}} = \left(\frac{1+i}{2}, \frac{1}{2}, \frac{1}{2} \right)$$

$$\|v_2\|^2 = (i)(-i) + \left(\frac{1+i}{2}\right)\left(\frac{1+i}{2}\right) + \left(\frac{1+i}{2}\right)\left(\frac{1+i}{2}\right) = 1 + \frac{1}{4}(1+1) + \frac{1}{4}(1+1) = 2$$

$$w_3 = \frac{v_3}{\|v_3\|} = \frac{\left(-i, \frac{1+i}{2}, \frac{1+i}{2}\right)}{\left\|\left(-i, \frac{1+i}{2}, \frac{1+i}{2}\right)\right\|} = \frac{\left(-i, \frac{1+i}{2}, \frac{1+i}{2}\right)}{\sqrt{2}} = \left(\frac{-i}{\sqrt{2}}, \frac{1+i}{2\sqrt{2}}, \frac{1+i}{2\sqrt{2}}\right)$$

10 Find the least square line fitted to the data (1,1), (2,2), (3,2), (4,3). Also, find the least square error. (Nov/Dec 2022)

Solution:

Let $y = ax + b$ (a linear function) then for the given data

$$\begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 2 \\ 3 \end{pmatrix} \Rightarrow AX = Y \text{ with } A = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix}; X = \begin{pmatrix} a \\ b \end{pmatrix}; Y = \begin{pmatrix} 1 \\ 2 \\ 2 \\ 3 \end{pmatrix}$$

$$A^* = [\bar{A}]^T = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$A^*A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 30 & 10 \\ 10 & 4 \end{bmatrix}$$

$$(A^*A)X = A^*Y \Rightarrow \begin{bmatrix} 30 & 10 \\ 10 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 23 \\ 8 \end{bmatrix}$$

$$\text{Solving, } a = \frac{3}{5} \text{ and } b = \frac{1}{2}$$

$$\therefore y = ax + b = \frac{3}{5}x + \frac{1}{2} \text{ is the least square line.}$$

$$\text{The Error } E = \|AX - Y\|^2 = \left\| \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} \frac{3}{5} \\ \frac{1}{2} \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 2 \\ 3 \end{pmatrix} \right\|^2 = \left\| \begin{pmatrix} 0.1 \\ -0.3 \\ 0.3 \\ -0.1 \end{pmatrix} \right\|^2 = 0.01 + 0.09 + 0.09 + 0.01 = 0.2$$

11 Find the least square line fitted to the data (1, 2), (2,3), (3,5), (4,7). Also, find the least square error. (April/May 2023)

Solution:

Let $y = ax + b$ (a linear function) then for the given data

$$\begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 5 \\ 7 \end{pmatrix} \Rightarrow AX = Y \text{ with } A = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix}; X = \begin{pmatrix} a \\ b \end{pmatrix}; Y = \begin{pmatrix} 2 \\ 3 \\ 5 \\ 7 \end{pmatrix}$$

$$A^* = [\bar{A}]^T = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$A^*A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 30 & 10 \\ 10 & 4 \end{bmatrix}$$

	$(A^*A)X = A^*Y \Rightarrow \begin{bmatrix} 30 & 10 \\ 10 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 51 \\ 17 \end{bmatrix}$ Solving, $a = \frac{17}{40}$ and $b = 0$ $\therefore y = ax + b = \frac{17}{10}x$ is the least square line. The Error $E = \ AX - Y\ ^2 = \left\ \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} \frac{17}{10} \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ 5 \\ 7 \end{pmatrix} \right\ ^2 = \left\ \begin{pmatrix} -0.3 \\ 0.4 \\ 0.1 \\ 0.2 \end{pmatrix} \right\ ^2 = 0.09 + 0.16 + 0.01 + 0.04 = 0.3$
12	Use the least square approximation to find the best fit with a linear function and hence compute error for the following data $(-3,9), (-2,6), (0,2)$ and $(1,1)$. (Apr/May 2022) Solution: Let $y = ax + b$ (a linear function) then for the given data $\begin{pmatrix} -3 & 1 \\ -2 & 1 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 9 \\ 6 \\ 2 \\ 1 \end{pmatrix} \Rightarrow AX = Y \text{ with } A = \begin{pmatrix} -3 & 1 \\ -2 & 1 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}; X = \begin{pmatrix} a \\ b \end{pmatrix}; Y = \begin{pmatrix} 9 \\ 6 \\ 2 \\ 1 \end{pmatrix}$ $A^* = [\bar{A}]^T = \begin{bmatrix} -3 & -2 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$ $A^*A = \begin{bmatrix} -3 & -2 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -3 & 1 \\ -2 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix}$ $(A^*A)X = A^*Y \Rightarrow \begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -38 \\ 18 \end{bmatrix}$ Solving, $a = -2$ and $b = \frac{5}{2}$ $\therefore y = ax + b = -2x + \frac{5}{2}$ is the least square line. The Error $E = \ AX - Y\ ^2 = \left\ \begin{pmatrix} -3 & 1 \\ -2 & 1 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} -2 \\ \frac{5}{2} \end{pmatrix} - \begin{pmatrix} 9 \\ 6 \\ 2 \\ 1 \end{pmatrix} \right\ ^2 = \left\ \begin{pmatrix} 0.5 \\ -0.5 \\ -0.5 \\ 0.5 \end{pmatrix} \right\ ^2 = 0.25 + 0.25 + 0.25 + 0.25 = 1$
13	Find the least square solution and least square error of the equation (Apr/May 2022) $\begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ Solution: $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$$A^*A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 5 \\ 5 & 6 \end{bmatrix}$$

$$A^*B = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

$$(A^*A)X = A^*B \Rightarrow \begin{bmatrix} 6 & 5 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

$$\text{Solving, } x_1 = \frac{4}{11} \text{ and } x_2 = \frac{4}{11}$$

$$E = \|AX_0 - B\| = \left\| \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{4}{11} \\ \frac{4}{11} \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\| = \left\| \begin{bmatrix} \frac{1}{11} \\ \frac{1}{11} \\ -\frac{3}{11} \end{bmatrix} \right\| = \frac{1}{\sqrt{11}} = 0.3$$

UNIT V EIGEN VALUE PROBLEMS AND MATRIX DECOMPOSITION

Eigen value Problems: Power method, Jacobi rotation method - Singular value decomposition - QR decomposition.

PART-A

1. Explain Power Method.

Solution:

The Power method is an iterative technique used to determine the dominant eigenvalue of a matrix that is, the eigenvalue with the largest magnitude.

By modifying the method slightly, we can also determine other eigenvalues.

2. Find the dominant eigen value $A = \begin{bmatrix} 2 & 3 \\ 5 & 4 \end{bmatrix}$ using power method up to one decimal place accuracy. Start with $X^{(0)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ (Nov/Dec 2022)

Solution:

$$AX^{(0)} = \begin{bmatrix} 2 & 3 \\ 5 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \end{bmatrix} = 9 \begin{bmatrix} 0.6 \\ 1 \end{bmatrix} = 9X^{(1)}$$

$$AX^{(1)} = \begin{bmatrix} 2 & 3 \\ 5 & 4 \end{bmatrix} \begin{bmatrix} 0.6 \\ 1 \end{bmatrix} = \begin{bmatrix} 4.2 \\ 7 \end{bmatrix} = 7 \begin{bmatrix} 0.6 \\ 1 \end{bmatrix} = 7X^{(2)}$$

The dominant Eigen value is 7

3. If the numerically dominant eigen value of $A = \begin{pmatrix} 5 & -1 & 7 \\ -1 & -1 & 1 \\ 7 & 1 & 5 \end{pmatrix}$ is 12, how can we find the one of other eigen value of A using power method?

Solution:

Consider $B = A - 12I$

Using power method on B, we can find a dominant eigen value of B.

Let it be say β , Then by properties of eigen vales of A, $\lambda - 12 = \beta \Rightarrow \lambda = \beta + 12$

4. Find all the eigen value of $A = \begin{pmatrix} 2 & 3 \\ 4 & 1 \end{pmatrix}$ by power method. Use $X^0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ (Apr/May 2022)

Solution:

$$AX^0 = \begin{pmatrix} 2 & 3 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \lambda X^1$$

	The dominant eigen value is 5. We know that, Sum of all eigen values = trace of A $5 + \lambda = 2 + 1 \Rightarrow \lambda = -2$ Therefore eigen values of A are -2,5
5	Explain Jacobi Method. Solution: ➤ The Jacobi method is suitable for finding the eigenvalues of a real symmetric matrices. ➤ A real symmetric matrix is systematically reduced to a diagonal matrix by Jacobi method. This method use the similarity transformed matrix which is simpler but has the same eigenvalues as the given matrix ➤ The transformation matrices, which are used, are orthogonal matrices. ➤ The advantage of using orthogonal matrices is that it minimizes errors in the process. ➤ Jacobi method can be used to find all eigenvalues simultaneously of any real symmetric matrix A. We know from matrix theory that; the eigenvalues of a real symmetric matrix A are real. ➤ This method reduces the given matrix to a diagonal form, where the diagonal elements are the eigenvalues of the given matrix. In this method, the given matrix A is transformed to a new matrix A_1 by the scheme $A_1 = P^{-1}_1 A P_1$ where P_1 is an orthogonal matrix. Therefore, $P^{-1}_1 = P^T_1$ This transformation introduces a zero at a non-diagonal position of A. Then another matrix A_2 is produced by the equation $A_2 = P^{-1}_2 A_1 P_2 = P^{-1}_2 P^{-1}_1 A P_1 P_2$ in which a new non-diagonal element is reduced to zero. Continuing this process of reducing the non-diagonal elements to zero one by one, we finally obtain a diagonal matrix. The eigenvalues are the diagonal elements.
6	State the fundamental principle for Jacobi method. Statement: If A is a real symmetric matrix and R is an orthogonal matrix then the transformation given by $B = R' A R$ transforms A into B and this transformation preserves the symmetry and the eigen values of B.
7	For $A = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$, find R such that $B = R' A R$ Solution: $\theta = \begin{cases} \frac{\pi}{4} & \text{if } a_{11} = a_{22} \text{ \& } a_{12} > 0 \\ -\frac{\pi}{4} & \text{if } a_{11} = a_{22} \text{ \& } a_{12} < 0 \end{cases}$ $a_{12} = 3 > 0 \quad \therefore \theta = \frac{\pi}{4}$ $R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos \frac{\pi}{4} & -\sin \frac{\pi}{4} \\ \sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$
8	For $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$, find R such that $B = R' A R$ $\theta = \begin{cases} \frac{\pi}{4} & \text{if } a_{11} = a_{22} \text{ \& } a_{12} > 0 \\ -\frac{\pi}{4} & \text{if } a_{11} = a_{22} \text{ \& } a_{12} < 0 \end{cases}$ $a_{12} = -1 < 0 \quad \therefore \theta = -\frac{\pi}{4}$

	$R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos\left(-\frac{\pi}{4}\right) & -\sin\left(-\frac{\pi}{4}\right) \\ \sin\left(-\frac{\pi}{4}\right) & \cos\left(-\frac{\pi}{4}\right) \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$
9	Write the procedure of Jacobi method for a 2 x 2 matrix. Solution: Let $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ 1. Assume the rotation matrix $R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ and find $\cot 2\theta = \alpha = \frac{a_{11} - a_{22}}{2a_{12}}$ 2. Find $\cot \theta = \beta = \alpha \pm \sqrt{\alpha^2 + 1}$ 3. Find $\sin \theta = \frac{1}{\sqrt{1 + \beta^2}}$, $\cos \theta = \frac{\beta}{\sqrt{1 + \beta^2}}$ if $\beta > 0$ 4. Calculate the Eigen values of A as $b_{11} = a_{11} \cos^2 \theta + a_{12} \sin 2\theta + a_{22} \sin^2 \theta$ $b_{22} = a_{11} \sin^2 \theta - a_{12} \sin 2\theta + a_{22} \cos^2 \theta$ $b_{22} = a_{11} + a_{22} - b_{11}$ The eigen values are b_{11}, b_{22} and the columns of R are the corresponding Eigen vectors.
10	Using Jacobi method, find the eigen vectors of $A = \begin{bmatrix} 6 & \sqrt{3} \\ \sqrt{3} & 4 \end{bmatrix}$ (April/May 2023) Solution: $R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}; \quad B = R'AR$ $\cot 2\theta = \alpha = \frac{a_{11} - a_{22}}{2a_{12}} = \frac{1}{\sqrt{3}}$ $\cot \theta = \beta = \alpha \pm \sqrt{1 + \alpha^2} = \sqrt{3} \text{ or } -\frac{1}{\sqrt{3}}$ Take $\beta = \sqrt{3} \Rightarrow \theta = \frac{\pi}{6}$ $\therefore b_{11} = 7; b_{12} = 3$ $\Rightarrow X_1 = \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix}; \quad X_2 = \begin{pmatrix} -1 \\ \sqrt{3} \end{pmatrix}$
11	Using Jacobi method, find the eigen values of $A = \begin{pmatrix} 1 & 3 \\ 3 & 4 \end{pmatrix}$ Solution: $\cot 2\theta = \alpha = \frac{a_{11} - a_{22}}{2a_{12}} = \frac{1-4}{6} = \frac{-1}{2}$ $\cot \theta = \beta = \alpha \pm \sqrt{1 + \alpha^2} = -1/2 \pm \sqrt{1 + 1/4} = \frac{-1 \pm \sqrt{5}}{2}$ take $\beta = \frac{\sqrt{5}-1}{2} > 0$ $\sin \theta = \frac{1}{\sqrt{1 + \beta^2}} = \frac{2}{\sqrt{10 + 2\sqrt{5}}}$ and $\cos \theta = \frac{\beta}{\sqrt{1 + \beta^2}} = \frac{\sqrt{5}-1}{\sqrt{10 + 2\sqrt{5}}}$

	$b_{11} = a_{11} \cos^2 \theta + a_{12} \sin 2\theta + a_{22} \sin^2 \theta = \frac{5+3\sqrt{5}}{2}$ $b_{22} = a_{11} + a_{22} - b_{11} = \frac{5-3\sqrt{5}}{2}$ Hence eigen values are $\frac{5+3\sqrt{5}}{2}$ and $\frac{5-3\sqrt{5}}{2}$
12	What is the minimum number of rotations required to bring into diagonal form?
	Solution: The minimum number of rotations required to bring A into diagonal form is $\frac{n(n-1)}{2}$, where n is the order of the matrix
13	Write short notes on QR decomposition of a Matrix.
	Solution: Every mxn matrix A can be factored in to the product of matrix Q, having orthonormal vectors for its columns and an upper(right) triangular matrix R. The product $A = QR$ is the QR decomposition of A. The QR decomposition follows immediately from the modified Gram-Schmidt process applied to the columns of A, provided those columns are linearly independent.
14	Write down the modified Gram-Schmidt iterative algorithm for QR decomposition of a matrix.
	Solution: The modified Gram-Schmidt iterative algorithm, with the k^{th} iteration given by the following steps: (i) Find $r_{kk} = \ X_k\ _2$ and $Q_k = \frac{X_k}{r_{kk}}$. ($r_{kk} \neq 0$) (ii) Find $r_{kj} = X_j^T Q_k$, $j = k+1, k+2, \dots, n$, (iii) Replace X_j by $X_j - r_{kj} Q_k$, $j = k+1, k+2, \dots, n$. If $r_{kk} = 0$, then choose Q_k to be any normalized vector which is orthogonal to $Q_1, Q_2, \dots, Q_{k-1}$.
15	State QR algorithm. (Apr/May 2022)
	Solution: The QR algorithm is a procedure for determining all eigen values of a real matrix A. This algorithm sequentially constructs matrices A_k by forming QR decompositions $A_{k-1} = Q_{k-1} R_{k-1} \text{ for } A_{k-1} \text{ and } A_k = R_{k-1} Q_{k-1} \text{ for } A_k$
16	State the properties of Singular value decomposition of a matrix. (April/May 2023)
	Solution: (i) The diagonal values of D are the square roots of the Eigen values of $A^T A$ and AA^T. (ii) The rank of the matrix A is equal to the number of positive singular values. (iii) The Euclidean norm of A is equal to the largest singular value.
17	Define singular value decomposition. (Nov/Dec 2022)
	Solution: For any real matrix A there exists a decomposition $A = U \Sigma V^T$ is called singular value decomposition of A, where U, V are orthogonal matrices and the block $\Sigma = \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}$. Here D is a diagonal matrix having as its main diagonal all the positive singular values of A (i.e positive square roots of nonnegative eigen values of $A^T A$).
18	Define normal matrix
	A normal matrix is a complex matrix that multiplied by its <u>conjugate transpose matrix</u> is equal to the product of the conjugate transpose by itself. That is, all normal matrices meet the following condition: $AA^H = A^H A$

	where A^H is the conjugate transpose matrix of A. However, in the real numbers field the previous condition is the same as saying that a matrix commutes with its transpose, that is: $AA^T = A^T A$ Since the conjugate transpose matrix of a real matrix is simply the transpose of the matrix.
19	Find Σ in the singular value decomposition of $A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 1 & 3 \end{pmatrix}$.
	Solution: Let $A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 1 & 3 \end{pmatrix}$ $A^T A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 3 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 3 \\ 1 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 2 & 6 \\ 2 & 2 & 6 \\ 6 & 6 & 18 \end{pmatrix}$ The characteristic equation of $A^T A$ is $\lambda^3 - 22\lambda^2 = 0 \Rightarrow \lambda = 22, 0, 0$ $D = [\sqrt{22}] \quad \Sigma = \begin{pmatrix} \sqrt{22} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
20	List some applications of eigen value problems
	1. The eigenvalues of a linear mapping is a measure of the distortion induced by the transformation and the eigenvectors are how the distortion is oriented. It is precisely this rough picture which makes PCA (Principal Component Analysis = A statistical procedure) very useful. 2. PageRank is an eigenvalue/ eigenvector problem. Your search engine goes to millions of websites and an eigenvalue problem is formulated in which the system matrix, P is called Markov Transition Matrix. The eigenvalue problem is given as $[P]\{W\} = 1 \{W\}$. 3. An engineering problem can be reduced to a form $AX = Y$ and then to an eigenvalue problem $[A]\{V\} = \lambda \{V\}$, whose solution i.e. $\det(A - \lambda I) = 0$ gives you eigenvalues (λ) and eigenvectors $\{V\}$. In mathematical sense, the system matrix $[A]$ can be thought of as a transformation matrix whose effect varies with directions and is maximum in certain direction. These maxima are called eigenvalues and the corresponding directions are known as eigenvectors.
PART-B	
1	Find the largest eigen value and eigen vector of $A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix}$, by using Power method. (Nov/Dec 2022)
	Solution: Let the initial eigenvector be $X^{(0)} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ $AX^{(0)} = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \\ 5 \end{pmatrix} = 7 \begin{pmatrix} 0.71 \\ 1 \\ 0.71 \end{pmatrix} = 7X^{(1)}$ $AX^{(1)} = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0.71 \\ 1 \\ 0.71 \end{pmatrix} = \begin{pmatrix} 3.84 \\ 6.42 \\ 3.84 \end{pmatrix} = 6.42 \begin{pmatrix} 0.6 \\ 1 \\ 0.6 \end{pmatrix} = 7X^{(2)}$

$$AX^{(2)} = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0.6 \\ 1 \\ 0.6 \end{pmatrix} = \begin{pmatrix} 3.4 \\ 6.4 \\ 3.4 \end{pmatrix} = 6.2 \begin{pmatrix} 0.55 \\ 1 \\ 0.55 \end{pmatrix} = 6.2X^{(3)}$$

$$AX^{(3)} = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0.55 \\ 1 \\ 0.55 \end{pmatrix} = \begin{pmatrix} 3.2 \\ 6.1 \\ 3.2 \end{pmatrix} = 6.1 \begin{pmatrix} 0.56 \\ 1 \\ 0.56 \end{pmatrix} = 6.1X^{(4)}$$

$$AX^{(4)} = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0.56 \\ 1 \\ 0.56 \end{pmatrix} = \begin{pmatrix} 3.12 \\ 6.06 \\ 3.12 \end{pmatrix} = 6.06 \begin{pmatrix} 0.51 \\ 1 \\ 0.51 \end{pmatrix} = 6.06X^{(5)}$$

$$AX^{(5)} = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0.51 \\ 1 \\ 0.51 \end{pmatrix} = \begin{pmatrix} 3.04 \\ 6.02 \\ 3.04 \end{pmatrix} = 6.02 \begin{pmatrix} 0.5 \\ 1 \\ 0.5 \end{pmatrix} = 6.02X^{(6)}$$

$$AX^{(6)} = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0.5 \\ 1 \\ 0.5 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ 3 \end{pmatrix} = 6 \begin{pmatrix} 0.5 \\ 1 \\ 0.5 \end{pmatrix} = 6X^{(7)}$$

$$AX^{(7)} = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0.5 \\ 1 \\ 0.5 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ 3 \end{pmatrix} = 6 \begin{pmatrix} 0.5 \\ 1 \\ 0.5 \end{pmatrix} = 6X^{(8)}$$

Sixth and seventh iterations are same.

Therefore The largest eigenvalue is 6 and the corresponding eigen vector is $\begin{pmatrix} 0.5 \\ 1 \\ 0.5 \end{pmatrix}$.

2 Using Power method ,Find the largest eigen value and corresponding eigen vector of

$$A = \begin{pmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{pmatrix} \text{ with initial vector } \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

(Apr/May 2022)

Solution:

Let the initial eigenvector be $X^{(0)} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

$$AX^{(0)} = \begin{pmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 9 \\ 13 \end{pmatrix} = 13 \begin{pmatrix} 0.23 \\ 0.69 \\ 1 \end{pmatrix} = 13X^{(1)}$$

$$AX^{(1)} = \begin{pmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{pmatrix} \begin{pmatrix} 0.23 \\ 0.69 \\ 1 \end{pmatrix} = \begin{pmatrix} 1.307 \\ 6.077 \\ 12.537 \end{pmatrix} = 12.537 \begin{pmatrix} 0.104 \\ 0.485 \\ 1 \end{pmatrix} = 12.537X^{(2)}$$

$$AX^{(2)} = \begin{pmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{pmatrix} \begin{pmatrix} 0.104 \\ 0.485 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.559 \\ 5.282 \\ 11.836 \end{pmatrix} = 11.836 \begin{pmatrix} 0.047 \\ 0.446 \\ 1 \end{pmatrix} = 11.836X^{(3)}$$

$$AX^{(3)} = \begin{pmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{pmatrix} \begin{pmatrix} 0.047 \\ 0.446 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.385 \\ 5.033 \\ 11.737 \end{pmatrix} = 11.737 \begin{pmatrix} 0.032 \\ 0.429 \\ 1 \end{pmatrix} = 11.737X^{(4)}$$

$$AX^{(4)} = \begin{pmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{pmatrix} \begin{pmatrix} 0.032 \\ 0.429 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.319 \\ 4.954 \\ 11.684 \end{pmatrix} = 11.684 \begin{pmatrix} 0.027 \\ 0.423 \\ 1 \end{pmatrix} = 11.684X^{(5)}$$

$$AX^{(5)} = \begin{pmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{pmatrix} \begin{pmatrix} 0.027 \\ 0.423 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.296 \\ 4.927 \\ 11.665 \end{pmatrix} = 11.665 \begin{pmatrix} 0.025 \\ 0.422 \\ 1 \end{pmatrix} = 11.665X^{(6)}$$

$$AX^{(6)} = \begin{pmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{pmatrix} \begin{pmatrix} 0.025 \\ 0.422 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.291 \\ 4.919 \\ 11.663 \end{pmatrix} = 11.663 \begin{pmatrix} 0.025 \\ 0.422 \\ 1 \end{pmatrix} = 11.663X^{(7)}$$

$$AX^{(7)} = \begin{pmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{pmatrix} \begin{pmatrix} 0.025 \\ 0.422 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.291 \\ 4.919 \\ 11.663 \end{pmatrix} = 11.663 \begin{pmatrix} 0.025 \\ 0.422 \\ 1 \end{pmatrix} = 11.663X^{(8)}$$

Therefore the largest eigenvalue is 11.663 and the corresponding Eigen vector is $\begin{pmatrix} 0.025 \\ 0.422 \\ 1 \end{pmatrix}$

3

Find all the Eigen value of $A = \begin{bmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{bmatrix}$ using power method

Solution:

Let $X_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ be an initial eigen vector

$$AX_0 = \begin{bmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 0 \\ 0.2 \end{bmatrix} = 5X_1$$

$$AX_1 = \begin{bmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 5.2 \\ 0 \\ 2 \end{bmatrix} = 5.2 \begin{bmatrix} 1 \\ 0 \\ 0.3846 \end{bmatrix} = 5.2X_2$$

$$AX_2 = 5.3846 \begin{pmatrix} 1 \\ 0 \\ 0.5429 \end{pmatrix} = 5.3846$$

continue this process,

then we get the eigen value $\lambda=6$, corresponding eigen vector = $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

$$\text{Let } B = A - \lambda I = A - 6I = \begin{bmatrix} -1 & 0 & 1 \\ 0 & -8 & 0 \\ 1 & 0 & -1 \end{bmatrix}, \text{ take } Y_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$BY_0 = \begin{bmatrix} -1 & 0 & 1 \\ 0 & -8 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = -1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = -1Y_1$$

$$BY_1 = \begin{bmatrix} -1 & 0 & 1 \\ 0 & -8 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix} = -2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = -2Y_2$$

$$BY_3 = \begin{bmatrix} -1 & 0 & 1 \\ 0 & -8 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix} = -2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = -2Y_3$$

Greatest eigen value of $B = -2$

$\therefore$ The smallest eigen value of $A = -2 + 5 = 4$

By the property of eigen value,

Sum of the eigen value = Trace of A

$$\Rightarrow \lambda_1 + \lambda_2 + \lambda_3 = 5 - 2 + 5$$

$$\Rightarrow 6 + 4 + \lambda_3 = 8 \Rightarrow \lambda_3 = -2$$

$\therefore$ The Eigen values are 6, 4, -2.

4

Find the numerically largest Eigen value of $A = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix}$ and the corresponding Eigen vector.

Solution:

Let the initial eigenvector be $X^{(0)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$$AX^{(0)} = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 25 \\ 1 \\ 2 \end{pmatrix} = 25 \begin{pmatrix} 1 \\ 0.04 \\ 0.08 \end{pmatrix} = 25X^{(1)}.$$

$$AX^{(1)} = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix} \begin{pmatrix} 1 \\ 0.04 \\ 0.08 \end{pmatrix} = \begin{pmatrix} 25.2 \\ 1.12 \\ 1.68 \end{pmatrix} = 25.2 \begin{pmatrix} 1 \\ 0.0444 \\ 0.0667 \end{pmatrix} = 25.2X^{(2)}.$$

$$AX^{(2)} = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix} \begin{pmatrix} 1 \\ 0.044 \\ 0.0667 \end{pmatrix} = \begin{pmatrix} 25.18 \\ 1.131 \\ 1.73 \end{pmatrix} = 25.18 \begin{pmatrix} 1 \\ 0.045 \\ 0.069 \end{pmatrix} = 25.18X^{(3)}$$

Therefore, the largest eigenvalue is 25.18 and the corresponding eigen vector is $\begin{pmatrix} 1 \\ 0.045 \\ 0.069 \end{pmatrix}$.

5

Find the largest Eigen value and Eigen vector of $\begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, by using Power method.

Solution :

Let $X_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ be an approximate eigen value

$$AX_1 = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = 1X_2$$

$$AX_2 = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 7 \\ 3 \\ 0 \end{pmatrix} = 7 \begin{pmatrix} 1 \\ 0.428 \\ 0 \end{pmatrix} = 7X_2$$

$$AX_3 = \begin{pmatrix} 3.5714 \\ 1.8572 \\ 0 \end{pmatrix} = 3.5714 \begin{pmatrix} 1 \\ 0.52 \\ 0 \end{pmatrix} = 3.5714X_4$$

$$AX_4 = \begin{pmatrix} 4.12 \\ 2.04 \\ 0 \end{pmatrix} = 4.12 \begin{pmatrix} 1 \\ 0.4951 \\ 0 \end{pmatrix} = 4.12X_5$$

$$AX_5 = \begin{pmatrix} 3.9706 \\ 1.9902 \\ 0 \end{pmatrix} = 3.9706 \begin{pmatrix} 1 \\ 0.5012 \\ 0 \end{pmatrix} = 3.9706X_6$$

$$AX_6 = \begin{pmatrix} 4.0072 \\ 2.0024 \\ 0 \end{pmatrix} = 4.0072 \begin{pmatrix} 1 \\ 0.4997 \\ 0 \end{pmatrix} = 4.0072X_7$$

$$AX_7 = \begin{pmatrix} 3.9982 \\ 1.9994 \\ 0 \end{pmatrix} = 3.9982 \begin{pmatrix} 1 \\ 0.500 \\ 0 \end{pmatrix} = 3.9982X_8$$

$$AX_8 = \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ 0.5 \\ 0 \end{pmatrix} = 4X_9$$

$\therefore$ The largest eigen value is 4, corresponding eigen vector is $\begin{pmatrix} 1 \\ 0.5 \\ 0 \end{pmatrix}$

6	Find the Eigen values and Eigen vectors of the matrix $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{pmatrix}$, by using Jacobi method. (Nov/Dec 2022)
	Solution: Given $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{pmatrix}$ is a symmetric matrix, Here, the numerically largest off-diagonal element is $a_{23} = a_{32} = -1$ and diagonal elements are $a_{22} = a_{33} = 3$ The rotation matrix is $P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}$ The angle is $\theta = \frac{1}{2} \tan^{-1} \left(\frac{2a_{23}}{a_{22} - a_{33}} \right) = \frac{1}{2} \tan^{-1} \left(\frac{-2}{2-2} \right) = \frac{1}{2} \tan^{-1}(\infty) = \frac{\pi}{4}$ The rotation matrix is $P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \frac{\pi}{4} & -\sin \frac{\pi}{4} \\ 0 & \sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$ $D = P^T A P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$ $D = P^T A P = \frac{1}{2} \begin{pmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{pmatrix} \begin{pmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix}$ $D = P^T A P = \frac{1}{2} \begin{pmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} \sqrt{2} & 0 & 0 \\ 0 & 2 & -4 \\ 0 & 2 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}$ The eigen values are 1, 2, 4 and eigen vectors are $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$.
7	Using Jacobi method, find the eigen values and eigen vectors of the matrix $A = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix}$ (Apr/May 2022)
	Solution: Given $A = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix}$ is a symmetric matrix, Here, the numerically largest off-diagonal element is $a_{13} = a_{31} = -1$ and $a_{11} = a_{33} = 2$

The rotation matrix is $P = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$

The angle is $\theta = \frac{1}{2} \tan^{-1} \left(\frac{2a_{13}}{a_{11} - a_{33}} \right) = \frac{1}{2} \tan^{-1} \left(\frac{2(-1)}{2-2} \right) = \frac{\pi}{4}$

The rotation matrix $P = \begin{pmatrix} \cos \frac{\pi}{4} & 0 & \sin \frac{\pi}{4} \\ 0 & 1 & 0 \\ -\sin \frac{\pi}{4} & 0 & \cos \frac{\pi}{4} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}$

$D = P^T A P = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}$

$D = P^T A P = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{3}{\sqrt{2}} \\ 0 & 2 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{3}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

The eigen values are 1, 2, 3 and eigen vectors are $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$.

8

Using Jacobi method, find the eigen values and eigen vectors of the matrix $\begin{pmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 3 & \sqrt{2} \\ 2 & \sqrt{2} & 1 \end{pmatrix}$
(April/May 2023)

Solution:

Given $A = \begin{pmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 3 & \sqrt{2} \\ 2 & \sqrt{2} & 1 \end{pmatrix}$ is a symmetric matrix

Here, the numerically largest off-diagonal element is $a_{13} = a_{31} = 2$

The rotation matrix is $P = \begin{pmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{pmatrix}$

$\theta = \frac{1}{2} \tan^{-1} \left(\frac{2a_{13}}{a_{11} - a_{33}} \right) = \frac{1}{2} \tan^{-1} \left(\frac{2 \times 2}{1-1} \right) = \frac{1}{2} \tan^{-1}(\infty) = \frac{1}{2} \left(\frac{\pi}{2} \right) = \frac{\pi}{4}$

The rotation matrix is $P = \begin{pmatrix} \cos \frac{\pi}{4} & 0 & -\sin \frac{\pi}{4} \\ 0 & 1 & 0 \\ \sin \frac{\pi}{4} & 0 & \cos \frac{\pi}{4} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}$

$$D_1 = P_1^T A P_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 3 & \sqrt{2} \\ 2 & \sqrt{2} & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$D_1 = P_1^T A P_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{3}{\sqrt{2}} & \sqrt{2} & \frac{1}{\sqrt{2}} \\ 2 & 3 & 0 \\ \frac{3}{\sqrt{2}} & \sqrt{2} & -\frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 3 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

The diagonal matrix is not obtained and then we proceed from step 1

Consider $\begin{pmatrix} 3 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & -1 \end{pmatrix}$. The largest off-diagonal element is 2 (i.e.,) $a_{12} = a_{21} = 2$

The rotation matrix is $P = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{2a_{12}}{a_{11} - a_{22}} \right) = \frac{1}{2} \tan^{-1} \left(\frac{2 \times 2}{3 - 3} \right) = \frac{1}{2} \tan^{-1}(\infty) = \frac{1}{2} \left(\frac{\pi}{2} \right) = \frac{\pi}{4}$$

$$P_2 = \begin{pmatrix} \cos \frac{\pi}{4} & -\sin \frac{\pi}{4} & 0 \\ \sin \frac{\pi}{4} & \cos \frac{\pi}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$D_2 = P_2^T D_1 P_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$D_2 = P_2^T D_1 P_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{5}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{5}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

D is the diagonal matrix. Therefore the eigen values are 5, 1, -1

To find eigen vectors:

$$P = P_1 P_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

	Therefore the eigen vectors are $\begin{pmatrix} \frac{1}{2} \\ \frac{1}{\sqrt{2}} \\ \frac{1}{2} \end{pmatrix}, \begin{pmatrix} -\frac{1}{2} \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{2} \end{pmatrix}, \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}$
9	Using Jacobi method, find the eigen values and eigen vectors of the matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
	Solution: Given $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $\theta = \frac{1}{2} \tan^{-1} \left(\frac{2a_{12}}{a_{11} - a_{22}} \right) = \frac{1}{2} \tan^{-1} \left(\frac{2(1)}{0-0} \right) = \frac{1}{2} \tan^{-1}(\infty) = \frac{\pi}{4}$ $P = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$ $D = P^T A P = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$ $D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ The Eigen values are 1, -1 The Eigen vectors are $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$
10	Construct the QR decomposition for the following matrix $A = \begin{pmatrix} -4 & 2 & 2 \\ 3 & -3 & 3 \\ 6 & 6 & 0 \end{pmatrix}$ (Apr/May 2022 , Nov/Dec 2022)
	Solution: Let $X_1 = \begin{pmatrix} -4 \\ 3 \\ 6 \end{pmatrix}, X_2 = \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix}, X_3 = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}$ $r_{11} = \ X_1\ _2 = \sqrt{(-4)^2 + 3^2 + 6^2} = 7.810250$ $Q_1 = \frac{X_1}{r_{11}} = \begin{pmatrix} -0.512147 \\ 0.384111 \\ 0.768221 \end{pmatrix}$ $r_{12} = X_2^T Q_1 = \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix}^T \begin{pmatrix} -0.512147 \\ 0.384111 \\ 0.768221 \end{pmatrix} = 2.432699$

$$r_{13} = X_3^T Q_1 = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}^T \begin{pmatrix} -0.512147 \\ 0.384111 \\ 0.768221 \end{pmatrix} = 0.128037$$

$$\text{Replace } X_2 \text{ by } X_2 - r_{12} Q_1 \text{ we get } X_2 = \begin{pmatrix} 3.2459016 \\ -3.934426 \\ 4.131148 \end{pmatrix}$$

$$\text{Replace } X_3 \text{ by } X_3 - r_{13} Q_1 \text{ we get } X_3 = \begin{pmatrix} 2.065574 \\ 2.9508197 \\ 0.0983607 \end{pmatrix}$$

$$r_{22} = 6.563686, Q_2 = \begin{pmatrix} 0.494524 \\ -0.599423 \\ 0.629395 \end{pmatrix}, r_{23} = -0.809221$$

$$r_{33} = 3.511234, Q_3 = \begin{pmatrix} 0.702247 \\ 0.702247 \\ 0.117041 \end{pmatrix}$$

$$\therefore A = \begin{pmatrix} -4 & 2 & 2 \\ 3 & -3 & 3 \\ 6 & 6 & 0 \end{pmatrix} = QR = (Q_1, Q_2, Q_3) \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ 0 & r_{22} & r_{23} \\ 0 & 0 & r_{33} \end{pmatrix}$$

$$= \begin{pmatrix} -0.512147 & 0.494524 & 0.702247 \\ 0.384111 & -0.599423 & 0.702247 \\ 0.768221 & 0.629395 & 0.117041 \end{pmatrix} \begin{pmatrix} 7.81025 & 2.432701 & 0.128037 \\ 0 & 6.5636686 & -0.809222 \\ 0 & 0 & 3.511234 \end{pmatrix}$$

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Construct the QR decomposition for the following matrix A =

$$A = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

(April/May 2023)

Solution:

$$\text{Let } X_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}, X_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}, X_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}, X_4 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$

Iteration 1:

$$\text{If } k=1 \quad r_{11} = \|X_1\|_2 = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$Q_k = \frac{X_k}{r_{kk}} \Rightarrow Q_1 = \frac{X_1}{r_{11}} = \begin{pmatrix} 0 \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$r_{12} = X_2^T Q_1 = \frac{2}{\sqrt{3}}; \quad r_{13} = X_3^T Q_1 = \frac{2}{\sqrt{3}}; \quad r_{14} = X_4^T Q_1 = \frac{2}{\sqrt{3}}$$

Replace X_2 by $X_2 - r_{12} Q_1$, we get $X_2 = \begin{pmatrix} 1 \\ -\frac{2}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix}$

Replace X_3 by $X_3 - r_{13} Q_1$, we get $X_3 = \begin{pmatrix} 1 \\ \frac{1}{3} \\ -\frac{2}{3} \\ \frac{1}{3} \end{pmatrix}$

Replace X_4 by $X_4 - r_{14} Q_1$, we get $X_4 = \begin{pmatrix} 1 \\ \frac{1}{3} \\ \frac{1}{3} \\ -\frac{2}{3} \end{pmatrix}$

If $k=2$ $r_{22} = \|\mathbf{X}_2\|_2 = \frac{15}{\sqrt{3}}$

$Q_k = \frac{X_k}{r_{kk}} \Rightarrow Q_2 = \frac{X_2}{r_{22}} = \frac{3}{\sqrt{15}} \begin{pmatrix} 1 \\ -\frac{2}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix} = \begin{pmatrix} \frac{3}{\sqrt{15}} \\ -\frac{2}{\sqrt{15}} \\ \frac{1}{\sqrt{15}} \\ \frac{1}{\sqrt{15}} \end{pmatrix}$

$r_{23} = X_3^T Q_2 = \frac{2}{\sqrt{15}}; \quad r_{24} = X_4^T Q_2 = \frac{2}{\sqrt{15}}$

Replace X_3 by $X_3 - r_{23} Q_2$ we get $X_3 = \begin{pmatrix} \frac{9}{15} \\ \frac{9}{15} \\ -\frac{12}{15} \\ \frac{3}{15} \end{pmatrix}$

Replace X_4 by $X_4 - r_{24} Q_2$ we get $X_4 =$

$$\begin{pmatrix} \frac{3}{5} \\ \frac{3}{5} \\ \frac{1}{5} \\ \frac{-4}{5} \end{pmatrix}$$

If $k=3$ $r_{33} = \|X_3\|_2 = \sqrt{\frac{7}{5}}$

$$Q_k = \frac{X_k}{r_{kk}} \Rightarrow Q_3 = \frac{X_3}{r_{33}} = \sqrt{\frac{5}{7}} \begin{pmatrix} \frac{9}{15} \\ \frac{9}{15} \\ \frac{-12}{15} \\ \frac{3}{15} \end{pmatrix}$$

$$r_{34} = X_4^T Q_3 = \frac{2}{\sqrt{35}}$$

Replace X_4 by $X_4 - r_{34} Q_3$ we get $X_4 =$

$$\begin{pmatrix} \frac{3}{7} \\ \frac{3}{7} \\ \frac{3}{7} \\ \frac{-6}{7} \end{pmatrix}$$

If $k=4$ $r_{44} = \|X_4\|_2 = \frac{3}{\sqrt{7}}$

$$Q_k = \frac{X_k}{r_{kk}} \Rightarrow Q_4 = \frac{X_4}{r_{44}} = \frac{\sqrt{7}}{3} \begin{pmatrix} \frac{3}{7} \\ \frac{3}{7} \\ \frac{3}{7} \\ \frac{-6}{7} \end{pmatrix}$$

$$\therefore A = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \frac{3}{\sqrt{15}} & \frac{3}{\sqrt{35}} & \frac{1}{\sqrt{7}} \\ \frac{1}{\sqrt{3}} & \frac{-2}{\sqrt{15}} & \frac{3}{\sqrt{35}} & \frac{1}{\sqrt{7}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{15}} & \frac{1}{\sqrt{35}} & \frac{1}{\sqrt{7}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{15}} & \frac{-4}{\sqrt{35}} & \frac{-2}{\sqrt{7}} \end{pmatrix} \begin{pmatrix} \sqrt{3} & \frac{2}{\sqrt{3}} & \frac{2}{\sqrt{3}} & \frac{2}{\sqrt{3}} \\ 0 & \frac{\sqrt{15}}{3} & \frac{2}{\sqrt{15}} & \frac{2}{\sqrt{15}} \\ 0 & 0 & \frac{\sqrt{7}}{\sqrt{5}} & \frac{2}{\sqrt{35}} \\ 0 & 0 & 0 & \frac{3}{\sqrt{7}} \end{pmatrix}$$

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Find the singular value decomposition of $\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}$.

Solution:

$$\text{Let } A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}; \quad A^T A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}$$

The characteristic equation of $A^T A$ is $\lambda^2 - 6\lambda = 0 \Rightarrow \lambda = 0, 6$

The eigen values are $\lambda = 0, 6$

The eigen vectors are $X_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, X_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

Normalized eigen vectors are $X_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, X_2 = \begin{bmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$

$$D = [\sqrt{6}]; \quad \Sigma = \begin{bmatrix} \sqrt{6} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$V = [V_1 | V_2]$$

$$V = [V_1 | V_2] = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$U_1 = AV_1 D^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} \end{bmatrix} = \begin{bmatrix} 0.5773 \\ 0.5773 \\ 0.5773 \end{bmatrix}$$

$$U = U_1 I = \begin{bmatrix} 0.5773 & 1 & 0 & 0 \\ 0.5773 & 0 & 1 & 0 \\ 0.5773 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.5773 & 1 & 0 \\ 0.5773 & 0 & 1 \\ 0.5773 & 0 & 0 \end{bmatrix}$$

The first three column of this matrix forms a maximal set of linearly independent column vectors. Deleting the last 2 column and applying Gram's Schmidt process, we get

$$U = \begin{bmatrix} 0.5772 & 0.9998 & 0.3331 \\ 0.5572 & 0 & 1.333 \\ 0.5572 & 0 & 0.3332 \end{bmatrix}$$

The required SVD is $A = U \Sigma V^T = \begin{bmatrix} 0.5772 & 0.9998 & 0.3331 \\ 0.5572 & 0 & 1.333 \\ 0.5572 & 0 & 0.3332 \end{bmatrix} \begin{bmatrix} \sqrt{6} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$.

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Find the singular value decomposition of $\begin{pmatrix} 2 & 2 & -2 \\ 2 & 2 & -2 \\ -2 & -2 & 6 \end{pmatrix}$.

Solution:

Let $A = \begin{pmatrix} 2 & 2 & -2 \\ 2 & 2 & -2 \\ -2 & -2 & 6 \end{pmatrix}$

$$A^T A = \begin{pmatrix} 2 & 2 & -2 \\ 2 & 2 & -2 \\ -2 & -2 & 6 \end{pmatrix} \begin{pmatrix} 2 & 2 & -2 \\ 2 & 2 & -2 \\ -2 & -2 & 6 \end{pmatrix} = \begin{pmatrix} 12 & 12 & -20 \\ 12 & 12 & -20 \\ -20 & -20 & 44 \end{pmatrix}$$

The characteristic equation of $A^T A$ is $\lambda^3 - 68\lambda^2 + 256\lambda = 0 \Rightarrow \lambda = 0, 4, 64$

The eigen values are 0, 4, 64

The eigen vectors are $X_1 = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}, X_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, X_3 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

Normalized eigen vectors are $X_1 = \begin{pmatrix} \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ \frac{-2}{\sqrt{6}} \end{pmatrix}, X_2 = \begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}, X_3 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \\ 0 \end{pmatrix}$

$$D = \begin{bmatrix} \sqrt{64} & 0 \\ 0 & \sqrt{4} \end{bmatrix} = \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

To Find $V = [V_1 | V_2]$

$$V_1 = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix} \quad V_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \\ 0 \end{bmatrix} \Rightarrow V = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} \\ \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 \end{bmatrix}$$

$$U_1 = AV_1D^{-1} = \begin{pmatrix} 2 & 2 & -2 \\ 2 & 2 & -2 \\ -2 & -2 & 6 \end{pmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} \frac{1}{8} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

$$U = U_1I = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 1 & 0 & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 & 1 & 0 \\ \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 1 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 \\ \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 \end{bmatrix}$$

The first three column of this matrix forms a maximal set of linearly independent column vectors. Deleting the last 2 column and applying Gram's Schmidt process, we get

$$U = (Q_1 \ Q_2 \ Q_3) = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{5}} \\ \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 \end{bmatrix}$$

$$\text{The required SVD is } A = U \Sigma V^T = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{5}} \\ \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 \end{bmatrix} \begin{bmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{-2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} & 0 \end{bmatrix}$$